Atlantium
Enlightened Water Technologies

UV Water Treatment
Atlantium Hydro-Optic™ Solutions

User Manual

Guide to the Atlantium Hydro-Optic UV System RZ300 Series

Standard Control Module

Models RZ300-11
RZ300-12
RZ300-13
RZ300-14

For General Applications

Supported Electrical Voltage: 400/440/480VAC

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System and accessory specifications are subject to change without notice.

IMPORTANT NOTICE

It is strictly forbidden to alter/change the system hardware in any way.

Any change made without written permission from Atlantium shall void the warranty.

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1 Introduction

Congratulations on purchasing an Atlantium **RZ300 Series** Hydro-Optic UV Water Treatment System. This user guide provides you with clear guidelines for preparing the site for installation, correct installation and software configuration procedures, and a good understanding of all system functions for maximum benefit from your Atlantium Unit.

The Atlantium **RZ300 Series** systems guarantee water bio-security at optimized energy efficiency, and meet EPA criteria for providing bio-security.

Atlantium systems are designed to provide applications for the following Industries:

- **Aquaculture**
- **Dairy**
- **Municipal**
- **Power Plants**
- **UPW**
- **Marine**
- **Bio-Pharma**
- **Bottled Water**
- **Beverages**
- **Beer**
- **Aquariums**
- **and more**

Applications

- **Water Disinfection** - meets US EPA criteria for providing bio-security
- **Chlorine Replacement** - provides a chemical-free alternative to chlorine for treating source water. It easily inactivates microorganisms including those that are resistant to chlorine and heat.
- **Non-Thermal Pasteurization** - replaces energy-intensive heat pasteurization of water - meets US FDA criteria for pasteurized equivalent water,.
- **Post-Active Carbon** - protects against breakthroughs and reduces bio-fouling downstream. Significantly reduces the frequency of aggressive cleaning, and lengthens the lifespan of RO membranes
- **Dechlorination / Chlorine Destruction -** efficient, clean, non-chemical alternative to activated carbon filter or sodium bio-sulfite. Reduces risk, better protection, lower operating expenses.
- **Ozone Destruction** – efficient, cost-effective solution.
- **Ultra-pure water (UPW)** - efficient, cost-effective solution that complies with the uncommonly stringent specifications for ***high-purity water.***
- And more.

Patented Innovative Technology

- Field proven worldwide, used by world-leading brands
- Based on Hydro-Optic engineering that optimizes UV efficiency and enables uniform dose distribution
- Medium Pressure High Intensity UV effective in cold & warm water, compromises microorganisms' repair mechanisms for long-term bio-security
- Guaranteed required dose for complete microbial inactivation.
- Sustainable operation provides the security and stability that the industry demands.

Guaranteed Dose Delivery

- Unique monitoring & control software ensures that required UV dose is being delivered at all times.
- Dedicated UVIS sensors per lamp for real-time monitoring
- Automatically adjusts lamp power based on real-time analyses of critical parameters such as water flow rate, lamp intensity and water transmittance to UV (UVT).
- A user-friendly interface displays the monitored data in real-time including the actual delivered dose as well as the other critical parameters, so you can verify that you are actually getting the dose that you need.

Lower Electrical Consumption

- Hydraulic and Fiber optic principles applied to UV system design enable:
- Optimized UV power and uniform dose distribution.
- Efficient use of power, with lower electrical consumption than low or medium-pressure-based UV systems.

Minimum Down Time

- True in-line system.
- Quick lamp replacement (4 minutes) and better safety.

Integrates with your Process Line

- Field proven installation.
- With no additional software, you can integrate the Atlantium system into your current process using Atlantium's customizable settings.

Regulatory Compliance

- Meets the most stringent regulatory criteria including FDA and EPA.
- Automatically generated reports at the push of ab button for proof of compliance and traceability.
- Provides validated minimum required dose and not average dose.

Efficiency and Easy Maintenance

- For total efficiency and dependable performance, the Atlantium system monitors the lamps for performance, ensuring that a lamp is replaced only when absolutely needed.
- By following the easy preventive maintenance programs you keep the system working at peak performance.
- The **Viewport** allows you to safely peek at what's going on inside.

1.1 Before You Begin

Before you begin using this product, or any installation or service operation, please read the following safety information:

- Attention to these warnings helps prevent personal injuries and damage.
- It is your responsibility to use the product in an appropriate manner.
- This product is designed for use solely indoors.
- You are responsible if the product is used for other than its designated purpose or in disregard of Atlantium instructions. Atlantium shall assume no responsibility in any way for such use of the product.
- The product is used for its designated purpose if it is used in accordance with its product documentation and within its performance limits.
- Using the product requires technical skills. It is therefore essential that only skilled and specialized staff or thoroughly trained personnel be allowed to use the product.
- Keep the basic safety instructions and the product documentation in a safe place and pass them on to the users.
- Applicable local or national safety regulations and rules for the prevention of accidents must be observed at all times.

1.2 Tags and Their Meaning

The following indicators are used in the product documentation to warn the reader about risks and dangers

.

	DANGER! Indicates a hazardous situation which, if not avoided, will result in death or serious injury.
	WARNING! Indicates a hazardous situation which, if not avoided, could result in death or serious injury.
	CAUTION! Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury.
	ATTENTION! Indicates the possibility of incorrect operation which can result in damage to the product.
	Indicates a hazardous situation involving electricity which, if not avoided, can result in death or serious injury.
	Indicates a hazardous situation involving Electrostatic Discharge (ESD), which, if not avoided, can result in damage to the product.
	Indicates a hazardous situation involving UV light which, if not avoided, can result in death or serious injury.
	Indicates a hazardous situation involving a hot surface which, if not avoided, can result in death or serious injury.
	Indicates a hazardous situation involving working in an enclosure with moving parts, which if not avoided, can result in serious injury to the hands.
	Indicates a hazardous situation involving mercury, a poisonous material which, if not avoided, can result in death or serious injury.

	Indicates a hazardous situation involving caustic chemicals which, if not avoided, can result in death or serious damage.
	Indicates that components or equipment are heavy and care is to be taken to avoid lifting incorrectly. Incorrect lifting can be dangerous to the personnel lifting and may result in dropping and damaging the components or equipment.
	Indicates that components or materials to be discarded are classified as hazardous waste and must be disposed of appropriately.
	Indicates information related to installation, safety or use of system.

2 About the RZ300 System

This Atlantium **RZ300-11 RZ300-12**, **RZ300-13**, and **RZ300-14** Hydro Optic Water Disinfection System includes a disinfection unit, a Ballast Module configuration, and Control Module. The quartz disinfection chamber takes advantage of ***Total Internal Reflection*** and reflects UV light back into the water to uniformly distribute the UV dose. Water flows unimpeded, resulting in low head-loss.

The specially designed short (Arc length) MPHI lamps deliver a broad spectrum disinfection that irrevocably damages DNA and proteins in microorganisms. Housing Atlantium Quick Connect MPHI lamps within a thick quartz tube significantly reduces risk of breakage and its consequences. A UV Intensity Sensor (UVIS) provides continuous measurement of each lamp's output.

In addition, a UVT Analyzer reports and tracks the water quality and a UV-proof Viewport allows you to view the inner workings. An optional cleaning system cleans the surface of the quartz tubes automatically.

2.1 The RZ300 Series Units

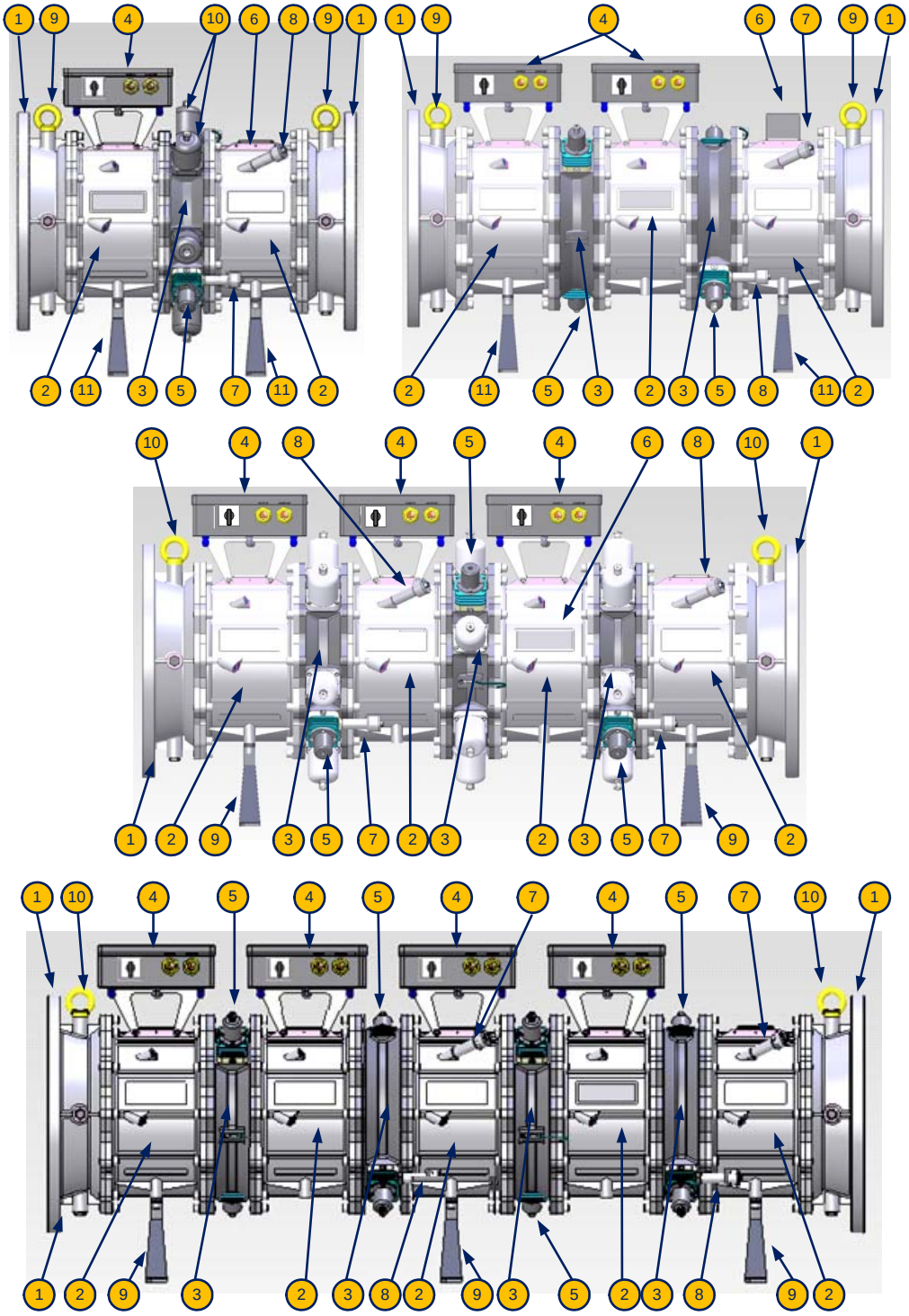


Figure 2-1: The **RZ300-11 RZ300-12**, **RZ300-13**, and **RZ300-14** Units

All of the system types may include the optional Ultrasonic Cleaner.

Table 2-1: Atlantium Unit Key

#	Item	Description
1	Flange interface	The unit connects to the facility piping via a flange interface as per customer order
2	Quartz Disinfection Chamber	The quartz pipe in which the disinfection occurs
3	Lamp Chamber	The chamber that holds the lamp and optional Cleaning system
4	Connection Box	The box that sits on top of each lamp unit and holds the electrical circuitry
5	Lamp Arc length (mm / inch): Lamp total length (arc + ceramic ends) (mm / inch):	Lamp encased in quartz tube assembly Medium Pressure High Intensity (MPHI)
6	Viewport	The UV-proof window to view the Unit working.
7	UVT Analyzer	The sensor that measures the UV transmittance through the water. Together with the UV Intensity sensor and the Control Module, it analyzes, tracks and reports the water quality.
8	UV Intensity Sensor	The sensor that measures the UV intensity in the Lamp chamber and feeds it to the real time dose calculation.
9	Lifting Support	A lifting element used when lifting the Unit.
11	Triangle Support Brackets	Brackets that are used to support and balance a unit anchored to customer supplied supports or surface that hold up the unit at least 750mm / 29.5inch above the floor, see ICD.

2.2 RZ300 Systems' Interface Control Drawings

For the ICDs of the systems in the **RZ300 Series**, consult the accompanying ICD files.

2.3 Atlantium System Footprint

The Atlantium system has a small, compact footprint that facilitates in-line installation in any process train.

Table 2-2: **RZ300-11/12/13/14** Physical Specifications

Item	Detail	
lamps **RZ300-11** **RZ300-12** **RZ300-13** **RZ300-14**	 1 2 3 4	
Atlantium Unit Construction Materials: Casing: Quartz Chamber:	 Electro-polished stainless steel 316 High grade fused silica (Quartz)	
Standard Pipe Flange Interface Options:	DIN2576 DN350 PIN16	14” 150-lbs ANSI
Unit Length (mm / inch) **RZ300-11** **RZ300-12** **RZ300-13** **RZ300-14**	 782 / 30.8 1100 / 43.3 1418 / 55.8 1736 / 68.4	 796.2 / 31.35 1104 / 43.9 1432 / 56.4 1754 / 69
Flange Inner ∅ Diameter (mm / inch)	350 / 13.8	355.6 / 14
Flange Outer ∅ Diameter (mm / inch)	505 / 19.9	533.4 / 21
Width at Widest Point (mm / inch)	505 / 19.9	505 / 19.9
Bolt Circle Diameter (mm / inch)	460 / 18.11	476.3 / 18.75
Bolt Hole Diameter (mm / inch)	22 / 0.87	28.4 / 1.12
Number of Bolts	16	12
Unit Volume (Liter / Gallon) **RZ300-11** **RZ300-12** **RZ300-13** **RZ300-14**	 60 / 15.85 83 / 22 105 / 27.7 127 / 33.5	 60 / 15.85 83 / 22 105 / 27.7 127 / 33.5
Weight (without water) (Kg / lb) **RZ300-11** **RZ300-12** **RZ300-13** **RZ300-14**	 140 / 309 184 / 406 228 / 503 249 / 549	 160 / 353 205 / 452 228 / 503 249 / 549
Service Clearance on each side (mm / inch)	750 / 29.5	750 / 29.5
Minimum Height above Floor Level (mm / inch)	750 / 29.5	750 / 29.5

Table 2-2: **RZ300-11/12/13/14** Physical Specifications

Item	Detail	
Maximum Working Pressure	7 Bars / 102 psi	7 Bars / 102 psi
Maximum Flow Rate	Application Dependent	Application Dependent
Distance between Triangle Support Brackets (mm / inch) **RZ300-11** **RZ300-12** **RZ300-13** **RZ300-14**	 318 / 12.5 636 / 25 954 / 37.6 636 / 25	 318 / 12.5 636 / 25 954 / 37.6 636 / 25
Distance between Bolt Holes on Supports	345 / 13.6	345 / 13.6
Bracket Nut / Bolt Size (mm / inch)	13 / 0.5	13 / 0.5

2.4 The Ballast Module

Ballast Module is a water-proof easy plug-n-play box that contains the ballast power supply and related circuitry. Its convenient plugs make it easy to service. Its modular design makes it serviceable and flexible and easier to fit into an existing plant. A Ballast Module is supplied for each lamp in the Atlantium system. Its heat sink frame complies with industry standards and contains one cooling fan on the top. It can be mounted on the wall or other suitable anchoring point. The location of the Ballast Module must be designated according to the maximum distance needed for positioning the **Cable Harness** of the Atlantium Unit. The **Cable Harness** is **5m / 16.4'**(also available in **10m / 32.8'** length). For systems with more than one lamp, the Ballast Modules can be positioned one above the other or next to one another. The Ballast Module must have at least 300 mm (11.8 inches) of clearance on its right and left sides for proper ventilation and 300mm (11.8 inches) of clearance on the bottom for cable routing.

Refer to the **Ballast Module ICD** accompanying this documentation.

For more details on mounting, see ***Installing the Ballast Module(s)*** on page **69**.

Table 2-3: Ballast Module Physical Details

Physical	Details
Dimensions (H x W x D) (mm/inch):	
Ballast Module B	313 / 12.3 x 228 / 9.0 x 548 / 21.6
Ballast Module C	486.7 / 19.16 x 348 / 13.7 x 114.5 / 4.51

Table 2-3: Ballast Module Physical Details

Physical	Details
Weight (Kg/Lb): Ballast Module B Ballast Module C	 21.5/ 47.4 6.5 / 14.3
Heat Load per Ballast Module: (For calculating space cooling requirements)	210W
Protection Category:	IP 56.

2.5 Electrical Usage

Utilize a Residual Current Device (RCD) that is Type A with 3000A surge resistance.

Atlantium recommends ABB P/N F204 A-xx/0.03 AP-R (xx represents the current rating according to the system load. Select the appropriate RCD accordingly. Calculating the value, takes into account the power consumption of the system (see the table in the section below) and the mains voltage of your locale.

The tables below detail the Atlantium Unit's electrical details and power consumption.

Table 2-4: **RZ300 Series** Electrical Details

Electrical	Details
Main power source	400 VAC 3~ + N + PE 50/60Hz 440Y/254 VAC 3~ + PE 50/60Hz 480Y/277 VAC 3~ + PE 50/60Hz
Second power source (only for 440-480 main source)	100-240 VAC 1ph 50/60Hz
Maximum allowed voltage tolerance	±10% from the nominal voltage

2.5.1 Power Consumption and Local Voltage Requirements

The table below details the power consumption or the Atlantium system according to local voltage.

Table 2-5: **RZ300-11/12** Power Consumption

# of lamps :			1		2	
Nominal lamp power (kW):			4.2		8.4	
Cleaner type	Feeding type	Feeding voltage (V)	Current consumption each phase (A)	Required circuit breaker	Current consumption each phase (A)	Required circuit breaker
No cleaner	Direct Feeding	400	6.7	3x10A	13.4	3x16A

Table 2-5: **RZ300-11/12** Power Consumption

# of lamps :			1		2	
Nominal lamp power (kW):			4.2		8.4	
Cleaner type	Feeding type	Feeding voltage (V)	Current consumption each phase (A)	Required circuit breaker	Current consumption each phase (A)	Required circuit breaker
No cleaner	Through Transformer (Power Module)	440	6.1	3x10A	12.1	3x16A
		480	5.6	3x10A	11.1	3x16A

Table 2-6: **RZ300-13** Power Consumption

# of lamps :			3					
Nominal lamp power (kW):			12.6					
System branch containment			Lamp 1+2 + optional cleaner		Lamp 3		Main feeding	
Cleaner type	Feeding type	Feeding voltage (V)	Current consumption each phase (A)	Required circuit breaker	Current consumption each phase (A)	Required circuit breaker	Current consumption each phase (A)	Required circuit breaker
No cleaner	Direct Feeding	400	20.0	3x25A	6.7	3x10A	20.1	3x25A

Table 2-7: **RZ300-13** Power Consumption

# of lamps :			3					
Nominal lamp power (kW):			16.8					
System branch containment			Lamp 1+2 + optional cleaner		Lamp 3		Main feeding	
Cleaner type	Feeding type	Feeding voltage (V)	Current consumption each phase (A)	Required circuit breaker	Current consumption each phase (A)	Required circuit breaker	Current consumption each phase (A)	Required circuit breaker
No cleaner	Through Transformer (Power Module)	440	12.1	3x10A	6.1	3x10A	18.2	3x25A
		480	11.1	3x16A	5.6	3x10A	16.7	3x20A

Table 2-8: **RZ300-14** Power Consumption

# of lamps :			4					
Nominal lamp power (kW):			16.8					
System Branch containment			Lamp 1+2 + optional cleaner		Lamp 3+4		Main feeding	
Cleaner type	Feeding type	Feeding voltage (V)	Current consumption each phase (A)	Required circuit breaker	Current consumption each phase (A)	Required circuit breaker	Current consumption each phase (A)	Required circuit breaker
No cleaner	Direct Feeding	400	13.4	3x16A	13.4	3x16A	26.8	3x32A
No cleaner	Through Transformer (Power Module)	440	12.1	3x16A	12.1	3x16A	24.2	3x32A
		480	11.1	3x16A	11.1	3x16A	22.2	3x32A

2.6 Surge Protection

Every Atlantium system **must** be protected by a device that prevents electrical surges and fluctuations from causing damage. There are three types of surge protection:

- UPS (Uninterruptible Power Supply) has an internal battery that allows the equipment plugged into it to continue to run in the event of a power outage until power is restored. It also has the benefit of performing the same functionality of surge protectors, protecting against spikes and surges. If needed, the UPS allows the system to be shut down cleanly if power is out for a lengthy period.
- Surge protection device protects your equipment from variations in electrical current, such as surges and spikes.
- OVPD module (Over Voltage Protection Device) - available from Atlantium. see below.

2.6.1 Surge Protector Devices

Surge protection devices are vital for the protection of industrial power networks. These devices are provided by Atlantium and uses a combination of 3-electrode gas discharge tubes and fast clamping diodes. Typical applications include industrial processing equipment, transmission systems, I/O cards, probes, actuators, and displays.

Table 2-9: Surge Protector Details

Atlantium PN	DATA SHEETS	Module PN	Description
EP0002300	DOC002300	8859660000	SURGE PROTECTOR 400V 3P 40KA TNC CONFIG
EP0002000	DOC002000	4983-DS277-404	SURGE PROTECTOR 440-480 3P 40KA
EP0002600	DOC002600	4983-DS120-402	SURGE PROTECTOR L + N, 115V
EP0002800	DOC002800	1352740000	SURGE PROTECTOR L + N, 230V

2.6.2 The OVPD Module

The Over Voltage Protection Device (OVPD Module) may be required in locations where fluctuations in the voltage may occur. If supplied, see the **OVPD Module ICD** accompanying the documentation. Consult your Atlantium representative.

A OVPD Module is not required for Ballast Module version C.

Table 2-10: OVPD Module Requirements Details

System Lamps Total Power	Mains Voltage	
	480VAC	440VAC
<7000W	An OVPD Unit is usually not required	An OVPD Unit is usually not required
<16800W	Requires an OVPD unit	An OVPD Unit is usually not required
≥16800W	Requires an OVPD unit	Requires an OVPD unit

Table 2-11: OVPD Module Details

Physical	Details
Dimensions (W x H x D) (mm/inch):	**400/15.74 x 600/23.62 x 282/11.10**
Weight (Kg/Lb.):	**20/44**
Minimum Clearance for Electrical Cables Routing on each of two sides (top/bottom) (mm/inch): **Total Clearance (mm/inch):**	**300/18.81** **1200/47.24**

Table 2-11: OVPD Module Details

Physical	Details
Electrical	
Voltage:	Up to 520V/300V*
Current:	Up to 42A or Up to 75A
Frequency:	DC to 60Hz

2.7 The Control Module

The Atlantium system is monitored and controlled from a **Control Module**. It can be mounted on a wall or suitable anchoring point up to 50 meters (164.04 ft.) from the Atlantium Unit. However, it should be located in close proximity to the Atlantium Unit where it is easy to reach the touch screen and read the monitoring information.

See the **Standard Control Module ICD** accompanying the documentation.

Table 2-12: Control Module Details

Physical	Details
Dimensions (W x D x H) (mm/inch):	400/15.75 x 300/11.82 x 144.4/6.12
Weight (Kg/Lb.):	0.9/1.9
Minimum Clearance for Electrical Cables Routing (mm/ inch):	50/2
Electrical	
Maximum Power Consumption:	Maximum 40W
Voltage:	1 phase 120-240 Voltage
Protection Category:	IP 54.

2.8 Power Module

The Power Module provides the step-down transformer required to interface between the voltage of the Atlantium Unit and the local voltage where relevant.

If supplied, see the **Power Module ICD** accompanying the documentation.

The Power Module is required for:

- In localities with electrical power distribution of 440-480VAC (such as in the USA and Mexico), where the Atlantium system includes the Ultrasonic Cleaner.

- In localities with electrical power distribution of less than 400VAC, such as 200VAC used in Japan.
- In localities with electrical power distribution of over than 480VAC, such as 600VAC used in Canada.

Table 2-13: Power Module Details

Physical	Details
Dimensions (W x D x H) (mm/inch):	18.97 / 482 x 10.35 / 263 x 19.68 / 500
Minimum area with open door:	15.67 / 398
Minimum Clearance for Electrical Cables Routing:	5 0/ 2
Weight (Kg/Lb.): **RZ300-13:** Without Ultrasonic Cleaner Power Module 1 Power Module 2 With Ultrasonic Cleaner Power Module 1 Power Module 2 **RZ300-14:** Without Ultrasonic Cleaner Power Module 1 Power Module 2 With Ultrasonic Cleaner Power Module 1 Power Module 2	 58 / 127.8 40 / 88.1 58 / 127.8 58 / 127.8 58 / 127.8 50 / 110.2 58 / 127.8 58 / 127.8
Electrical	
Voltage:	3 phase x 200 / 230 / 400 / 440 / 480 Voltage
Protection Category:	IP 56.

2.9 Environmental Requirements

The Atlantium system is to be situated indoors in an area that is adequately cooled and with sufficient air flow. However, there have been many successful Atlantium installations in less friendly environments. Consult your Atlantium representative to choose the most suitable location.

Table 2-14: Atlantium System Environmental Details

Environmental	Details
Maximum ambient air temperature:	45°C (113°F)
Maximum ambient air temperature in the Ballast Modules environment:	45°C (113°F)
Maximum atmospheric relative humidity:	90%
Water Temperature: Maximum – high temperature: For water temperature above 70°C (160°F) consult with Atlantium service engineers Minimum – cold temperature:	99°C(210.2°F) No limit

2.10 Regulatory Compliance

The Atlantium System complies with the following regulatory standards:

- Long Term 2 Enhanced Surface Water Treatment Rule of January 5th, 2006
- 3rd party validated by HDR/HydroQual, Inc.
- EMC directive 2014/30/EU
- European Low Voltage Directive (LVD), 2014/35/EU for electrical safety
- Council Directive 98 / 83 / EC of 3 November 1998 for the quality of water intended for human consumption
- ISO 9001:2008 Quality Management Standard – Development, Design, Production and Sales of Water Disinfection Systems
- US Federal Performance Standards 40 CFR 141.720
- All components have CE or UL 508 NRTL (MET) certification
- Resolucion exenta No 2.327 del 31 de diciembre de 2010, en el marco de la Ley General de Pesca y Acuicultura
- GOST Standards Institute, Russian Federation
- National Institute of Public Health, Poland

3 Getting Ready for Installation

Using the guidelines and information in this section, you can prepare the piping and electrical setup for installing the Atlantium Hydro Optic Water Treatment System. If you have any questions, consult with an Atlantium application engineer.

3.1 Planning

Consider where to locate the unit from the perspective of your process. Prepare your installation plan according to the planning points below, as well as the other guidelines in this manual.

3.1.1 Access

- **Premises Access**
  - Take into account that sufficient door space is required to transport the Atlantium Unit into the facility where it is to be placed.
- **Service Access**
  - Take into consideration that sufficient space is required for maintenance personal to access the Atlantium Unit for periodic component replacement.

3.1.2 Hydraulics

- **Piping**
  - Align the upstream and downstream piping with the designated location for the Atlantium Unit.
  - Prepare the flanges/ferrules and piping as specified according to ***The Piping Infrastructure*** on page 33. To avoid a step in the connection, make sure that the internal diameters where the Atlantium Unit and your piping match.
  - **Do not** use sealing material that contains graphite when sealing the flanges/ ferrules and gaskets as it accelerates the corrosive process on stainless-steel components, including those of the Atlantium system.
- **Mechanics**
  - Prepare the supports needed for the Atlantium Unit, noting the minimum level above the floor is at least 750mm / 29.5 inch between the floor and the bottom of the provided triangle support brackets.
  - Check that the piping on the inlet and outlet sides has adequate support structures so that there are no mechanical stresses that may endanger the Unit.

- Check the upstream and downstream piping for vibrations. Make sure that the piping to be attached to the Atlantium Unit is anchored and reinforced to protect the Unit against vibration. A high vibration level can cause damage to any hydraulic system, including the Atlantium unit. High vibration can be caused by an unbalanced pump or by sudden valve close.

- **Hydrodynamics**
  - Water hammer can cause damage to any hydraulic system, including the Atlantium Unit. Make sure that this phenomenon does not occur in your facility's water line. To prevent water hammer, consult your system engineer and check your facility system procedures, including:
    - When starting the water flow when the lines are empty, keep valves open so that the air in the pipes has a simple release pathway.
    - Sudden valve closures can cause water hammer. Make sure that valves close gradually, enough to avoid this problem.
    - Ensure that all pumps that impact the Atlantium system employ soft start-up procedures.
    - The Atlantium Unit must be totally filled with water. Assure that no air can be trapped in the unit and that air can be released. Verify through the Unit's viewport after installation that there are no bubbles.

3.1.3 Flow Data

- Evaluate your flow trends to determine whether you require a flow meter or flow switch. If the flow varies, a **flow meter** is required to detect and report on the flow coming into the unit. Make sure that there are to be **no** branches on or off the pipes between the flow meter and the Unit inlet. If the flow rate is static, a **flow switch** can be used instead.

3.1.4 Application

- Verify that the dose-related parameters, such as the UVT and the flow, are the same as those defined in the Atlantium application scope of supply.
- Determine what a low flow situation means to your process and whether you need to shut down the Unit (low flow indicates upstream problems) or recirculate (low flow normal) when the flow goes below the normal range.

3.2 Typical Installation Set-Up

Use the diagram below as a basis for your installation drawings. Be sure to include service access as you fit the Atlantium system into your facility (i.e., nearby pipes, electrical fixtures, support beams, etc.)

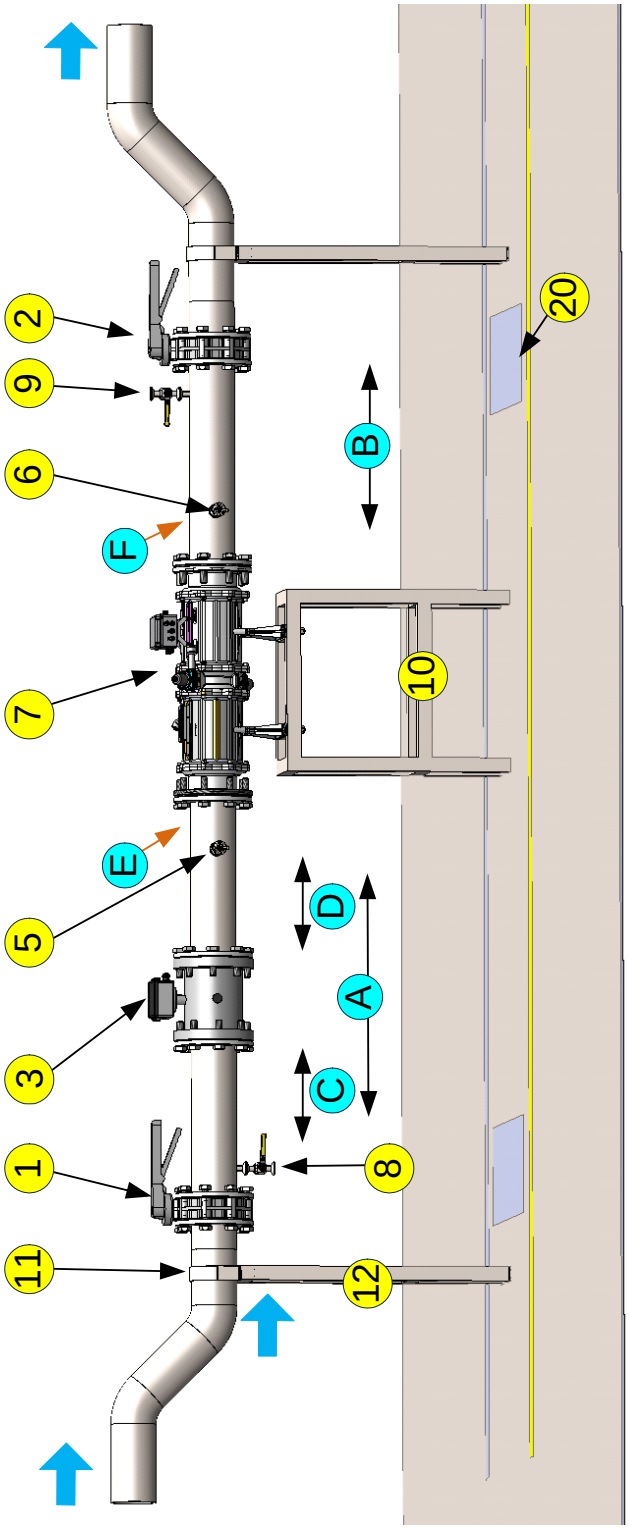


Figure 3-1: Typical Atlantium System Setup (No Bypass)

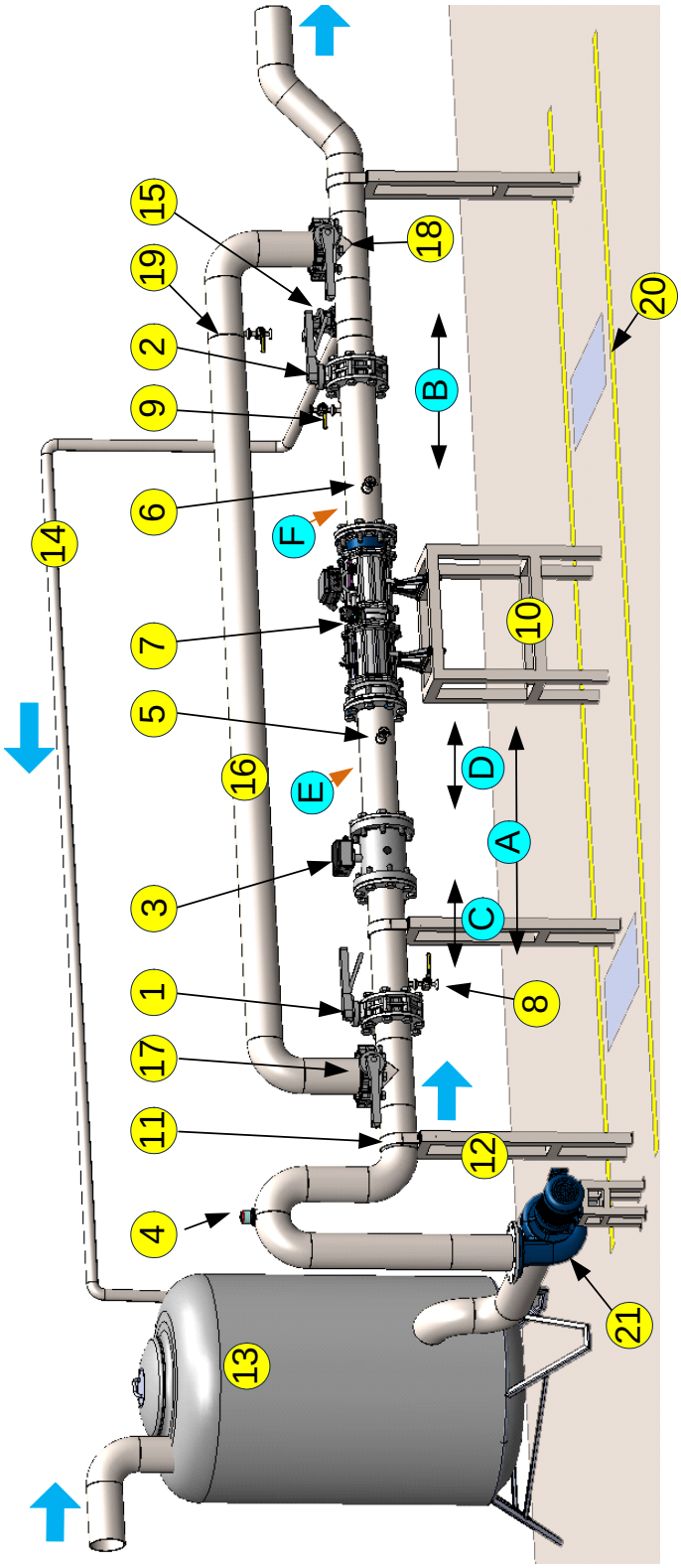


Figure 3-2: Typical Atlantium System Setup (With Bypass)

Table 3-1: Typical Atlantium System Component Setup Key

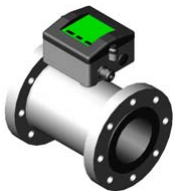
#	Description		Atlantium P/N or recommended and available at Atlantium
	Direction of water flow in the pipes		-
1	Inlet isolation Valve ■ Isolating Valve on the Inlet side		-
2	Outlet isolation Valve ■ Isolating Valve on the Outlet side		-
3	■ Flow Meter flow measurement device with 4-20mA analogue output signal ■ To be installed on the same line as the Atlantium system, with no branching before or after the flow meter ■ Follow the manufacturer's Installation instructions		Available Flow Meters: KROHNE OPTIFLUX 6000 KROHNE OPTIFLUX 2300
4	Air Release valve - To prevent air bubbles where necessary ■ Automatic Air Release valve is used to release air bubbles trapped within the line ■ Install at a high position to allow air bubbles to accumulate and release from this point		-
5/6	Inlet/Outlet Sampling Points ■ Position aseptic sampling valves **horizontally** (90^0) on the pipe center, within 50cm of the Unit ■ Aseptic sampling valve must be a dedicated valve for sampling only and resistant to alcohol and flame		■ Aseptic Sampling Valve Kit for Stainless Steel Pipe - KTB005700 ■ Aseptic Sampling Valve Kit for Plastic Pipe - KTB0011500
7	Generic Atlantium Unit		-

Table 3-1: Typical Atlantium System Component Setup Key

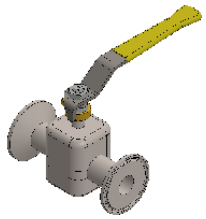
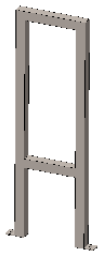
#	Description		Atlantium P/N or recommended and available at Atlantium
8	Draining/CIP Port - For performing CIP on the Atlantium Unit ■ The Draining port should be positioned close to the Inlet valve and between that and the Unit, ***on the bottom*** of the pipe ■ It is also used as one of the CIP ports and connects to a hose	 	■ CIP Kit - SAB012900 ■ CIP Valve Kit for Stainless Steel - KT0011600 ■ CIP Valve Kit for Plastic - KT0011700 (the valve itself is not part of the kits) See ***CIP Ports and CIP Kits*** on page **3-37.**, for information on CIP equipment.
9	CIP Port - For performing CIP on the Atlantium Unit ■ The CIP port should be positioned between the Unit and the Outlet valve, ***on top*** of the pipe, at the ***highest*** point before the isolation valve, in order to allow the pipes to be completely filled with solution		
10	System Supports ■ Supports are required to assure no mechanical stress on the Atlantium Unit		-
11	Support brackets ■ Brackets are used to hold the Atlantium Unit and pipes attached to the support mounts ■ Is to include internal flexible material to absorb vibrations and assure that the pipes and Atlantium Unit are mounted tightly and securely		-
12	Pipe Supports ■ Pipe Supports are required to assure no vibrations in the pipeline		-
13	Water tank		-
14	Recirculation Line >5m3/h (>20gpm (Optional) ■ A Recirculation line is highly recommended to keep water flowing constantly, thereby assure a high disinfection level		-

Table 3-1: Typical Atlantium System Component Setup Key

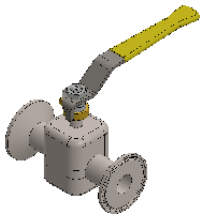
#	Description		Atlantium P/N or recommended and available at Atlantium
15	Recirculation Line Isolation valve (Optional)		-
16	By-Pass (Optional) ■ If the facility uses quartz corrosive CIP chemicals, such as caustic soda, use a By-Pass line to prevent damage to the Atlantium Unit		-
17	By-Pass Inlet isolation Valve (Optional)		-
18	By-Pass Outlet isolation Valve (Optional)		-
19	By-pass draining port (Optional) ■ The Draining port should be positioned on the bypass line, at the lowest point of the pipe	 	-
20	Drain ■ It is recommended to install a drain close to the draining valve		-
21	Pump ■ To prevent water hammer, the pump must be able to start up gradually.		-

Table 3-2: Typical Atlantium System Distances Setup Key

#	Description
A	Straight horizontal pipe between the Inlet isolation valve and the Atlantium Unit. For most applications, a minimum 10D (3000mm/ 118inch) of straight horizontal pipe between the Inlet isolation valve and the Atlantium Unit. For Power Plants, and other applications with space constraints, consult an Atlantium Application Engineer to find appropriate solutions.
B	Straight horizontal pipe between the Atlantium Unit and the Outlet isolation valve. For most application, a minimum 5D (1500mm/59inch) of straight horizontal pipe between the Atlantium Unit and the Outlet isolation valve. For Power Plants, and other applications with space constraints, consult an Atlantium Application Engineer to find appropriate solutions.
C	Some manufacturers require positioning the Flow Meter at a minimum of 5D (mm/inch) of straight horizontal pipe immediately preceding it on the Inlet side. Refer to the manufacturer's instructions.
D	Some manufacturers require positioning the Flow Meter at a minimum of 2D (600mm/24inch) of straight horizontal pipe immediately following it before the Atlantium Unit. Refer to the manufacturer's instructions.
E	Position the Inlet Sampling Point Maximum Distance from the Atlantium System Inlet .5 meters/1.5ft.
F	Position the Outlet Sampling Point Maximum Distance from the Atlantium System Outlet .5 meters/1.5ft.

3.3 Electrical Requirements

The electrical infrastructure is to be installed and in place prior to the system's delivery.

For information on the footprint of the components of the Atlantium system, see the section, ***The RZ300 Series Units*** on page 13. Use the Electrical ICD provided by Atlantium to help you plan the electrical infrastructure. See also ***Power Consumption and Local Voltage Requirements*** on page 19 and ***Connecting the Ballast Module Cables*** on page 72.

Take into account:

For locations with 440Y/254 VAC 3~ +PE 50/60HZ
or 480Y/277 VAC 3~ +PE 50/60HZ

The Atlantium System requires a balanced voltage supply (WYE configuration). If you do not have such a connection, you must create an appropriate power feeding using a 1:1 isolation transformer (where the ground is connected to the middle point of the star configuration at the secondary side of the transformer).

- Evaluate the quality of your power and the likelihood of blackout/brownouts, electrical surges or interruptions. Determine if you need power surge protection and or uninterrupted power supply if any.

- To verify that you order the correct lengths for cables (i.e., Cable Harness, power cables, data cable, etc.), measure the distance between the proposed location of the relevant Atlantium components and between them and the Power source.
  - Planning the placement of the Ballast Modules, **Control Module** and the path of the various cables must take into consideration the length of the **Cable Harness** and the other cables. Cables must be placed in a path that does not interfere with passage ways.
  - The power mains connections must be in the vicinity of the Ballast Modules, Control Module.
  - The power cable and connecting plug are to be obtained locally.
- If your region experience periodic power outages, consider including uninterrupted surge protection devices.
- Prepare cables of sufficient length to connect from the auxiliary equipment (i.e., the flow meter/flow switch, etc.) to the Atlantium system.
- See ***Preparing for the Electrical Wiring to the Mains*** on page **3-32.**

For Systems Utilizing Seawater

- Minimize the corrosion potential of the treated water by ensuring that there is sufficient grounding.
- Do not leave the Atlantium Unit standing with seawater. Either rinse with fresh water and leave the Atlantium Unit standing with fresh water or empty.

3.3.1 Preparing for the Electrical Wiring to the Mains

You must prepare the elements listed below for connecting the Atlantium system to the mains.

For preparing all electrical connections, be sure to use the *Electrical ICD* drawing provided to you by Atlantium with this manual. If no electrical drawings were provided please contact your Atlantium representative.

- Mains system area circuit breaker according to the total systems' power consumption to be connected to this line. See ***Electrical Usage*** on page **2-18.**
- For sites with 440/480VAC , plan to include an Over Voltage Protection Device (OVPD) (available from Atlantium) if needed. See ***The OVPD Module*** on page **2-21.**
- Install the Required Circuit Breakers, as defined in the Electrical ICD.

- If you intend to lengthen or replace the mains power cable connected to the Atlantium Unit with a longer cable, prepare the cables according to your location electrical distribution system requirements.

  - **For locations with 400 VAC: 5-wire system that contains 3 phases, Neutral and Ground (The Neutral and Ground are mandatory)**
  - **For locations with 440Y/254 VAC 3~ +PE 50/60HZ**
    or 480Y/277 VAC 3~ +PE 50/60HZ
    The Atlantium System requires a balanced voltage supply (WYE configuration). If you do not have such a connection, you must create an appropriate power feeding using a 1:1 isolation transformer (where the ground is connected to the middle point of the star configuration at the secondary side of the transformer).

- **The Control Module's Mains Cable**

  - For sites with 400VAC 3 phase + Neutral, the Control Module's Mains feeding voltage is 230 VAC (L1 to N or L2 to N or L3 to N).
  - For sites with 440/480VAC 3 phase, the Control Module cannot be fed from one phase of the 3 phase supply. The Control Module must be connected to a separate mains feeding, which is in the voltage range of 100-240 VAC.

3.4 The Piping Infrastructure

To review how to prepare the infrastructure, read through the typical installation section before you begin and refer to the section, ***About the RZ300 System*** on page 12.

Install the piping infrastructure before the Atlantium system arrives, but to avoid built-in mechanical stresses perform the final pipe adaptations and welds when the unit is actually situated in its permanent position.

ATTENTION!

- When sealing the flanges/ferrules and gaskets, **do not** use sealing material that contains graphite as it accelerates the corrosive process on stainless-steel components, including those of the Atlantium system.
- Be sure to align the piping on the inlet and outlet sides of the designated location for the Atlantium Unit with the unit to avoid mechanical stress on the Unit.
- Be sure to secure the piping around the designated location for the Atlantium Unit so that there is no vibration.
- Be sure to position adequate supports on the piping line needed to hold up and stabilize the process piping on the inlet and outlet side of the proposed location of the Atlantium unit.
- Determination of support requirements and design should be made by an on-site mechanical engineer.

3.5 Flow Measurement / Detection

The Atlantium unit requires a flow signal to calculate the dose and operate properly. The flow signal is a measurement of the actual flow or simply a flow/no-flow indicator. The flow signal is collected from either a flow meter or a flow switch.

- A flow meter is required in facilities where the flow rate is variable for any reason.
- A Flow switch can be used in facilities where the flow is at a fixed, steady rate.

The Flow meter or flow switch does not necessarily have to be installed in close proximity to the Atlantium Unit. It may be more remotely positioned, provided that it is located on the inlet side between it and the Unit and there is **no** branching or splits of the water flow.

ATTENTION!

The Atlantium system cannot be operated when there is no water or no water flow. Any damage caused due to operating the system without water or without water flow invalidates the Atlantium warranty.

3.5.1 Flow Meter

In most cases, a flow meter with a 4-20 mA output is required to measure the flow and automatically adjust the output of the UV system. The flow meter must be able to measure the expected maximum water flow of the specified Atlantium unit. The power source for the flow meter must come from an external source according to the manufacturer's instructions.

ATTENTION!

Be sure to provide the power source for the flow meter as part of the preparations for installing the Atlantium Unit.

Install the flow meter in strict accordance with the instructions supplied by its manufacturer.

Flow meters are available from Atlantium, such as:

- **KROHNE OPTIFLUX 6000** flow meter for high-purity water applications
- **KROHNE OPTIFLUX 2300** flow meter – OPTIFLUX 2000 flow meter, includes **KROHNE IFC 300** signal converter

Consult your Atlantium representative.

Examples of Flow Meters

Figure 3-3: KROHNE OPTIFLUX Flow Meters

Flow Switch

The flow switch tells the Atlantium Unit when water is or is not flowing. The recommended flow switch is Dry Contact with minimum flow sensing of 0.2m^3/hr (1gpm).

Install it in a vertical (12:00 o'clock) orientation on top of the pipe. Follow the manufacturer's instructions.

Figure 3-4: Examples of Flow Switch

3.6 Sampling Valve Components and Kits

Sampling valve kits are available from Atlantium.

One Accessory kit is required for each Sampling point.

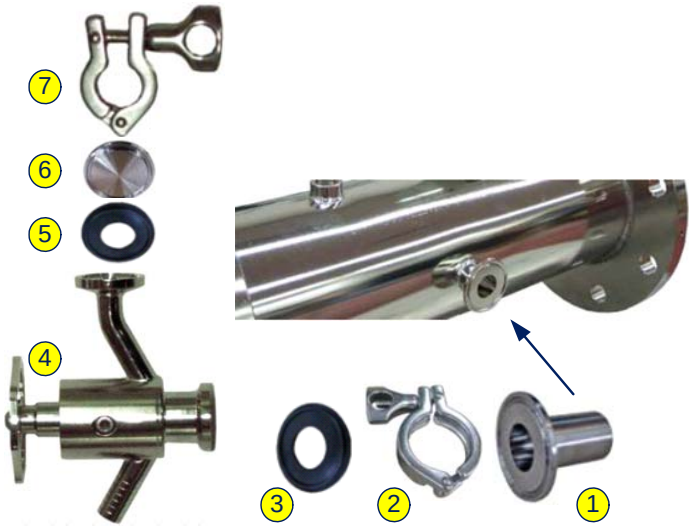


Figure 3-5: Sampling Valve Assembly

The table below details them. For information on installing the Sampling valves, see ***Installing the Inlet/Outlet Sampling Points*** on page **51**.

Table 3-3: Sampling Valve Components and Kits

	Catalog number	Description		Image	Kit for Stainless Steel Pipe KTB005700	Kit for Plastic Pipe KTB005750
1	FAC003700	Long welding ferrule 1” Adapter	For welding the tri-clamp port to the pipe		**2 Pcs**	-
1	FAB008100	Threaded Male Adapter 1/2” BSP	For connecting the tri clamp port to the plastic pipe		-	**2 Pcs**
2	FA0000045	Clamp 1.5", ST.ST 304	For connecting the valve to the welded port		**2 Pcs**	**2 Pcs**
3	FA0000046	Gasket 1", FJ31.05-E	For sealing the valve connected to the pipe		**2 Pcs**	**2 Pcs**
4	COB000500	Aseptic sampling valve, ST.ST 316L	For taking water samples		**2 Pcs**	**2 Pcs**
5	FA0000048	Gasket 1/2”, FJ31/25-E	For sealing the valve's flushing port		**2 Pcs**	**2 Pcs**
6	FA0000049	Solid End Cap, 1/2", ST.ST 304	For covering the valve's flushing port		**2 Pcs**	**2 Pcs**
7	FA00000047	Clamp 1/2”, ST.ST 304	For sealing the flushing port of the valve		**2 Pcs**	**2 Pcs**
	FA0008300	Round Brush - 15/60/150	For cleaning the valve during the process		**2 Pcs**	**2 Pcs**

3.7 Draining Valves

It is important to plan for draining the system. Install one draining valve at the lowest point of the pipeline, before or after the Atlantium unit. If you plan to install CIP ports, one of them can double as a draining valve. If there is a Bypass line used occasionally to bypass the Unit in any circumstance, add an additional draining valve to assure that the bypass line does become a potential source of standing water. A bypass line must be empty when not in use.

3.8 CIP Ports and CIP Kits

The CIP (Cleaning-In-Place) recirculation system is used for the periodic cleaning and sterilization process of the Atlantium unit's inner surfaces and quartz sleeve. The CIP Cart includes a reservoir, a pump, a supply hose, a return hose, and a pocket to hold these hoses. See the diagram below. Refer to **CIP Cart ICD drawing ICB012900**.

The CIP Cart:

For systems with 400VAC: catalog number: SAB012900

For systems with 440/480VAC: catalog number: SAB012950

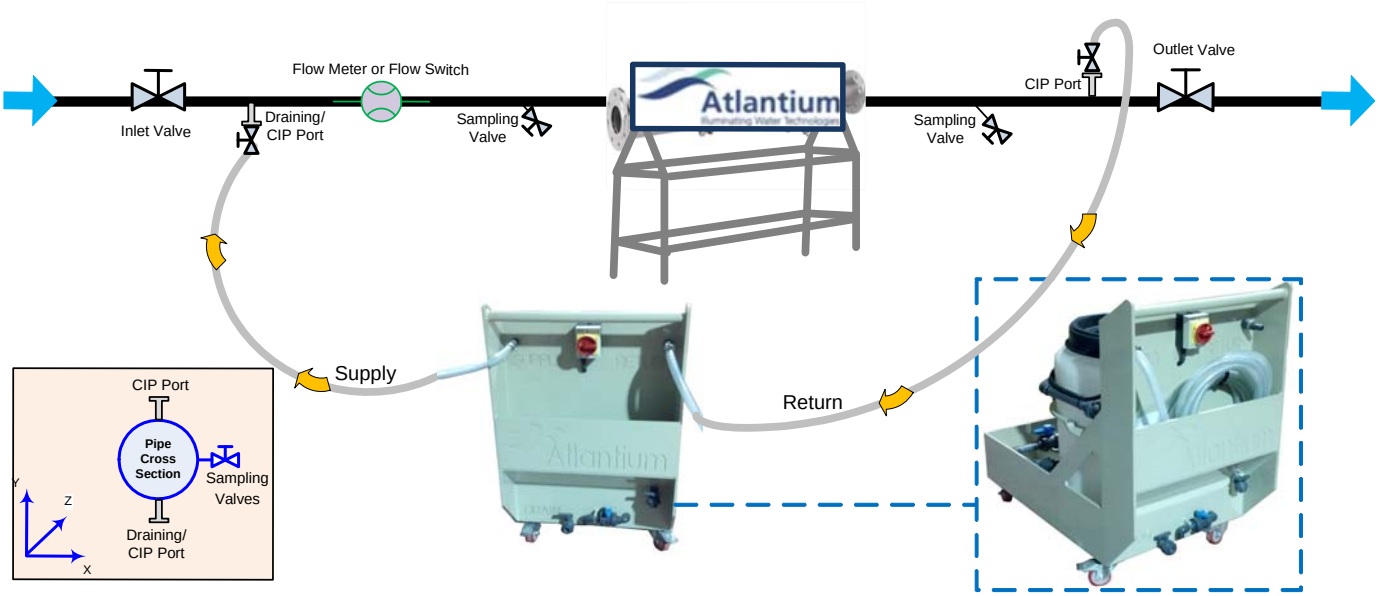


Figure 3-6: CIP Recirculation System Connections

3.9 CIP Accessories and CIP Kits

The CIP (Cleaning-In-Place) recirculation system is used for the periodic cleaning and sterilization process of the Atlantium unit's inner surfaces and quartz sleeve.

Two Accessory kits are required, one for each CIP port.

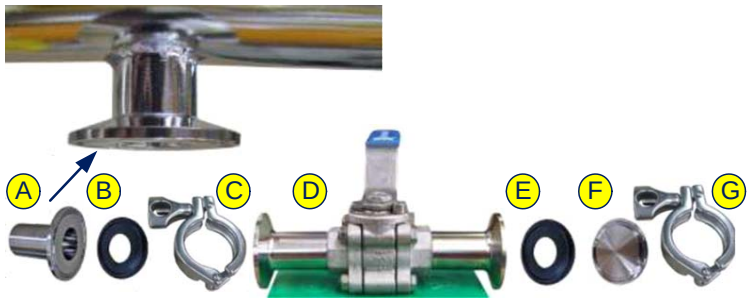


Figure 3-7: CIP Valve Kit

Table 3-4: CIP Accessory Kits

	Catalog number	Description	Purpose	Image	Kit for Stainless Steel Pipe KT0011600	Kit for Plastic Pipe KT0011500
A	FAC003700	Long welding ferrule 1” Adapter	For welding the tri-clamp port to the pipe		2 Pcs	-
A	FAB008100	Threaded Male Adapter 1/2” BSP	For connecting the tri clamp port to the plastic pipe		-	2 Pcs
B,E	FA0000046	Ferrule Gasket 1", FJ31.05-E	For sealing the valve connected to the pipe		2 Pcs	2 Pcs
C,G	FA0000045	Clamp 1.5", ST.ST 304	For connecting the valve to the welded port		2 Pcs	2 Pcs
F	FAC004100	Solid End Cap, 1 1/2", ST.ST 304, EG16A3A 1.54L	For covering the CIP port while not in use		2 Pcs	2 Pcs

Table 3-4: CIP Accessory Kits

	Catalog number	Description	Purpose	Image	Kit for Stainless Steel Pipe KT0011600	Kit for Plastic Pipe KT0011500
D		Valves	Two high-purity chemical resistant brass ball valve (aseptic ball valves), 1 inch diameter, with tri-clamp fitting connection mechanism (A3 Standard) on both sides.		**2 Pcs Not Included Purchased by Customer**	**2 Pcs Not Included Purchased by Customer**
	FAB005700	Hose Adaptor 1” to 3/4"	To attach the CIP hose		**2 Pcs**	**2 Pcs**

3.9.1 CIP Safety Equipment

Required but not included in Atlantium's CIP Kits: protective equipment such as goggles, clothing and (chemical-safe) gloves, according to the safety standards specified at your facility.

3.10 Connection to Facility Control

Options are available to set up control of the Atlantium system via your facility control system. Control setup can be made via any of the following:

- **Modbus** - RTU on RS485 (network) serial connection directly to the Facility control (**Appendix A**, ***Modbus Communication Protocol***)
- **Signal control** - utilizing System Ready, External On/Off, and General Alarm functions, etc. (See ***Cabling Components*** on page **5-74.**)

With any of the above control options, you can still utilize the function of the **Control Module** of the Atlantium system. Whichever control facility is utilized, the system communicates to the other control systems such that the last command issued determines the current status.

4 Safety Overview

The Atlantium System has been designed according to the highest safety standards, ensuring the safety of operating personnel, the environment and the treated water.

DANGER!

Improper use of controls or adjustments or performance of procedures other than those specified herein could result in significant hazards.

Therefore, personnel operating or servicing this system must be thoroughly familiar with all safety requirements and operating procedures and are to adhere to them during use of this system.

To guard against injury, observe basic safety precautions, including the following:

All persons operating the Atlantium system must be aware of proper use and the potential hazards of violating safeguards. Be certain that all personnel carefully review the safety information and the procedures specified in this manual. Only authorized individuals with appropriate training and knowledge, including local regulations, should operate, assist in the operation of or provide maintenance to the system.

All service and repair of the Atlantium system must be performed by Atlantium field service technicians or other factory-authorized personnel ONLY.

UV Exposure: The system generates ultraviolet (UV) light within the Unit, which can cause serious eye damage or blindness if you stare at it directly when it is working. Use the Viewport only. Do not look directly into the lamp enclosure during system operation, examining, or servicing the system's internal components during operation, or when lamp's caps are open, or if energized testing is required.

If the viewport is broken, DO NOT look directly into it. Be sure to request a replacement part immediately. Do not open the viewport's metal cover until the replacement is installed.

Persons potentially subject to UV light exposure – technicians who are servicing the Unit – must wear appropriate eye protection whenever the system is operating.

The Viewport is specifically designed with a UV filter to prevent UV light from escaping so it is safe to look inside. Nevertheless, personnel should not stare at it or any other brilliant light source for any but short intervals to check on Unit operations. The Viewport cover should also be preserved intact.

During maintenance procedures that are conducted with the UV lamps turned on:

- **Use Caution floor signs at a distance of 1m/40inch from the Atlantium Unit to warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Wear appropriate eye protection, such as containing polycarbonate lens that meet EN166, CAN/CSA-Z94.3-02 and/ or Z87.1 standards.
- **DO NOT look directly into openings that emit UV light.**

Protect Hands: **Do not touch the UV lamp with bare hands.**

Wait at least 10 minutes until the lamp is cooled down before touching the lamp or starting the replacement procedure.

Use appropriate protective gloves both to protect your hands and to avoid skin oils that leave fingerprints and/or harm the UV lamp.

Lamps can reach a temperature of 1,000C° under operating conditions.

Wait ten minutes to allow the lamp to cool down fully before replacing it.

Keep materials that are sensitive to heat, or which contain solvents, a safe distance away from the lamp, Unit, and electrical connections.

Electric Shock: **Before replacing a UV lamp or other components, and during any maintenance requiring lamps to be turned off, make sure the switches on the Connection Boxes of the relevant lamps are turned OFF and place a sign on the Atlantium Unit and Control Module alerting others NOT to touch the switches or the screen during maintenance so that no one can inadvertently turn it on while maintenance is in progress.**

ATTENTION!

Prevent Damage: Handle the UV lamp assembly by holding only the contact housings (white ceramic casing).

When handling the UV lamp, always place it on a flat surface or table, so that it is not accidentally damaged.

If the body of the UV lamp is touched accidentally, clean the fingerprints off with alcohol (at least 70%) and wipe dry with a soft, clean, lint-free cloth (usually provided with the lamp). Do not use cleaning rags or materials that can leave a residue.

Do not use a UV lamp that shows any scratches, cracks, or other damage.

Mercury Poisoning: Lamps contain mercury, which is a hazardous substance. Always wear gloves and safety glasses when handling or replacing a UV lamp.

Inhalation of vaporized mercury compounds can be harmful to the lungs, kidneys, and nervous system. Mercury that penetrates the skin or is ingested can also be harmful.

If mercury is inhaled, penetrates the skin, or is ingested accidentally, seek emergency medical treatment immediately.

If a UV lamp brakes during handling and releases mercury, the following precautions should be observed to minimize the risk of exposure to mercury:

- Always wear gloves and safety glasses when handling or replacing a UV lamp.
- Leave the area immediately to avoid inhalation of the mercury vapor.
- Thoroughly ventilate the area for at least 30 minutes or until the mercury vapor concentration is in compliance with applicable federal and local health and safety regulations.
- After handling a broken UV lamp, carefully remove gloves, and then wash hands thoroughly with soap and water. Follow all applicable federal and local health, safety, and environmental regulations.
- Do not turn on or operate the system until all mercury contamination has been cleaned up and removed.
- Establish a Hazard Response Plan to deal with all related hazards and provide it to all personnel associated with the Atlantium system.

Refer the Chemical Safety information at: http://www.inchem.org/documents/icsc/icsc/eics0056.htm

ATTENTION!

Lamp Compatibility: Use original Atlantium UV lamps ONLY. **Other lamps are incompatible.** They **void the warranty** and may damage the system.

Proper Disposal of Lamps

Expired lamps are classified as hazardous waste and must be disposed of appropriately.

- A non-ruptured UV lamp that is no longer used must be disposed of according to specific local or federal environmental and hazardous waste regulations.
- Broken lamps and parts contaminated by the mercury have to be vacuum-packed and disposed of. Broken lamps, as well as contaminated packing material and other parts have to be considered as special waste, which may only be removed by authorized waste disposal companies.

4.1 Electrical Hazards and Safety Considerations

DANGER!

UNDER NO CIRCUMSTANCES

SHALL ELECTRICAL WORK BE CARRIED OUT WITHOUT VERIFYING THAT ALL MAIN CIRCUIT BREAKERS Of THE ELECTRICAL LINE FEEDING THE ATLANTIUM SYSTEM COMPONENTS ARE SET TO THE **OFF** POSITION AND THAT **NO** ELECTICAL FEED IS LIVE.

DO NOT SET ANY MAIN CIRCUIT BREAKER OF THE ELECTRICAL LINES FEEDING THE ATLANTIUM SYSTEM COMPONENTS TO THE ON POSITION UNTIL **ALL** ELECTRICAL WIRING IS PROPERLY CONNECTED. UNCONNECTED WIRES/CABLES IN A LIVE ELECTRICAL FEED IS A HAZARDOUS SITUATION THAT CAN RESULT IN DEATH OR SERIOUS INJURY.

- **DO NOT DRILL INTO THE CONNECTION BOXES OR ALTER THEM IN ANY WAY!!!!**
- **The Connection Box already contains entry glands for ALL of the cabling needs.**
- **Making holes in a Connection Box will destroy its waterproof seal, endangering the components inside.**
- **Do Not draw electricity from the Atlantium Unit's Connection Box as it will damage the Unit's electrical configuration.**

The Atlantium Hydro-Optic Water Disinfection system incorporates high-voltage internal components, which can cause serious injury or fatal electrical shock if not used or serviced properly. High voltage components can retain a charge for some period of time even after the system has been turned off.

- Never attempt to remove any system covers or to dismantle any parts, except when performing the maintenance procedures detailed in ***System Maintenance*** on page 105.
- No portion of the Atlantium system is be opened or removed by anyone other than a trained and authorized technician.
- Do not spray or pour any type of fluid directly into the Unit or electrical components. Moisture causes damage to the equipment and electrical shock may result.
- Do not operate the system if the power cables or harnesses are frayed or otherwise damaged.

CAUTION!

DO NOT enter **Technician Mode** in the **Control Module** unless instructed to do so by Atlantium certified personnel or by an Atlantium System User Manual, for a specific task.

If you are in Technician Mode, **DO NOT WALK AWAY!**

NEVER LEAVE THE ATLANTIIUM SYSTEM WHEN IT IS OPERATING IN TECHNICIAN MODE!!

A SYSTEM LEFT IN TECHNICIAN MODE WITHOUT SUPERVISION CAN RESULT IN PERMANENT DAMAGE!!

Utilize best practices regarding electrostatic discharge (ESD).

Avoid direct contact with the electronic circuit boards of the **Control Module,** which is sensitive to electrostatic discharge (ESD).

Wear an electrostatic discharge (ESD) Wrist Strap when performing procedures where contact with a circuit board is possible.

4.2 Keep the Unit Full of Water

ATTENTION!

The Atlantium system cannot be operated when there is no water. Any damage caused due to operating the system without water invalidates the Atlantium warranty.

4.3 Chemical Use for Cleaning in Place (CIP)

ATTENTION!

- The chemicals selected and the concentrations used must be approved by Atlantium's Application staff, and the ship's quality assurance and safety officials. The CIP process must comply with the written protocols and procedures set in place by those departments. Refer to the section, ***Selecting the Correct Chemicals for CIP*** on page **107**
- Use and handle all chemicals in strict accordance with their manufacturer's instructions, product information sheets, and material safety data sheets (MSDS).
- NaOH (Caustic Soda) as it causes irreversible damage to quartz. It is strictly forbidden to use it without a specific analysis of temperatures, concentration, frequency and duration and written approval from Atlantium's application engineer.

Refer the Chemical Safety information at: http://www.inchem.org/documents/icsc/icsc/eics0056.htm

4.4 Anti-Corrosion Spray Safety

WARNING!

When using OKS 2101 Spray:

Refer to the OKS 2101 Safety Data Sheet according to Regulation (EC) No. 1907/2006 - GB.

Use the following protective gear:

- Safety glasses with side-shields conforming to EN166
- Protective gloves that satisfy the specifications of EU Directive 89/686/EEC and the standard EN 374 derived from it
- Respiratory protection mask with filter type A-P

Follow these precautions for safe handling:

- Do not use in areas without adequate ventilation.
- Do not breathe vapors or spray mist.
- In case of insufficient ventilation, wear suitable respiratory equipment.
- Avoid contact with skin and eyes.
- Keep away from fire, sparks and heated surfaces.
- Persons with a history of skin sensitization problems or asthma, allergies, chronic or recurrent respiratory disease should not be employed in any process in which this mixture is being used.
- Smoking, eating and drinking should be prohibited in the application area.
- Wash hands and face before breaks and immediately after handling the product.
- Do not get in eyes or mouth or on skin.
- Do not get on skin or clothing.
- Do not ingest.
- Do not use sparking tools.
- These safety instructions also apply to empty packaging which may still contain product residues.
- Pressurized container: protect from sunlight and do not expose to temperatures exceeding 50 °C. Do not pierce or burn, even after use.

4.5 System Safety Features

The Atlantium Hydro-Optic Water Disinfection system includes the following safety features:

4.5.1 UV Protective Seal

Each lamp is enclosed within a sealed lamp chamber that prevents the UV light from escaping. This means that with regular operation there is no ambient UV escaping from the Unit.

While servicing the UV Unit or checking the Intensity sensor, UV blocking safety glasses should be worn by service staff or others working in the area to prevent exposing the eyes to the UV light.

4.5.2 Lamp On/Off Switch

CAUTION!

An On/Off switch is located on the Connection Box on top of the Atlantium Unit.

In the event of an emergency, as well as for maintenance purposes, each lamp can be shut down immediately by turning this switch to the Off position.

To restore operation, rotate the button clockwise and then the system restarts.

4.5.3 Software Safety Controls on Control Module

CAUTION!

When the system is started, the Atlantium software performs a series of internal self-testing routines prior to starting system operation.

If the system **does not** pass these tests satisfactorily, UV lamp operation is disabled until the problem is resolved.

If problems arise during normal operation, appropriate error messages with varying levels of severity are generated on the touch-screen of the Control Module.

Control Module screens that provide setup options and compliance reports are password protected.

ATTENTION!

DO NOT leave the Control Module unattended, even for a short time, while setting change screens are accessible. An untrained person could damage the operation of the Atlantium system by making unauthorized setting changes.

4.6 Atlantium Training

As part of the Commissioning and Hand-Off process, Atlantium provides hands-on training on all aspects of the Atlantium system for your designated personnel. Should any component be replaced with upgraded features, Atlantium offers training as needed and is ready to provide additional training for new personnel or any other needs related to its systems.

5 Installing the Atlantium System

This section provides details about Installation the Atlantium system.

For preparing all electrical connections, be sure to review the *ICD* drawings provided to you by Atlantium with this manual. If no electrical drawings were provided please contact your Atlantium representative.

5.1 Installation Overview

Hardware setup to be implemented by your facility personnel:

Phase 1 Unpacking and checking the contents of the Atlantium system packaging (See below)
Phase 2 Installing Sampling valves and CIP ports (See page **51**)
Phase 3 Positioning the Atlantium Unit and connecting to the Piping Infrastructure (See page **55**)
Phase 4 Mounting the **Control Module(s)**, Ballast Module(s), etc. (See page **60**)
Phase 5 Cabling the **Control Module(s)**, Ballast Module(s), etc. (See page **72**)

Commissioning to be implemented by Atlantium personnel:

Phase 6 Starting up the system (See page **99**)
Phase 7 Configuring the **Control Module** (See page **57**)
Phase 8 Tuning the System (See page **147**)
Phase 9 Completing the System Setup and Testing (See page **100**)

The Installation Process Flow diagram below maps out the installation process.

5.2 Installation Process Flow Diagram

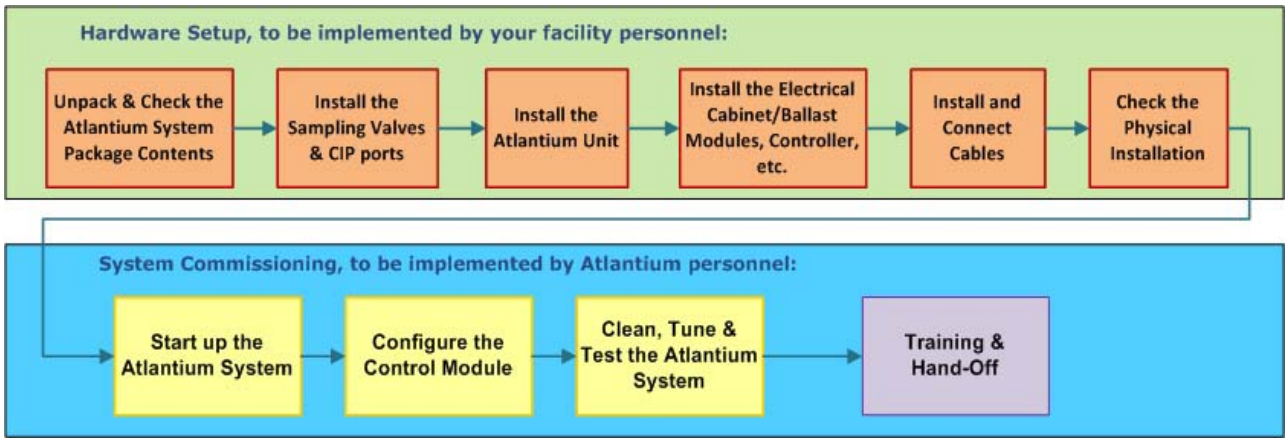


Figure 5-1: Installation Flow Diagram

5.3 Setting the Main Circuit Breakers to OFF Position

Before beginning work to connect the wires/cables of the Atlantium system components, turn the main circuit breakers of the electrical lines feeding

the Atlantium system components to the **OFF** position. **Verify that no electrical feed to the Atlantium system components is live.**

DANGER!

UNDER NO CIRCUMSTANCES

SHALL ELECTRICAL WORK BE CARRIED OUT WITHOUT VERIFYING THAT ALL MAIN CIRCUIT BREAKERS OF THE ELECTRICAL LINES FEEDING THE ATLANTIUM SYSTEM COMPONENTS ARE SET TO THE OFF POSITION AND THAT NO ELECTICAL FEED IS LIVE.

DO NOT SET ANY MAIN CIRCUIT BREAKER OF THE ELECTRICAL LINES FEEDING THE ATLANTIUM SYSTEM COMPONENTS TO THE ON POSITION UNTIL ALL ELECTRICAL WIRING IS PROPERLY CONNECTED. UNCONNECTED WIRES/CABLES IN A LIVE ELECTRICAL FEED IS A HAZARDOUS SITUATION THAT CAN RESULT IN DEATH OR SERIOUS INJURY.

5.4 Unpacking And Checking Package Contents

The Atlantium system is packaged and shipped in specially packed crates that contain:

- The Atlantium Unit
- Ballast Module(s)
- Control Module

- Power Module (for Ultrasonic Cleaner, if included)
- OVPD Module (if included)
- Documentation
- Peripheral accessories as per your order and listed on your delivery documents are crated according to their size and weight. Small accessories may be crated together.

Each of the crates is connected to a forklift-friendly wooden pallet. The Atlantium crates must be lifted off the transport vehicle using a forklift or crane by a qualified operator.

➲ To unload and check the system:

Step 1. Using the forklift or crane, lift each of the crates off the transport vehicle and lower them to a level ground surface.

Step 2. Walk around and inspect each crate for damage. Note if there is any separation in the joints or any other indication of any damage in transit. If there is any indication of damage, take a photograph of it and use extra caution when proceeding to the next step.

Step 3. Unclamp and remove the panels of the crates.

Step 4. As you remove the wrapping and shipping protections, verify that the contents of the crates are correct according to the delivery documents. The Atlantium Unit contains a plate with the model number and other product information. Verify that the correct product version is supplied.

- Immediately report to Atlantium and to the insurance carrier any damage to the crates or to the Atlantium system found before opening the crates or during every step of unpacking, installing or testing of the system.
- If you ordered more than one Atlantium system, check the serial numbers against the delivery documents to verify which components comprise each individual system.

5.5 Installing Sampling Points and CIP/Draining Ports

Both the Sampling Points and CIP/Draining Ports must be installed on the inlet and outlet sides of the Atlantium Unit.

5.5.1 Installing the Inlet/Outlet Sampling Points

Aseptic sampling valves are to be located indoors in clean environment without away from direct contact with fan or A/C air flow. They each must be a dedicated valve for sampling only and resistant to alcohol and flame. For information on the Sampling valve's components, see ***Installing Sampling Points and CIP/Draining Ports*** on page **51**. See ***Taking Microbial Samples*** on page **9-105.** for information on using the Sampling valves.

On the inlet and outlet sides of the water line, connect the aseptic sampling valves.

➲ To install an inlet/outlet sampling point:

Step 1. On the Inlet side, install the **Long welding ferrule** or **threaded male adapter (1).**

- Position aseptic sampling valves **(1 & 2** in **Figure 5-6** on page **54) horizontally** (90^0) on the pipe center, within 50cm of the Unit.
- **For Stainless Steel. Pipes:**
  - Weld the 1 inch **ferrule (1)** clamp fitting at a horizontal orientation (3 or 9 o'clock).

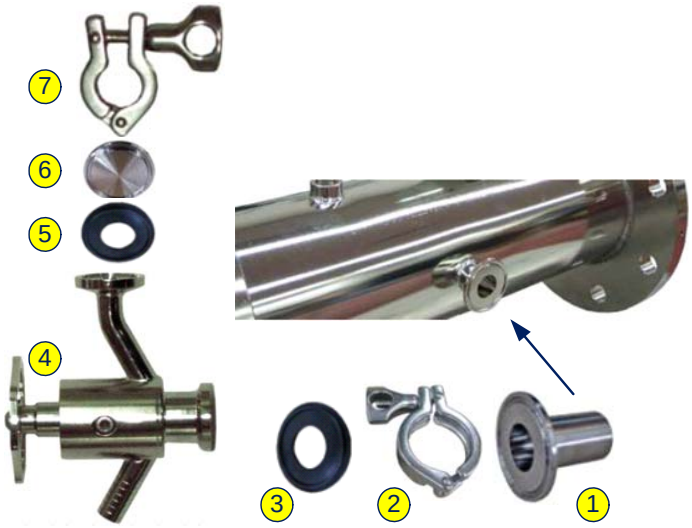


Figure 5-2: Sampling Valve Assembly

- Perform a polishing process to achieve a smooth area on the external and internal part of the pipes.
- Passivate the pipes to avoid corrosion.

■ **For Plastic Pipes:**

- Thread the **male adapter (1)** clamp fitting to the port.

■ Repeat for the Outlet side sampling point.

Step 2. Fit the gasket **(3)** into the **welding ferrule** or **threaded male adapter (1).**

Step 3. Fit the aseptic **sampling valve (4)** on over the gasket **(3).**

Step 4. Fit the clamp **(2)** around the aseptic **sampling valve (4)**/gasket **(2)**/**welding ferrule** or **threaded male adapter (1)** and close and lock it.

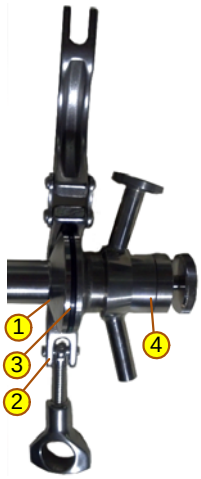


Figure 5-3: The clamp is placed around the aseptic **sampling valve** /gasket /**welding ferrule** or **threaded male adapter**

Closing Off the Aseptic Sampling Valve When Not in Use

Step 5. Into the spout of the sampling valve, fit the 1/2”gasket **(6)** and fit the cap **(7)** over it.

Step 6. Fit the 1/2” clamp **(5)** around the sampling valve spout/ gasket **(6)** /cap **(7)** and close and lock it.

Figure 5-4: The clamp is closed

Figure 5-5: The clamp is closed

5.5.2 Installing the CIP/Draining Ports

The CIP (Cleaning-In-Place) recirculation system is used for the periodic cleaning and sterilization process of the Atlantium unit's inner surfaces and quartz sleeve. For more information on performing CIP, see ***Cleaning In Place (CIP)*** on page **113**.

A CIP KIT includes the CIP Cart with water reservoir and a pump, capable of pumping up to 4 cubic meters per hour. It includes a supply hose: ID 0.75" / 19mm (5 meters / 15 feet long) and a return hose: ID 0.75" / 19mm (5 meters / 15 feet long) and pocket to hold these hoses. See the diagram below.

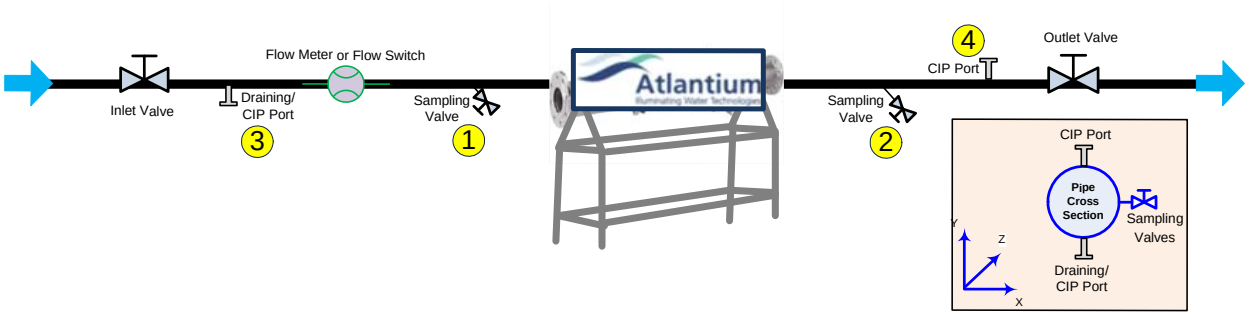


Figure 5-6: CIP Ports and Sampling Valves Orientation

Position the CIP/draining valves **(3 & 4)** on both the inlet and outlet sides of the Atlantium unit. To enable connection of the CIP kit, use high-quality, sanitary equipment, including a 1 inch tri-clamp/ferrule on both sides. A CIP/Draining valve is to be located 1D after the Inlet valve and 1D before the Outlet valve. When not using the CIP [ports. close the valve outlet using the tri-clamp cover.

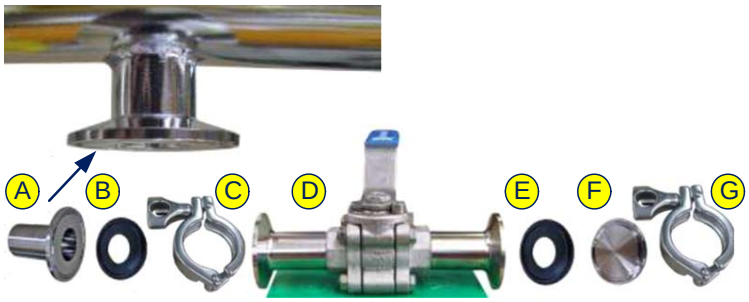


Figure 5-7: CIP Valve Kit

➲ To install the CIP ports:

Step 1. Weld or thread the 1 inch ferrule at a vertical orientation:

For municipal applications, prepare 1" screw ports for the valves and insert the valves:

- **Draining Valve:** at a 6:00 o'clock orientation on one side of the Atlantium unit.
- **CIP Valve:** at a 12:00 o'clock orientation on other side of the Unit.

For all application except municipal:

- **For Stainless Steel. Pipes:**
  Weld the 1 inch clamp fittings:
  - **Draining port:** at a 6:00 o'clock orientation on one side of the Atlantium unit.
  - **CIP port:** at a 12:00 o'clock orientation on other side of the Unit.
  - Perform a polishing process to achieve a smooth area on the external and internal part of the pipes.
  - Passivate the pipes to avoid corrosion.
- **For Plastic Pipes:**

- Thread the clamp the valve fittings to the port..

Attaching the Valve to the ferrule

Step 2. Fit the gasket **(B)** into the **welding ferrule** or **threaded male adapter (A).**

Step 3. Fit the aseptic **valve (D)** on over the gasket **(B).**

Step 4. Fit the clamp **(C)** around the aseptic **valve (D)**/gasket **(B)**/**welding ferrule** or **threaded male adapter (A)** and close and lock it.

Closing Off the CIP/draining Ports When Not in Use

Step 5. Into the ferrule, fit the gasket **(E)** and fit the cap **(F)** over it.

Step 6. Fit the clamp **(G)** around the CIP port/ gasket **(E)** /cap **(F)** and close and lock it.

Preparing the CIP Hoses

Step 7. On one end of a hose **(A)**, using an appropriate screwdriver, loosen the band **(C)**.

Step 8. Connect the hose to one of the adapters on the cart **(B)**. Push the hose up a far as you can.

Step 9. Using the screwdriver, tighten the band **(C)**.

Step 10. On the other end of a hose **(D)**, using an appropriate screwdriver, loosen the band **(F)**.

Step 11. Connect the hose to one of the hose adapters **(E)**. Push the hose up a far as you can.

Step 12. Using the screwdriver, tighten the band **(F)**.

Step 13. Repeat with the second hose to the adapter on the other side of the cart.

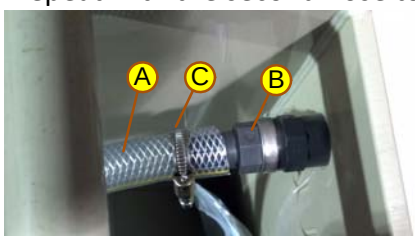


Figure 5-8: Hose Connection to One Side of Cart

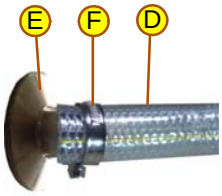


Figure 5-9: Hose Connection to Hose Adapter

5.6 Positioning the Atlantium Unit

Read through this typical installation section ***before*** you begin.

Make sure to use your installation drawing during this procedure. The Unit's triangle support brackets must rest on a strong, stable, level surface and anchor the Unit.Trace the path to be taken to position the Unit to its planned location and clear it of all obstacles.

Atlantium systems can be installed horizontally or vertically in your piping system. For a vertical installation, the Atlantium Unit can be positioned against a wall or on skids.

5.6.1 Required Equipment

Supply the following tools and equipment:

- Lift or crane
- Lifting strap
- Bolts, washers & lock washers
- Supports for inlet/outlet piping and for the Unit as shown on your installation plan

The Atlantium unit is heavy - refer to Table 2-3, ***Ballast Module Physical Details***, in **Section 2**, page 18. Two people are required to get the Atlantium Unit to the designated installation location - a qualified lift or crane operator, at least one other person to guide the Unit as it is transported, and others as needed.

➲ To position the Atlantium unit:

Figure 5-10: Lifting Strap Locations

Step 1. Note the direction of the water flow that is to occur in the piping process into which you intend to place the Atlantium Unit and the Flow indicator marked on the unit. Use this information to decide at which angle to lift the Unit for transport. This is to help you to prepare for placing the Unit at the location.

Step 2. To lift the Unit from its packing crate, loop a lifting strap the end fittings on each side of the Unit.

Step 3. Join the straps together with a third central strap that is being connected to the lift or crane.

Step 4. Ensure that all straps are connected securely to the Unit, to each other and to the lift/crane.

Step 5. Using the lift/crane, lift the Unit by the straps, out of the crate and position it for safe transport (near-ground level is preferred).

Step 6. Using the lift/crane and the designated person to guide, carry the Unit to its intended location.

Step 7. Install the Atlantium Unit:

Horizontal Installation

a At the designated location, check the installation plan, including the distances between the triangle support brackets.

b Position the customer-provided supports /surface so that the Unit's triangle support brackets are at least 750mm / 29.5 inch above the floor. (Refer to the **RZB300** ***Systems' Interface Control Drawings*** provided by Atlantium.)

c Carefully lower the Unit to its position.

d Using the appropriate bolts, washers and lock washers, bolt the triangle support brackets to the customer-provided corresponding supports.

Vertical Installation

a Position the Atlantium Unit vertically so that the Flow label is point in up
To verify if the position is correct, check that the **RZB300** Unit is positioned so that the view port is on the **top** end

b Position the Unit's triangle support brackets against the wall or skids.

c Carefully lower the Unit and using the appropriate bolts, washers and lock washers, bolt the triangle support brackets to the customer-provided corresponding supports.

Figure 5-11: FlowLabel Points Up

Step 4. To connect the fitting on one side of the Unit to the fittings of the facility's pipes, insert an o-ring between the Unit's and the facility's pipes. In some cases, a reducer or expander may be necessary.

Step 5. On one side of the Unit to the flanged fittings of the facility's pipes with eight appropriate **(8)** sets of bolts/nuts/flat and lock washers - do not tighten the hardware.

Step 6. Repeat the step above for the other side of the Unit.

Step 7. Tighten the fastening hardware so that the torque pressure around the fittings is even, and the Unit is firmly connected to the facility's water line.

Step 8. For installations with multiple units, repeat this procedure for the second unit.

Step 9. To check for any leakage, open the Inlet and Outlet valves to allow water to flow through the newly installed Unit and its surrounding process piping.

Step 10. If you find any leakage, tighten the connections.

Step 11. Close the Inlet and Outlet valves and use the Draining valve, drain the Atlantium system piping from Inlet valve to Outlet valve.

Figure 5-12: Generic Flange fitting to the Water Line

5.7 Installing Surge Protection

Verify that you have a protective device that prevents electrical surges and fluctuations from causing damage. You must have at least one of these three types of surge protection:

- UPS (Uninterrupted Power Supply) has an internal battery that allows the equipment plugged into it to continue to run in the event of a power outage until power is restored. It also has the benefit of performing the same functionality of surge protectors, protecting against spikes and surges. If needed, the UPS allows the system to be shut down cleanly if power is out for a lengthy period.
- Surge protection device protects your equipment from variations in electrical current, such as surges and spikes.
- OVPD module (Over Voltage Protection Device) - if you have purchased this from Atlantium, see below for directions on mounting and connection it.

5.7.1 Installing a Surge Protection Device

The diagram to the right shows the generic wiring for a surge protection device. See the manufacturer's installation instructions with the surge protection device to be installed. For more information about surge protection devices available via Atlantium, see ***Surge Protector Devices*** on page **20**.

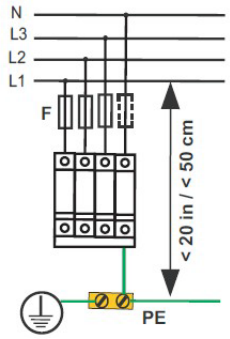


Figure 5-13: Surge protection device wiring - 400VAC

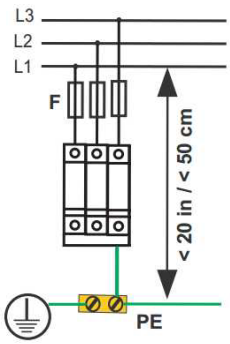


Figure 5-14: Surge protection device wiring - 440/480VAC

5.7.2 Mounting and Connecting the OVPD Module (if included)

If your system includes the OVPD Module (see ***The OVPD Module*** on page **21**), mount it in an appropriate location in your electric box and connect it according to the directions in this section. refer to the ***Electrical ICD*** drawing provided to you by Atlantium with this manual.

➲ To connect the OVPD

Step 1. To connect the mains to the OVPD module, thread the mains cables (customer provided) through the openings at the bottom of the OVPD module **(B)** and connect them to the terminal block for the mains **(C)** according to the diagram below.

Step 2. To connect the cables coming from the circuit breaker before the Atlantium Unit, thread the cables through the opening at the top of the OVPD module **(A)** and connect them to the terminal block for these cables **(D)** according to the diagram below.

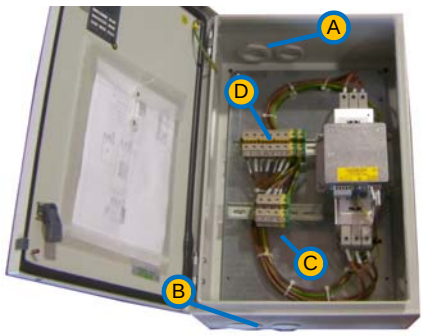


Figure 5-15: OVPD Connections

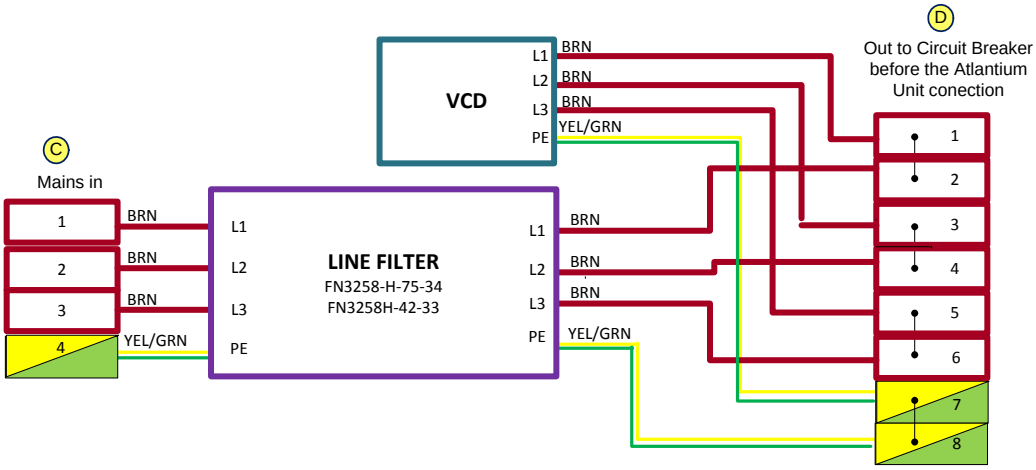


Figure 5-16: Typical Atlantium System Connections for the OVPD

5.8 The Control Module

Table 5-1: Control Module Components

#	Item
1	Mounting Holes
2	Data Cable + Gland
3	Electrical Cable + Gland
4	Electrical Connectors
5	Glands for Peripheral Component Connections
6	Peripheral Component Connections
7	Data Cable and Modbus Connectors

Figure 5-18: ControlModule

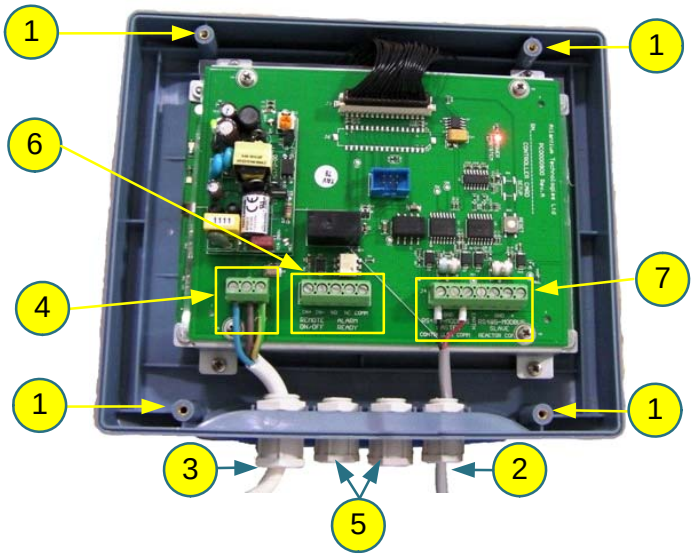


Figure 5-17: Control Module Interior

5.8.1 Mounting the Control Module

The **Control Module** can be mounted on a wall or suitable anchoring point. The **Control Module** comes with an electrical 1.6M/5.25ft. long (1.5m/4.92ft. from the PG gland) and a Data cable (for length options, see the table below). However it can be placed up to 50M/ 164.04 ft. from the Atlantium Unit, in a place that it is easy to reach the touch screen and read the monitoring information. A longer Data Cable can be ordered for this purpose.

ATTENTION!

- When mounting the **Control Module**, take into consideration the length of the cables.
- **Do not Stretch Electrical Cables - Avoid tension on the electrical cables. They are to be installed within conduits and must not be stretched.**

➲ mount the Control Module on the wall:

Step 1. Using the **Control Module**'s mounting bracket, on the wall space designated for the **Control Module**, make four marks according to the its holes **(1)**.

Step 2. The mounting holes are 8.5mm (0.33") in diameter. Using the appropriate drill bit, drill the four holes.

Step 3. In each hole, insert a plastic anchor.

Step 4. Position the **Control Module**'s mounting bracket so that its four holes **(1)** fit over the drilled holes.

Step 5. Using appropriate screws and with a screwdriver, screw in the four screws in a diagonal pattern (one corner, then the opposite corner).

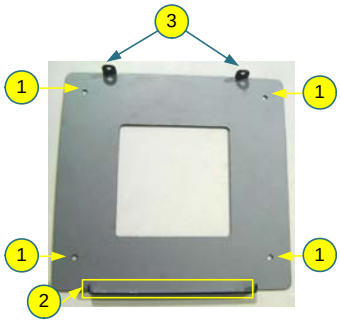


Figure 5-19: Control Module's Mounting Bracket

Step 6. To secure the Unit firmly, return to each screw in the diagonal patter and tighten the screws securely.

Step 7. Before proceeding to mount the **Control Module**:

- To connect the Modbus cable and cables for any peripheral equipment to be part of the system and/or to replace the Data cable with a longer one, see ***Control Module Connections*** on page **62**
- If you plan to use a longer power cable, see ***Mounting the Power Module*** on page **67**.

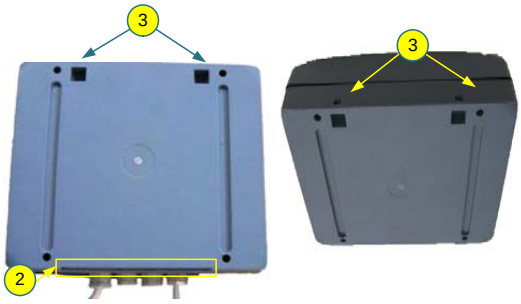


Figure 5-20: Control Module's Mounting Interface

Step 8. To mount the **Control Module**, place it onto its mounting bracket so that its grove at the bottom **(2)** rests on the lip at the bottom of its mounting bracket **(2).**

Step 9. Press the **Control Module** back onto the mounting bracket such that the top of the **Control Module** fits under the mounting bracket's tabs **(3)** at the top. The screw holes **(3)** on the top of the **Control Module** should line up with the mounting brackets tabs **(3).**

Step 10. Using the provided screws, and with a screwdriver, fasten the screws at the top **(3)** so that the **Control Module** is firmly attached to its mounting bracket.

➲ To mount the Control Module on skids:

Step 1. Set up the skids with spacing that matches the spacing measurements for the holes in the **Control Module**'s mounting bracket **(1)**. Refer to **Figure 5-19** above.

Step 2. Position the **Control Module**'s mounting bracket so that its four holes **(1)** fit over corresponding holes on the skids.

Step 3. Using appropriate screws and with a screwdriver, screw in the four screws in a diagonal pattern (one corner, then the opposite corner).

Step 4. To secure the Unit firmly, return to each screw in the diagonal patter and tighten the screws securely.

Step 5. Before proceeding to mount the **Control Module**:

- To connect the Modbus cable and cables for any peripheral equipment to be part of the system and/or to replace the Data cable with a longer one, see ***Control Module Connections*** on page **62**
- If you plan to use a longer power cable, see ***Control Module Connections*** on page **62**.

Step 6. To mount the **Control Module**, place it onto its mounting bracket so that its grove at the bottom **(2)** rests on the lip at the bottom of its mounting bracket **(2).**

Step 7. Press the **Control Module** back onto the mounting bracket such that the top of the **Control Module** fits under the mounting bracket's tabs **(3)** at the top. The screw holes **(3)** on the top of the **Control Module** should line up with the mounting brackets tabs **(3).**

5.8.2 Control Module Connections

The **Control Module** includes the following connections:

- Power Cable
- External On/Off Signal
- General Alarm Signal
- Data Cable
- Modbus

Avoid direct contact with the electronic circuit board which is sensitive to electrostatic discharge (ESD). Utilize best practices regarding ESD when replacing the power cable.

➲ To connect peripheral equipment:

Opening the Control Module's Back

Step 1. Using the appropriate screwdriver, unfasten the screws on the **Control Module (A)** and carefully remove the back.

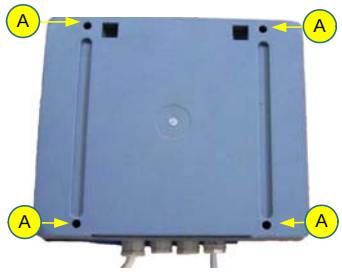


Figure 5-21: Control Module's Back

Connecting the Signals

Step 2. To connect an **External On/Off** signal, thread the signal cable through one of the holes with plastic glands on the bottom **(5)** and connect the wires to the appropriate connectors as show to the right **(6)** and as shown in **Figure 5-23** as terminals **4** and **5** (corresponding to **6** in **Figure 5-21**).

Step 3. To connect a.n **General Alarm** signal, thread the signal cable through one of the holes with plastic glands on the bottom **(5)** and connect the wires to the appropriate connectors as show to the right **(6)** and as shown in **Figure 5-23** us terminals **6**, **7** and **8** (corresponding to **6** in **Figure 5-21**).

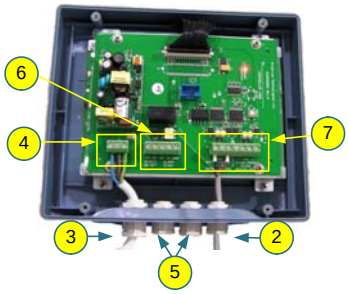


Figure 5-22: Control Module's Circuit Board

Connecting the Modbus

Step 4. To connect to an external facility control via Modbus, thread the signal cable through one of the holes with plastic glands on the bottom **(5)** and connect the wires to the appropriate connectors as show to the right **(7)** and as shown in **Figure 5-23** as terminals **12**, **13**, **14** and **15** (corresponding to **6** in **Figure 5-23**). For more information, see **Appendix A**, ***Modbus Communication Protocol*** . Add jumpers to any unused pins as shown in the diagram below.

Adding a Longer Data Cable

Step 5. To replace the Data cable with a longer one:

- **a** Unfasten the Data cable from the connectors as show to the above **(7)** as terminals **9**, **10** and **11** (**- GND +**) (corresponding to **6** in **Figure 5-21**).
- **b** Loosen the plastic nut of gland **(2)** and pull it through and out of the plastic gland at the bottom.
- **c** Thread the longer Data cable through its plastic gland **(2)** and fasten it to the appropriate connectors mentioned above.

5.8.3 PIN Assignments

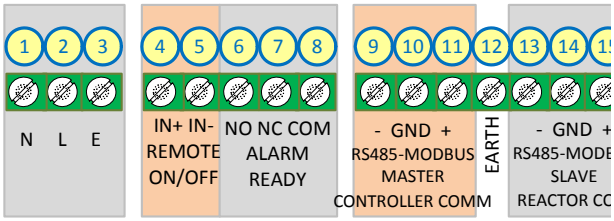


Figure 5-23: Control Module Terminal Connectors

Schematic Diagrams

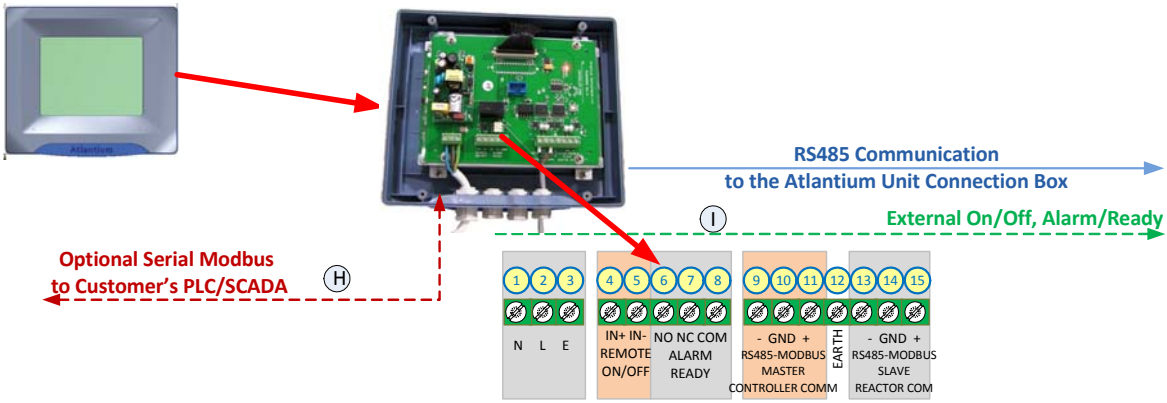


Figure 5-24: Control Module Terminal Connectors Schematic Drawing

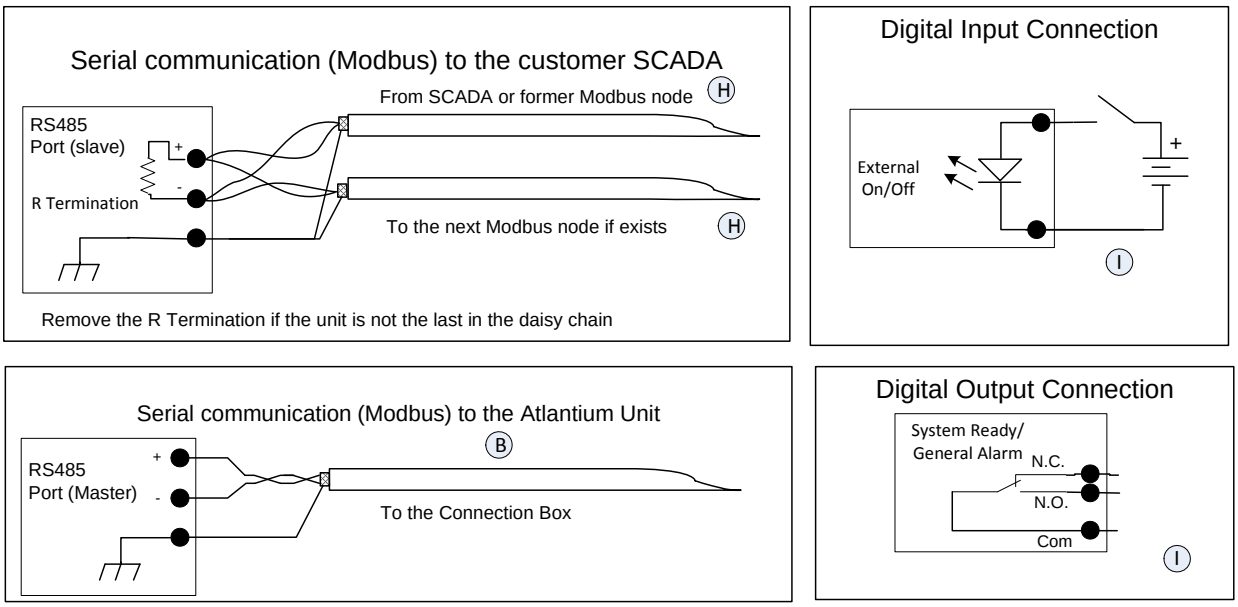


Figure 5-25: Control Module Terminal Connectors Wiring Drawings

Table 5-2: Peripheral Equipment Cable Details

#	Item	Sizes	Cat. No.	Description	Diameter	Supplied by
B.	Control Module RS485 data communication cable	5m/16.4ft. 10m/ 32.8ft.	HSL013200 HLS014500	Shielded twisted pair, 22AWG, 300V(General cables C0720A)	0.25 inch	Atlantium
G.	Flow Meter 4-20mA or Flow Switch cable Dry contact (See ***Cabling Components*** on page **5-74.**)	Customer defined	-	Connect to Connection Box no.1 through gland size 7. Wires size according to Flow Meter / Flow Switch manufacturer's instructions	0.25 inch	Customer
H.	RS485 Data communication cable	Customer defined	-	Shielded twisted pair, 22AWG, 300V	0.25 inch	Customer
I.	Digital output, Digital inputs, Analog outputs	Customer defined	-	Shielded cable, pairs of 22AWG, 300V	According to the number of pairs	Customer
R.	Laptop Control Module interconnect	Customer defined	-	Standard ETHERNET cable	-	Customer

Table 5-3: Control Module Connector Pins

#	Item	
1, 2, 3	Power cable connection.	
4	External On/Off IN+	On: 20-28VDC Off: <1VDC See ***External On/Off - Additional Notes*** below.
5	External On/Off IN-	
6	General Alarm/System Ready Dry Contact Normally Open **NO**	
7	General Alarm/System Ready Dry Contact Normally Closed **NC**	
8	General Alarm/System Ready Dry Contact Common **COM**	
	Data Cable Connectors	
9	RS485 / Modbus Master TX-	Cable is provided already attached. For information on attaching the cable to the Connection Box of Lamp 1. .
10	RS485 / Modbus Master Ground	
11	RS485 / Modbus Master TX+	
12	Earthing connector	
13	RS485 / Modbus Slave TX- to connect to an external control system	The **Control Module** is provided including a **100R, <5% , 0.25W, Through hole resistor** between **#13** and **#15**. This resistor is to be removed in **Control Modules** connected by daisy-chain, but is to be left attached in the last **Control Module** of the daisy-chain.
14	RS485 / Modbus Slave Ground	
15	RS485 / Modbus Slave TX+	

5.8.3.1 External On/Off - Additional Notes

For regular wiring:

- Positive voltage must be connected to pin 4 – IN+
- Negative (GND) must be connected to pin 5 – IN-

Table 5-4: External On/Off

Control Module Screen More Settings	Physical Connection	Physical Connection	
	IN+	IN-	Lamp Control
NC Unchecked	0 VDC	0 VDC	Off
NC Unchecked	+24 VDC	0 VDC	On
NC Checked	0 VDC	0 VDC	On
NC Checked	+24 VDC	0 VDC	Off

5.8.4 Closing the Control Module's Back

➲ To close the Control Module's back:

Step 1. Tighten all of the glands' plastic nuts.

Step 2. Return the **Control Module**'s back cover to the back of the **Control Module**. Make sure that the waterproof seal is positioned correctly between the back and front parts of the **Control Module.**

Step 3. Using the appropriate screwdriver, refasten the screws in the **Control Module (A** in **Figure 5-21)**.

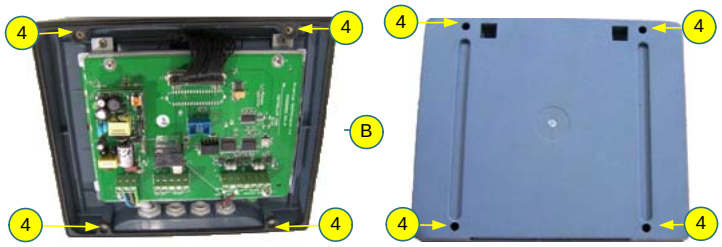


Figure 5-26: Control Module's Circuit Board

5.9 Mounting the Power Module

For systems requiring the Power Modules (refer to***Power Module*** on page **21** and ***Electrical Requirements*** on page **31**) the Power Modules can be mounted on a wall or suitable anchoring point up to 10 meters (32.8 ft.) (according to the cable length you ordered from Atlantium) from the Atlantium unit. A Power Module can be positioned near a Ballast Module with a minimum of 300 mm (11.8 inches) of clearance on each side for proper ventilation.

- **When mounting the Power Module, take into consideration the length of the cables.**
- **Do not Stretch Electrical Cables - Avoid tension on the electrical cables. They are to be installed within conduits and must not be stretched.**
- **The customer must provide the Power cable to the mains.**

To mount the Power Module on the wall:

Step 1. On the wall space designated for the Power Module, make four marks according to the spacing measurements for mounting orientation on the ICD drawing accompanying the documentation.

Step 2. The mounting holes **(B)** are 8.5mm (0.33") in diameter. Using the appropriate drill bit, drill the four holes.

Step 3. In each hole, insert a plastic anchor.

Step 4. Uncrate the Power Module package and carefully lift it out.

Step 5. Open the door of the Power Module **(A)** and locate the mounting holes **(B)**.

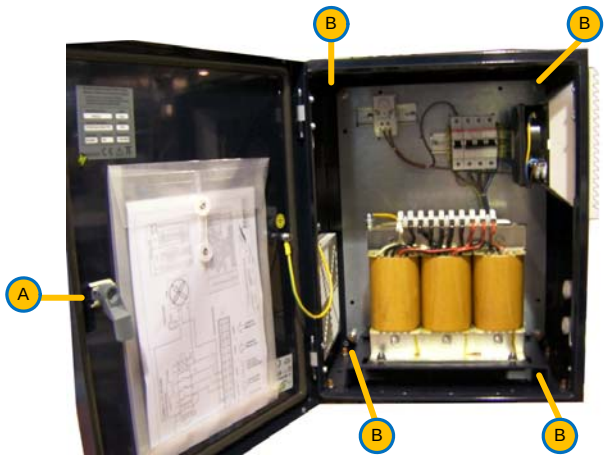


Figure 5-27: Power Module - Mounting Holes

Step 6. Remove the protective covers from each hole.

Step 7. Position the Power Module so that its four holes fit over the drilled holes.

Step 8. Using appropriate screws and with a screwdriver, screw in the four screws in a diagonal pattern (one corner, then the opposite corner).

Step 9. To secure the Unit firmly, return to each screw in the diagonal pattern and tighten the screws securely.

To mount the Power Module on skids:

Step 1. At the location designated for the Power Module, set up the skids with spacing that matches the spacing measurements for mounting orientation on the appropriate ICD drawing accompanying the documentation.

Step 2. The mounting hole **(B)** are 8.5mm (0.33") in diameter. Using the appropriate bolts and nuts, with a screwdriver, screw in the four bolts and add the nuts leaving around 5.6mm (0.20-0.24') sticking out.

Step 3. Uncrate the Power Module package and carefully lift it out.

Step 4. Open the door of the Power Module **(A)** and locate the mounting holes **(B)**.

Step 5. Remove the protective covers from each hole.

Step 6. Position the Power Module so that its four holes fit over the corresponding holes on the skids.

Step 7. Using appropriate bolts and nuts and with a screwdriver, screw in the four bolts adding the nuts in a diagonal pattern (one corner, then the opposite corner).

Step 8. To secure the Unit firmly, return to each bolt in the diagonal patter and tighten them securely.

5.10 Installing the Ballast Module(s)

Each lamp chamber has a corresponding Ballast Module. The location for mounting the Ballast Module(s) must take into account the length of the cables to be attached from the Atlantium Unit. If you need to calculate space cooling for the location in which you are installing the Ballast Module(s), refer to the heat load per Ballast Module shown below.

Heat Load per Ballast Module: (For calculating space cooling requirements)	210W

ATTENTION!

- **Do not Stretch Electrical Cables - Avoid tension on the electrical cables. They are to be installed within conduits and must not be stretched.**

Ballast Modules must be installed vertically and positioned at least one meter above ground level. Ballast Modules must be at least 300mm (12") from each other or from any other object around them to allow proper ventilation. Ballast Modules must not be installed within closed cabinets.

You can mount them next to each other horizontally or one above another vertically. Atlantium recommends you arrange the Ballast Modules in consecutive order from left to right according to the number marked on the bottom of the Ballast Module.

Consult the ***Ballast Module's ICD*** accompanying this manual.

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BALLAST MODULE B

(Active Cooling version)

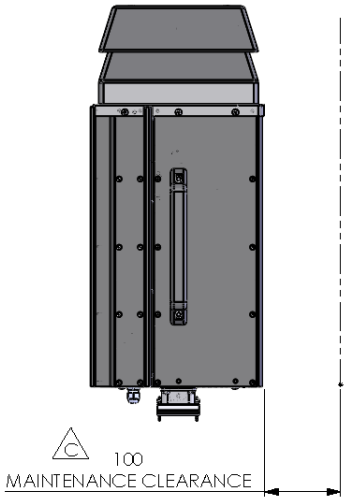


Figure 5-28: Ballast Module B Depth

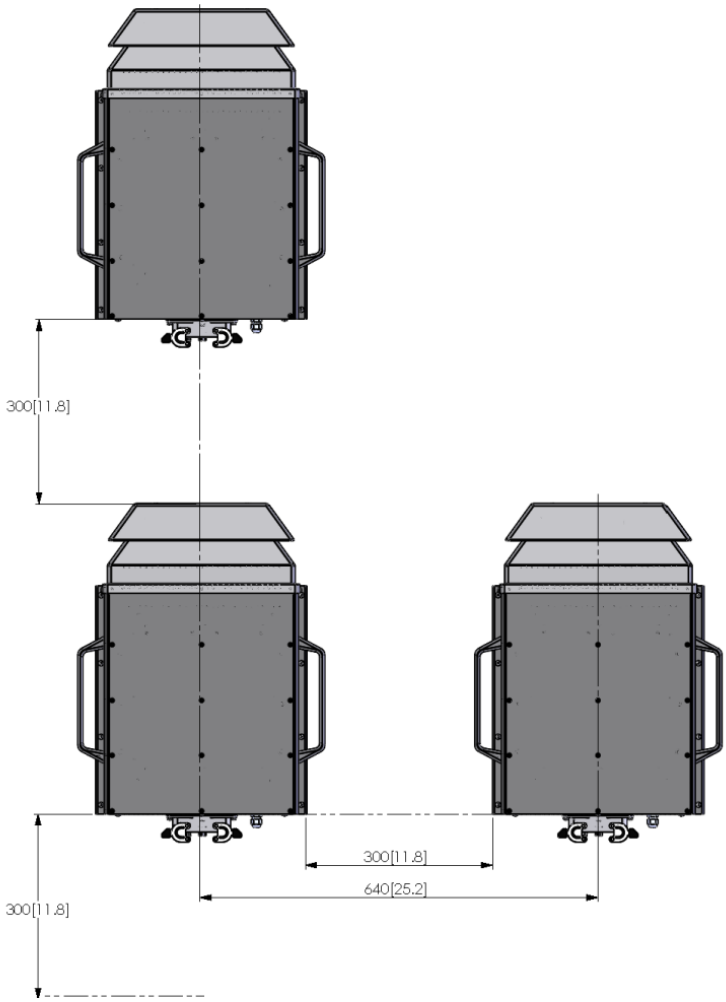


Figure 5-29: Multiple Ballast Module B Installation Diagram

BALLAST MODULE C

(Power Unit version)

Figure 5-30: Ballast Module C Depth

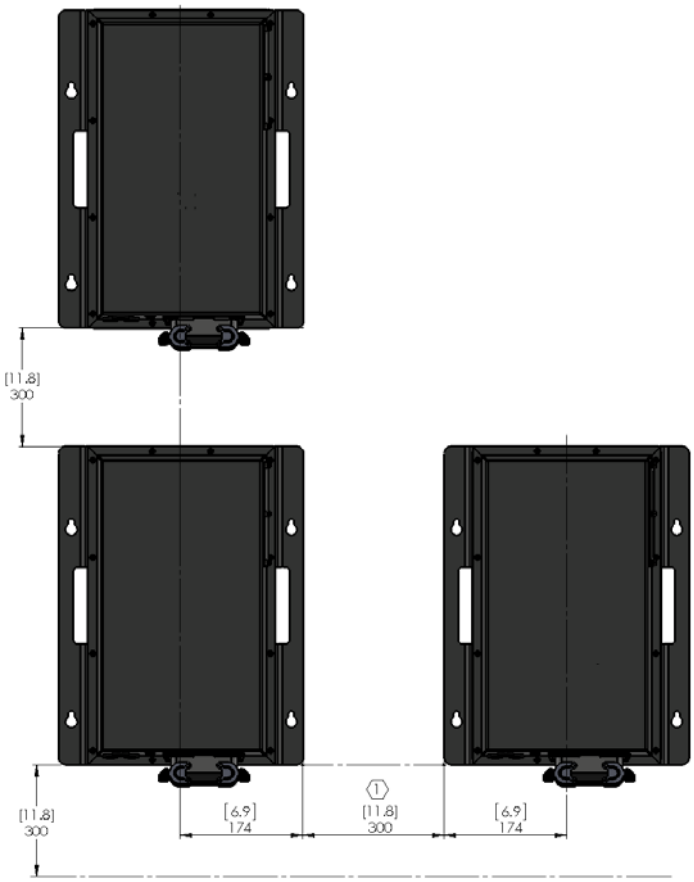


Figure 5-31: Multiple Ballast Module C Installation Diagram

➲ To mount a Ballast Module on the wall:

The Ballast Module is heavy! Be sure that the wall on which you intend to mount the Ballast Module is strong enough to carry its weight.

Step 1. On the wall space designated for the Ballast Modules, make four marks according to the spacing measurements for mounting orientation on the ICD drawing accompanying the documentation.

Step 2. The keyhole slots for mounting are 7mm (0.28”) in diameter for screws size M6 OR 1/4” bolts. Using the appropriate drill bit, drill the four holes.

Step 3. In each hole, insert a plastic anchor and with the appropriate screwdriver, screw in the screw leaving around 5.6mm (0.20-0.24’) sticking out of the wall.

Step 4. Uncrate the Ballast Module package and carefully lift out the Unit.

Step 5. Fit the keyhole slots onto the protruding screw heads and slide the Unit downward until the top of the keyhole slots rest on the screws.

Step 6. To secure the Unit firmly, tighten the four screws.

➲ To mount a Ballast Module on skids:

Step 1. At the location designated for the Ballast Modules, set up the skids with spacing that matches the spacing measurements for mounting orientation on the ICD drawing accompanying the documentation.

Step 2. The keyhole slot for mounting is 7mm (0.28”) in diameter for screws size M6 OR 1/4” bolts.

Step 3. Using the appropriate screwdriver, screw in the four bolts and add the nuts leaving around 5.6mm (0.20-0.24’) sticking out.

Step 4. Uncrate the Ballast Module package and carefully lift out the Unit.

Step 5. Fit the keyhole slots onto the protruding bolt heads and slide the Unit downward until the top of the keyhole slots rest on the bolts.

Step 6. To secure the Unit firmly, tighten the four bolts and their nuts.

5.11 Connecting the Ballast Module Cables

One side of the **Cable Harness** is attached to the connection box of each lamp of the Atlantium Unit. Refer to the ***Electrical ICD*** drawing provided to you by Atlantium with this manual for the connection of the other end of the **Cable Harness**.

Table 5-5: Ballast Module Details

#	Item
A	Single Connector with lock latch
B	48VDC for Fan - 440/480VAC only
C	Pressure Relief
D	Fuse Housing (for Atlantium Technicians only)
E	Connector width 13.11cm / 5.159inch.

Figure 5-33: Connector Detail

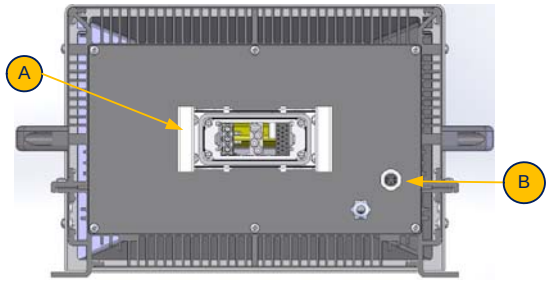
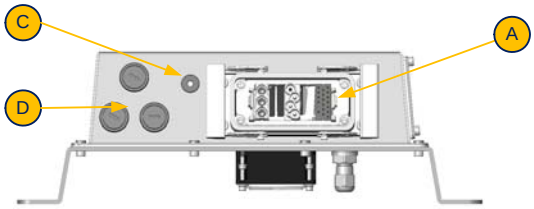


Figure 5-32: Ballast Module A-Waterproof Box



Ballast Module A-Waterproof Box

To connect the cable:

Step 1. According to the electrical drawing of the Installation plan, bring the **Cable Harness** attached to the Connection Box of Atlantium Unit up to the Ballast Module.

Step 2. On the Ballast Module, pull the lock latch of the connector to the open position and remove the protective cover.

Be sure to save the protective cover of the Ballast Module connector in an accessible location for the future in case you need to dismantle the Ballast Module for any reason.

Step 3. Connect the lamp cable lead to the Lamp connection port (**A** in the figures above) (the connector is keyed so it only matches in one position) and pull the lock latch closed to secure the connection.

Step 4. For system with electrical supply 440/480VAC only, connect the 48VDC cable for the fan (**B** in **Figure 5-33** above).

5.12 Cabling Components

According to your system configuration, connect the cables of the mandatory flow signal, and optional Inlet, Outlet/Ready and Bypass Valves, etc. to the Connection Box of lamp 1 on the Atlantium Unit. On a side of the Connection box, there are 5 holes marked **9** in the diagram below to be used to thread these cables.

WARNING!

DO NOT DRILL INTO THE CONNECTION BOXES OR ALTER THEM IN ANY WAY!!!!

The Connection Box already contains entry glands for ALL of the cabling needs.

Making holes in a Connection Box will destroy its waterproof seal, endangering the components inside. Do Not draw electricity from the Atlantium Unit's Connection Box as it will damage the Unit's electrical configuration.

Any person involved in handling the Connection Box card or its components is required to wear an electrostatic discharge (ESD) Wrist Strap.

Connect the ESD Wrist Strap to any bolt on the body of the Atlantium Unit or to the metal bracket under the Connection Box using a banana plug or an alligator clip.

The directions below are for all **RZ300 Series** systems, in which the first Connection Box is labeled number **1**.

Where there is reference to the first Connection Box, for systems that contain only one Connection Box the single Connection Box is the first Connection Box.

5.12.1 Connection Box Card 1300+1500

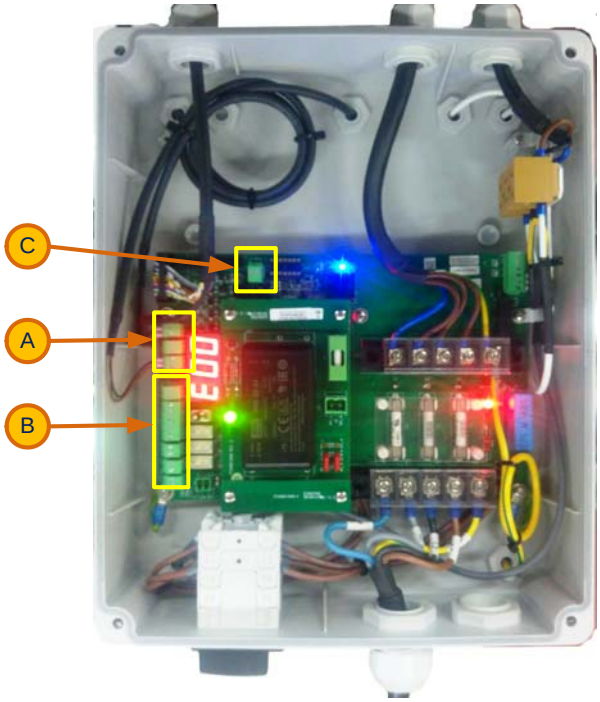


Figure 5-34: Connection Box Card 1300+1500 for Systems with 400VAC

Figure 5-35: Connection Box Card 1300+1500 for Systems with 440/480VAC

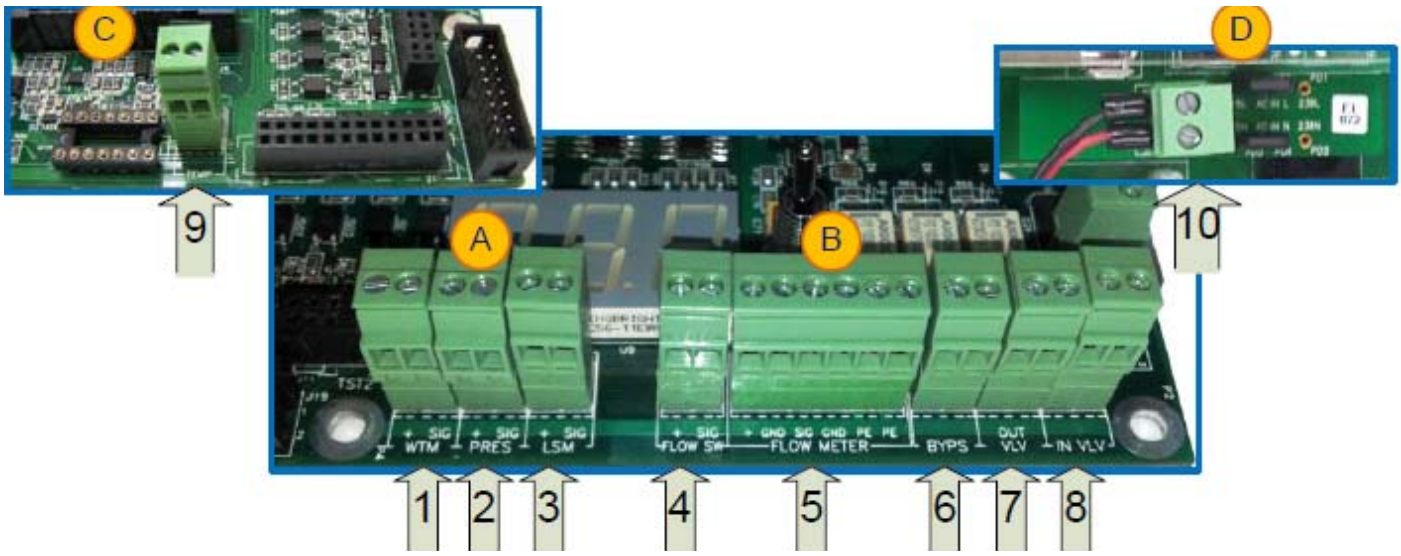


Figure 5-36: Signal Connections on Connection Box Card 1300+1500

Table 5-6: Connections Key - Connection Box Card 1300-1500

#	Text on the Circuit Board	Description	Signal type	Value	Notes
A					
1	WTM	UVT Analyzer input	Analog	4-20mA	already assembled
2	PRES	Pressure Transmitter input (Optional)	Analog	4-20mA	Optional
3	LSM	UVIS Sensor input	Analog	4-20mA	already assembled
B					
4	FLOW SW	Flow switch input	Digital	Dry contact / Open collector	Either Flow Switch or Flow Meter Optional *
5	FLOW METER	Flow meter input	Analog	4-20mA	
6	BYPS STR	Bypass valve output	Dry contact		Optional *
7	OUT VLV	Output valve output	Dry contact		Optional *
8	IN VLV	Inlet valve output	Dry contact		Optional *
C					
9	TEMP	Temperature sensor input	Analog		already assembled
D	For 440/480VAC Only				
10	115L J3 115N	115V connector with two pins	Analog		Already Assembled

5.12.2 Schematic Diagrams

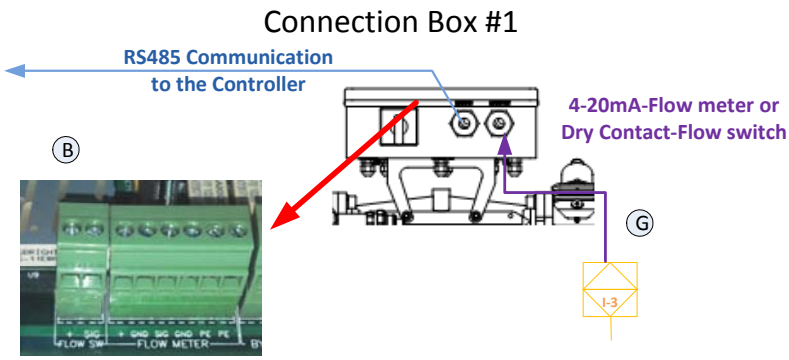


Figure 5-37: Signal Connections on Connection Box Card - Schematic Drawing

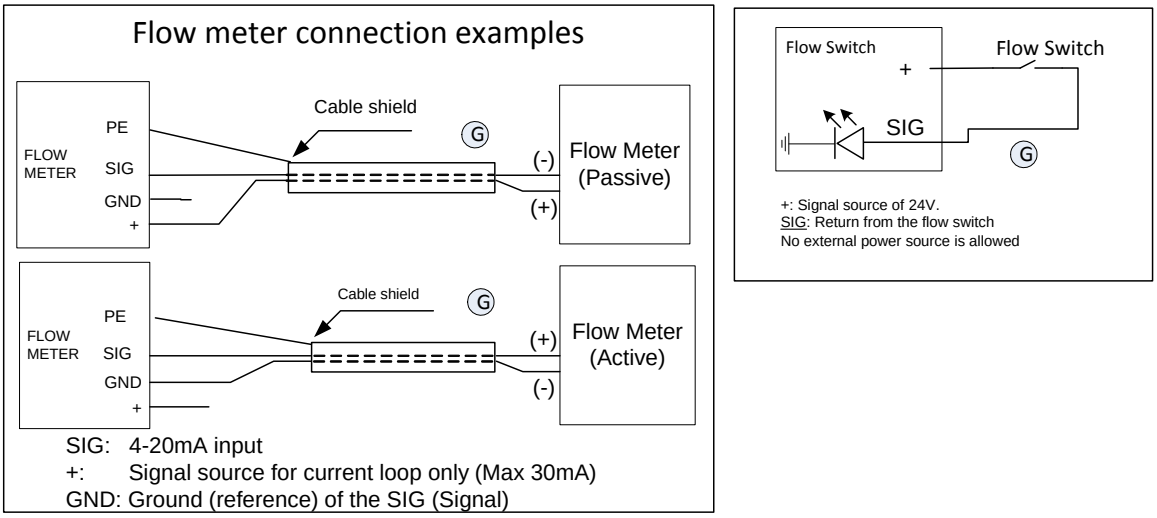
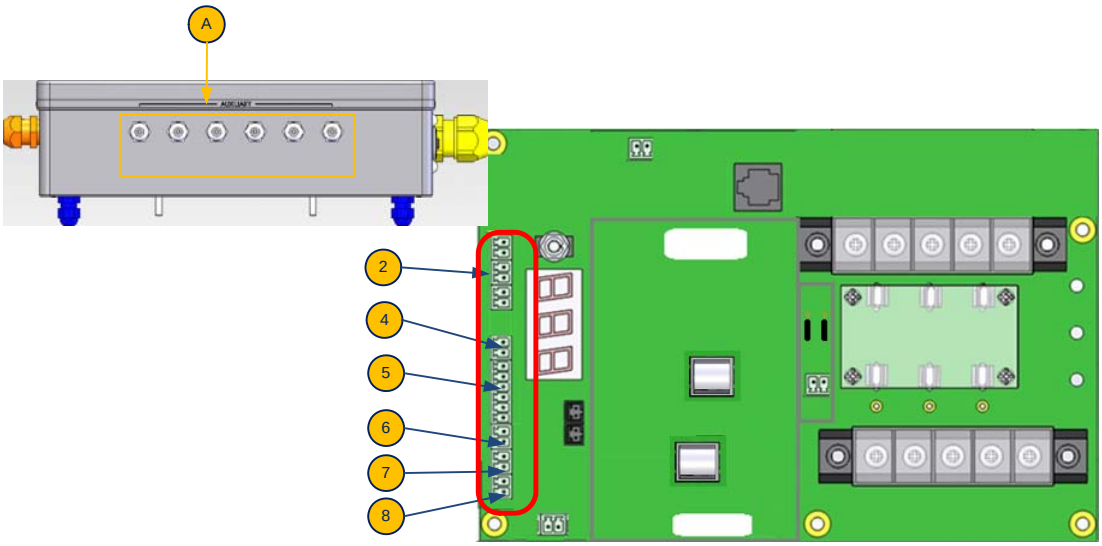


Figure 5-38: Signal Connections on Connection Box Card - Schematic Drawing

For details regarding the cables marked **G**, refer to Table 5-2, “Peripheral Equipment Cable Details,” on page 65.

➲ To cable signal components:



Step 1. Connection Box with Card 1300-1500 with 115V connector On the first **Connection Box**, remove the covers of holes (marked **A** in the diagram below) as needed for the auxiliary components in your system configuration.

Step 2. Add plastic glands to each hole to be used.

Step 3. For the (optional) **Inlet Valve** connection, the maximum voltage is **24V max**. The maximum current is **2A max**. The maximum switching capacity is **50W max**. On the first Connection Box, insert the **Inlet Valve** cable into one of the holes marked **A** and thread it up to the Dry Contact connectors labeled **IN VLV** (marked **8** in the diagram above.)

Step 4. For the (optional) **Outlet Valve** connection, the maximum voltage is **24V max**. The maximum current is **2A max**. The maximum switching capacity is **50W max**. On the first Connection Box, insert the **Outlet Valve** cable into one of the holes marked **A** and thread it up to the Dry Contact connectors labeled **OUT VLV** (marked **7** in the diagram above.)

Step 5. For the (optional) **Bypass Valve** connection, the maximum voltage is **24V max** The maximum current is **2A max**. The maximum switching capacity is **50W max**. On the first Connection Box, insert the **Bypass Valve** cable into one of the holes marked **A** and thread it up to the Dry Contact connectors labeled **BYPS STR** (marked **6** in the diagram above.)

Step 6. To connect the **Flow Signal:**

- For a **Flow Meter**, on the first Connection Box, insert the cable into one of the holes marked **9** and thread it up to the connectors labeled **FLOW METER** (marked **5** in the diagram above.) the standard current is 4-20mA. Connect the wires according to the manufacturer's instructions. The connectors are marked **PE**, **PE**, **SIG**, **GND**, and **+**. Typically, the cable is comprised of three wires and the shield. A wire connects to the

+ port and a second wire connects to the **SIG** port. The GND wire connects to the **GND** port. The Shield connects to one of the **PE** ports.

ATTENTION!

Use the **+** pin for the signal source (current loop) **ONLY**.

Do not use the **+** pin as the power source for the flow meter. For more information see ***Flow Measurement/Detection*** on page **34**.

- For a **Flow Switch**, on the first Connection Box, insert the cable into one of the holes marked **9** and thread it up to the connector labeled **FLOW SW** (marked **4** in the diagram above.) Connect the wires according to the manufacturer's instructions. The ports are marked **+** and **SIG**.

Step 7. Fasten the plastic nut of each of the cable glands.

5.12.2.1 For Systems with 440/480VAC - Connecting the Ultrasonic Cleaning Module to the Power Module

The Ultrasonic Cleaning Module's power cable must be connected to the Power Module.

➲ To connect the Ultrasonic Cleaning Module to the Power Module:

Step 1. Open the door of the Power Module and locate the following:

- **C** - the glands for threading the Power cables
- **D** - The circuit breakers to which the Power cable coming from the Ultrasonic Cleaning Module are to be connected (see the diagram below.)
- **E** - The terminal block to which the Power cable from the mains (provided by the customer) is to be connected according to the relevant diagram above.

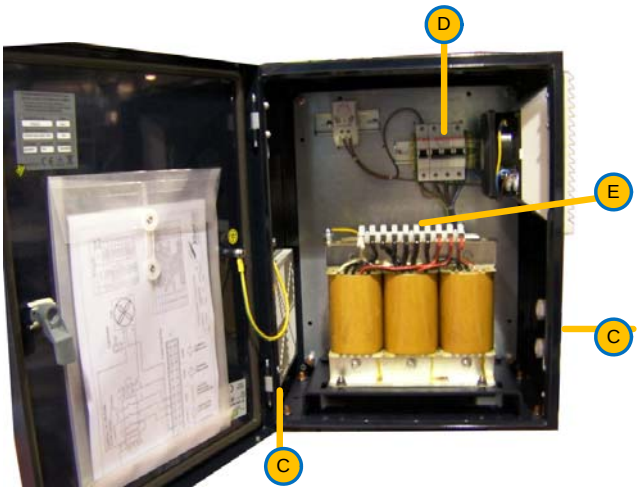


Figure 5-39: Power Module - Mounting Holes

Step 2. Thread the other end of the cable (marked **Y** in **Figure 5-43** and **Figure 5-44** above) through an available gland **(C)** in the Power Module frame.

Step 3. Connect it to the circuit breakers **(D)** according to the following list and as shown in the diagram below.

- **L** to **4** in the Power Module's terminal as marked in the figure below.
- **N** to **N** in the Power Module's terminal as marked in the figure below.
- **PE** to **PE** in the Power Module's terminal as marked in the figure below.

Regarding connecting the Power Module to the Mains, See ***Connecting the Atlantium System to the Mains*** on page **5-82.**

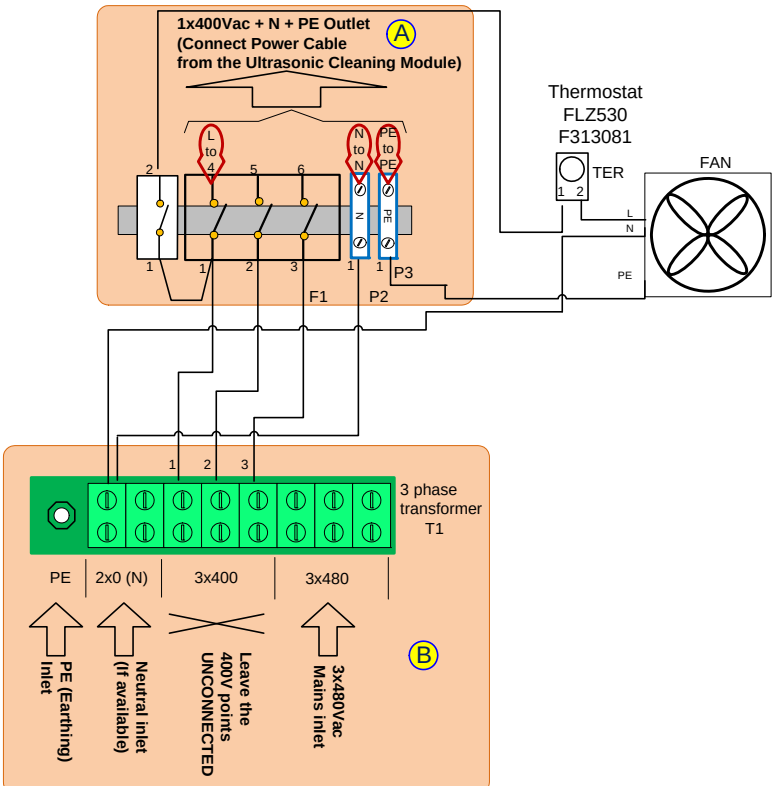


Figure 5-40: Typical Ultrasonic Cleaning Module Connections to Power Module

5.13 Modbus Communication Connections

Modbus is an application layer messaging protocol for client/server communication between devices connected on different types of buses or networks. Modbus is often used to connect a supervisory computer with a remote terminal unit (RTU) in supervisory control and data acquisition systems.

The Atlantium system can be controlled directly from your facility control system via Modbus - RTU on RS485 (network) serial connection.

To accomplish this:

- The Modbus Settings must be configured in the **Control Module**. See ***Configuring the Modbus*** on page **6-94.**

- The Facility control system communication properties must be configured to match that of the Atlantium system. See ***Setting Up the Facility System Communication Properties*** on page **5-81.**
- For the terminal block connectors for the Modbus in the **Control Module,** see Table **5-3** ***Control Module Connector Pins*** above.
- Signalling communication incoming to the facility control system is according to the Modbus Registers. **Appendix C**, ***System Messages*** .

ATTENTION!

To ensure communications are operational, be sure to connect the cables **TX to RX** and **RZ to TX**.

5.13.1 Setting Up the Facility System Communication Properties

To properly configure communications between your Atlantium system and control network, set up the communication properties and **COM1** port settings at the facility control system as follows:

Node Address:	2 to 254 ***
Available Commands	Function codes
Function Codes: *	0x03
Supports IEEE Floating Point: **	NO (register/xxx
Baud Rate:	4800, 9600, 19200, 38400, 57600, 115200 ***
2 Wire	
Parity:	None
# of Stop Bits:	1
# of Data Bits	8

(*) Read all the registers in the system at once with **0x03** (includes coils, discreet coils, holding registers and input registers).

(**) Divide the desired parameter by xxx to calculate the correct signal point. See the Divide column in the table in the section, ***Modbus Communication Protocol*** on page **166**.

(***) Must be the same in the **Control Module** setting

To view the Modbus Registers, which govern the signaling communication from the Atlantium system to the facility control system, see **Appendix C**, ***System Messages*** .

5.14 Connecting the Atlantium System to the Mains

DANGER!

ALL MAIN CIRCUIT BREAKERS OF THE ELECTRICAL LINE FEEDING THE ATLANTIUM SYSTEM COMPONENT MUST BE SET TO THE OFF POSITION AND LOCKED WITH AN APPROPRIATE LOCK TO PREVENT ACCIDENTAL CHANGE TO THE ON POSITION.

DO NOT REMOVE THE LOCK OR SET ANY MAIN CIRCUIT BREAKER TO THE ON POSITION UNTIL ALL ELECTRICAL WIRING IS PROPERLY CONNECTED.

UNCONNECTED WIRES/CABLES IN A LIVE ELECTRICAL FEED IS A HAZARDOUS SITUATION THAT CAN RESULT IN DEATH OR SERIOUS INJURY.

The infrastructure of the electrical connections is to have been prepared according to the ***Electrical ICD*** drawing provided to you by Atlantium with this manual.

For installations that require the OVPD unit (440-480VAC only), the customer must provide the Power cable to the mains.

5.14.1 Mains Connection Phases

Phase 1 Connect the **Control Module** to the mains.

Phase 2 Connect the Atlantium Unit's 100-240VAC connection to the mains (if exists).

Phase 3 Connect the Atlantium Unit's Power cable to the mains either directly or via the Power Module.

ATTENTION!

- **Do not Stretch Electrical Cables - Avoid tension on the electrical cables. They are to be installed within conduits and must not be stretched.**

5.14.2 Connecting the Control Module's Power Cable

The **Control Module** comes with a power cable of 2m/6.5ft. length. For the electrical specifications, See ***The Control Module*** on page **2-22.** Prepare the appropriate plug for the electrical standards of your region. Considering the distance between the **Control Module** and your electrical box, you may want to replace the electrical cable with a longer one.

➲ To connect the Control Module's power cable:

- Connect the appropriate plug for the electrical standards of your region.
- **Do not** plug in the **Control Module** to an outlet until after the hardware installation is complete and the system is ready to be initialized. (See ***Initializing the Atlantium System*** on page **7-99.**)

5.14.2.1 Replacing the Control Module's Power Cable

You can order a longer cable from Atlantium or acquire one that conforms to the specifications in **Table 5-2** on page **65** from a local source.

Avoid direct contact with the electronic circuit board which is sensitive to electrostatic discharge (ESD). Utilize best practices regarding ESD when replacing the power cable.

➲ To connect a longer power cable to the Control Module:

Step 1. Using the appropriate screwdriver, unfasten the screws on the **Control Module (A)** and carefully remove the back.

Step 1. Unfasten the Power cable from the connectors as show to the right **(4)** under **4A PWR** (corresponding to **4** in **Figure 5-21**) numbers **1**, **2** and **3** (and listed in **Figure 5-21** below **(1**, **2** and **3)**.

Step 2. Loosen the plastic nut of gland **(3)** and pull it through and out of the plastic gland at the bottom.

Step 3. Thread the longer Power cable through its plastic gland **(3)** and fasten it to the appropriate connectors mentioned above.

Step 4. Connect the Power cable's two power leads to **N and L** and connect the ground lead to **E**.

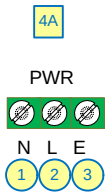


Figure 5-41: Control Module Terminal Connectors

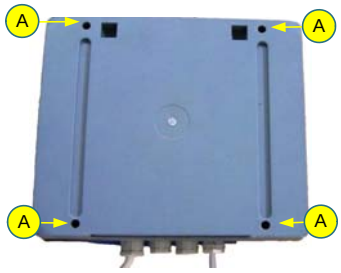


Figure 5-42: Control Module's Back

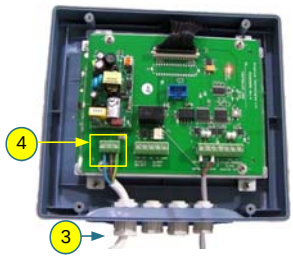


Figure 5-43: Control Module's Circuit Board Power Connections

5.14.3 For Systems with 440/480VAC - Connecting the 100-240VAC Wires to the Mains

If your system is with 440/480. it may include the secondary 100-240VAC connection, the wires are to be connected L-N for 100-240VAC or L-L for 200VAC (Non-Industrial network). Refer to the ***Electrical ICD*** drawing provided to you by Atlantium with this manual.

Please follow the Electrical ICD drawing provided to you by Atlantium with this manual.

Prepare the appropriate plug for the electrical standards of your region. Considering the distance between the Power Modules and your electrical box.

A certified electrician is required to perform these procedures. Follow your local regulations.

Do not plug in the Unit's Power cable to an outlet until after the hardware installation is complete and the system is ready to be initialized. (See ***Initializing the Atlantium System*** **on page 7-99.**)

5.14.3.1 For Systems with 440/480VAC & Ultrasonic Cleaner - Connecting the Mains to the Power Module

If your system with 440/480VAC includes the Ultrasonic Cleaner, it includes a Power Module. Connect according to the diagram and instructions below.

Each Power Module must be connected to the mains with a customer provided cable, that contains **according to local regulations**. In calculating the length of this cable, consider the distance between the Power Module and your electrical box.

Also see the ***Electrical ICD*** drawing provided to you by Atlantium with this manual.

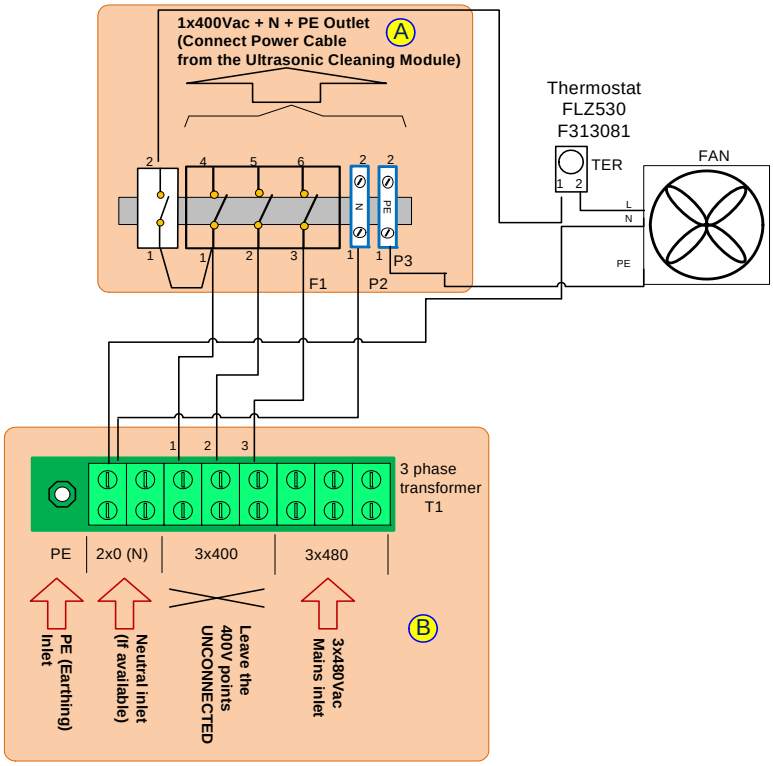


Figure 5-44: Typical Power Module Connection to the Mains

➲ To connect the Power cables and the Power Module:

Step 1. Open the door of the Power Module and locate the following:

- **C** - the glands for threading the Power cables
- **D** - The circuit breakers to which the Power cable coming from the Atlantium Unit and the Ultrasonic Cleaning Module are to be connected (see the diagram above)
- **E** - The terminal block to which the Power cable from the mains (provided by the customer) is to be connected according to the relevant diagram above.

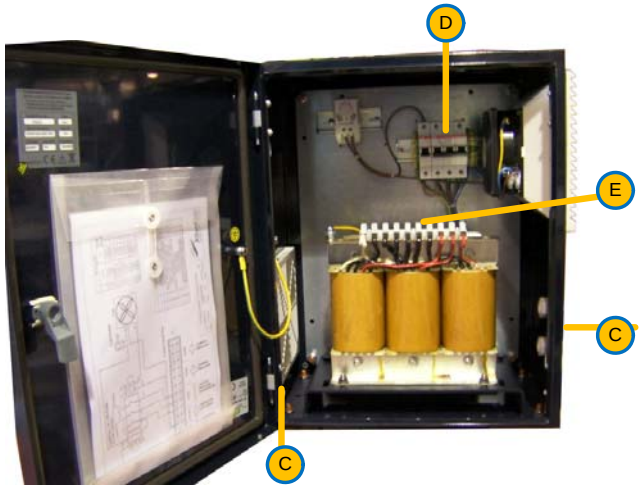


Figure 5-45: Power Module - Mounting Holes

Step 2. Use **A** in the diagram above and Figure 5-44, ***Typical Power Module Connection to the Mains***, on page 84 to connect a Power cable from a Connection Box to a Power Module and the Power cable from the Ultrasonic Cleaning Module.

Step 3. Use **B** in the diagram above to connect a Power cable that is to connect to the mains.

Step 4. Thread cables into an available convenient gland.

Step 5. Connect the power cables to the mains according to local regulations.

5.14.4 For Shipboard Installations

- When the Atlantium Unit is installed shipboard, all metal components must be grounded to the nearest possible grounding point on the ship's metal structure. Be sure to prepare a ground point next to the designated location for the Atlantium Unit.

5.14.5 Connecting the Atlantium Unit's Power Cable

This section applies to systems with 400VAC and for systems with 440/4800VAC without Ultrasonic Cleaner

For systems with Ultrasonic Cleaner, Power Module(s) are included. If your system includes the Ultrasonic Cleaner, see ***Connecting the Atlantium System to the Mains*** on page 82

The Atlantium Unit comes with a power cable of (5m/15ft.) or 10m/30ft (according to you order.) Prepare the appropriate cables and plugs for the facility mains connection according to the electrical standards of your region. Considering the distance between the Atlantium Unit and your electrical box, you may want to replace the electrical cable with a longer one.

- **A certified electrician is required to perform these procedures. Follow your local regulations.**
- **Do not** connect the Unit's Power cable to an outlet until after the hardware installation is complete and the system is ready to be initialized. (See ***Initializing the Atlantium System*** on page **7-99.**)
- **Do not** connect the Unit's Power cable to an outlet until after the hardware installation is complete and the system is ready to be initialized. (See ***Initializing the Atlantium System*** on page **7-99.**)

➲ To connect the Atlantium Unit's power cable:

- Connect the appropriate plug for the electrical standards of your region and connect it to a mains outlet.

- The power cables supplied with the unit has 5 wires: 3 Phases (L1, L2, L3), Neutral (N) and Ground (PE). However, for the 440/480V installation, the Neutral wire is redundant and is not to be connected on either side of the cable.

5.15 Installation Check List

When you are ready to schedule the Atlantium technician to commission the Atlantium system, complete the check list below.

Send this check list together with digital photos of every angle of the installed system to your Atlantium representative. It generally requires at least 2 weeks after the check list has been completed to schedule the Atlantium commissioning.

Table 5-7: Atlantium System Installation Check List

Item	Done
■ The System is installed: ❑ Vertically - Water is flowing upward ❑ Horizontally	
■ The Atlantium Unit, Ballast Module and other components are located in their final position	❑
■ The Atlantium Unit and all of its components are not located where they may be in contact with direct water or chemical dripping	❑
■ Ballasts are installed in a vertical position.	❑
■ All Ballast Module connections are complete.	❑
■ The electrical cables between components are not stretched.	❑
■ All of the System components are securely installed and have sufficient service clearance on all sides (according to the ICDs).	❑
■ All power preparations are complete and Power connects are wired as instructed in this guide.	❑
■ Correct voltage was measured where the Power connects to the Ballast Module	❑
■ Voltage rate _____ Number of phases _____	

Table 5-7: Atlantium System Installation Check List

Item	Done
■	❑
■	❑
■ The Atlantium Unit is connected to the water pipes according to the Installation plan developed by facility engineers and Atlantium personnel. Proper pipe lengths (Inlet - 10D, Outlet - 5D) (Inlet - 10D, Outlet - 5D) are installed.	❑
■ Water flows through the Atlantium Unit - The Unit has been checked while full of water and no bubbles or leaks exist in the system. The system is balanced, supported and pressure stable (no vibration, etc.). No known water hammers or pressure above 10 bars are on the line.	❑
■ The **Control Module** connects is complete. All supporting features (e.g., General Alarm, External On/Off, etc.) and wiring were installed according to the Installation plan based on the instructions in this guide.	❑
■ All Connection Box connections are complete. All supporting features (e.g., flow meter or flow switch, auto valves open/close) and wiring were installed as instructed in this guide.	❑
■ For systems with **TRACS**, network connection was installed between **TRACS** the **Control Module**.	❑
■ **You are scheduling an electrician to be on site and available during the System commissioning (a must or the installation cannot be completed).**	❑
■ To be sent to Atlantium with this checklist, attach photos of the following: ❑ Atlantium Unit ❑ Ballast Module(s), **Control Module** and other components ❑ Accessories (Flow meter or flow switch, CIP ports, Sampling valves, etc.)	

Customer Approval

Name of Customer's authorized representative:

_____

Position:

_____

Signature of Customer's authorized representative:

_____

Date:

_____

6 Configuring the Control Module

The Standard **Control Module** controls the operations and dose measurements of the Atlantium system.

The System uses a validated algorithm to measure the dose and display it in real time. Depending on how you choose to run your process, you can configure the Control Module to maintain a specific dose or to simply use a particular power level and inform you if it does not reach the dose you designate.

The **Control Module** provides a real-time display of operational data.

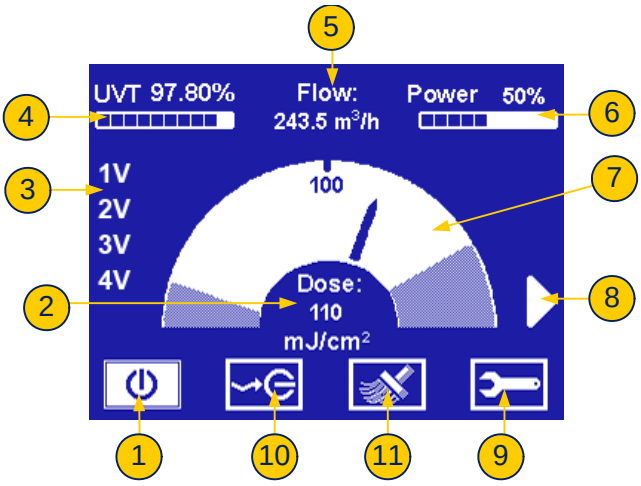


Figure 6-1: Initial Main Operation Screen

Table 6-1: Main screen components

#	Item
1	**Lamp Power Button** - upon Control Module initiation, is on the Off position. After being pressed, turns white, showing that the lamps are working. It can be used to shut lamps down in an orderly manner.
2	**Dose** - the **Op**(erate) **At d**ose is the dose the user wants the Unit to achieve and maintain. Generally, it should be set about 10-20% over the minimum dose to assure stability and on-spec operation in the case of small variations of water quality or flow. You should confer with the Atlantium application engineer in setting this number based on the capacity of the Unit, variability at the site and power considerations. When the actual dose is below the **Op**(erate) **At d**ose, the number rounds down. Otherwise, it rounds up.
3	Numbers **1-4** represents up to four lamps included in the System. **V** indicates lamps in operation. **X** indicates lamps in not operating. When the System is in an initial state, no indicator appears.
4	**UVT** - scale exhibits the current quality of the water flowing through the system, and the amount of the UV light absorbed by the water, compared to the amount UV available for disinfection. If the UVT percentage factor is high, the clarity factor of the water is high, and vice versa.
5	**Water Flow** - Number shows the current rate of water flowing through the Unit, based on a flow signal (for variable flows) or a constant volume (for on/off steady flow applications).
6	**Power** - displays the current power level. The Unit's electronic ballast varies continuously (not in limited steps) to maintain the dose and respond to changing conditions.
7	**Dose Gauge**- represents the total system dose. The gauge has two different linear functions - from the dose setting down and from the dose setting up. The minimum and maximum allowable dosage levels are user-defined.

Table 6-1: Main screen components

#	Item
8	Press to access the System Status screen.
9	Press to access Settings and Technician protected screens via Password entry screen.
10	**External Operation button** - appears on the main operating screen only if the External On/Off feature is activated in the main Settings screen. If the feature is enabled, the system can be turned on or off from the facility's External On/Off switching mechanism. When the feature is disabled, the system may be turned off only from this screen or on the (optional) **TRAC Software**.
11	**Cleaning System button** - appears on the main operating screen only if the Cleaning feature is included in the system. Enables/disables the optionally-installed Cleaning mechanism for cleaning the Unit's quartz tubes.

Plan your **Control Module** configuration with your Atlantium Application Engineer to assure that the system performs properly and achieve its objectives. Review the Configuration Printout with the Atlantium Technician who is commissioning the site to make sure the settings are understood and correct. The Atlantium Technician configures the initial system settings via the Control Module. The User can afterwards change settings as experience suggests improvements in the operations and processes.

➲ To access the Settings screen:

Step 1. From the **Main Operations** screen, press . The **Password** screen appears.

Step 2. Using the numbered keys, enter the password (default password is **1357**) and press **OK**. The Settings screen appears.

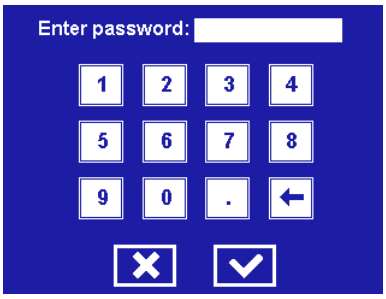


Figure 6-3: Password Screen

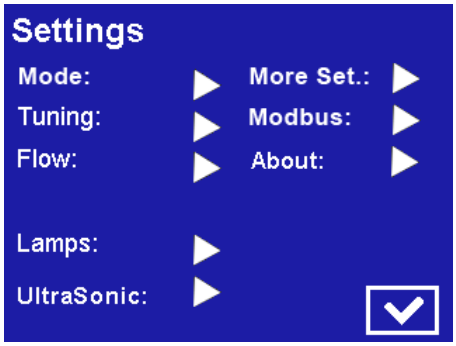


Figure 6-2: Settings Screen

6.1 Configuring the Operation Mode

The **Control Module** provides Atlantium system has two operational modes. Determine which suits your objectives and facility processes: of control:

- **Power Mode** - The system is set to operate at a particular power level. Under all circumstances, the system operates at the specified designated power level. Power mode is the right choice if maintaining a particular dose is not the objective. It is often chosen where energy consumption concerns are paramount or when the maximum disinfection power is needed at all times and concern that low flows might give high doses or that high flows might give lower doses is not an issue (i.e., maintaining a particular dose is not the objective).
- **Dose Mode** - The system is set to operate at a particular dose. It maintains the designated dose as conditions allow and generally is set to alarm when the dose is not achieved.

➲ To set the Operation Mode:

Step 1. Access the **Control Module's** Settings screen as detailed above and for **Mode**, press ▶. The Mode Setting screen appears

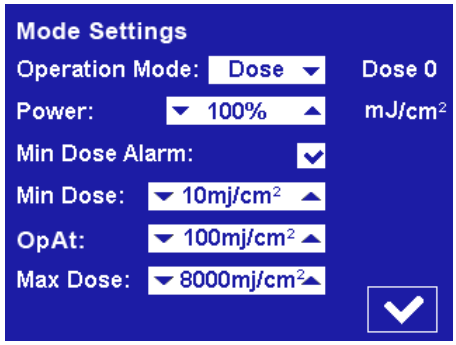


Figure 6-4: Dose Settings Screen

- **Dose mode**:
  - **a** For **Operation Mode**, use the ▲▼ arrows to select **Dose**.
  - **b** Ignore the **Power%** parameter
  - **c** To enable an alarm to be triggered should the Dose reach the minimum allowed, press the **Min Dos Alarm** box until a check-mark appears.
  - **d** For **Min Dose**, press the ▲▼ arrows to set the lowest allowable dose in mJ/cm^2. This sets the linear function for the left side of the dose gauge and is the dose the Unit maintains. If it is not maintained, it dips into the shaded area, and, if configured to send an alarm, it does so.
  - **e** For **Op**(erate) **at**, press the ▲▼ arrows, to set the operating dose. This generally includes a safety factor so that the Unit does not operate at the minimum margin and does not send false alarms when the water quality or flow quivers or changes rapidly.
  - **f** For **Max Dose**, press the ▲▼ arrows, to set the maximum UV dose limit. This sets the linear function for the right side of the gauge and defines when the arrow goes into the shady area. In cases where there is low flow, unusual water chemistry or the potential for overheating, high dose can indicate impending issues.
  - **g** Press ☑ **OK**. The settings are saved and you are returned to the Settings screen.
- **Power mode**:
  - **a** For **Operation Mode**, use the ▲▼ arrows to select **Pwr**.

b Set the **Power %** using the ▲▼ arrows.

c If you want to be alerted to the dose going beneath a value, set it here as the **Min Dose**. Press the ▲▼ arrows, to set the lowest acceptable dose in mJ/cm^2. This configures the left side of the gauge, but the system does not change the power or operation to achieve or maintain this dose. It is simply for information. If you set the **Minimum Dose alarm** (see below) it also triggers an alarm if the Unit falls below this dose.

d Ignore the **Operate at** parameter.

e For **Max Dose**, press the ▲▼ arrows, to set the maximum UV dose limit. This sets the linear function for the right side of the gauge.

f To enable the alarm trigger if the unit goes below the minimum allowed, press the **Min Dos Alarm** box until a check-mark appears.

g To save your changes and return to the Settings screen, press ☑ **OK**.

6.2 Configuring the Flow Settings

The Flow signal and related settings are critical for determining the dose and assuring that the system is working within design guidelines. The accuracy and tolerance of the flow meter measurement should match the **Control Module** interpretation of the flow signal.

➲ To configure the flow settings:

Step 1. Access the **Control Module's** Settings screen as detailed above.

Step 2. For **Flow**, press ▶. The Flow Settings screen appears.

Step 3. For **Flow Input**, use the ▲▼ arrows to select either **Meter** or **Switch**.

- If you selected **Switch**:
  - Mark the electronic polarity as **Normally Closed** or **Normally Open**.
  - The constant flow number is set by the Atlantium Technician in **Technician 2 Screen**.
- If you chose **Meter**:

a To set **Gain**, you need the meter's maximum-design flow rate (appears on the flow meter datasheet). You need to enter flow in m^3/h, so if it is written in any other unit of measurement, you must convert it to m^3/h.

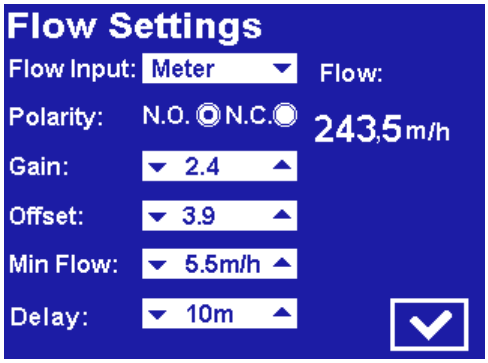


Figure 6-5: Flow Settings Screen

b In the **Gain** field, using the ▲▼ arrows, enter the meter's maximum-allowed flow rate divided by 100. (If the max flow rate is 100 m^3/h, enter the value **1**. Convert any value into m^3/h values to make this calculation.)

c To set **Offset**, turn off the flow meter and note the reading on the screen in the right hand corner. If it is not zero, using the ▲▼ arrows, enter the reading. This assures that the system registers zero in the dose calculation when the flow meter is turned off and compensates for any measurement anomalies.

d For **Min Flow**, using the ▲▼ arrows, enter the minimum flow for standard operations. Set it, making sure to convert your calculations if necessary to use the same units of measurement as are on the screen. Set this number based on process and unit considerations. The Unit cannot run without flow and in some cases, low flow may be indicative of some upstream issue. Once the Low flow number has been reached, the unit reduces power to the lowest possible power setting until the Delay has passed, and then shuts down. If the flow increases before the delay period elapses, the Unit returns to its normal operation. If the Unit shuts down, if the **Restart on Flow** is enabled (see **Step 8.** on page **94**.), it resumes operations when normal flow is restored, otherwise it needs to be turned on manually to return to normal operations.

Step 5. For **Delay**, using the ▲▼ arrows, enter the number of minutes the system should wait before shutting down in the event of flow below the minimum.

Step 6. To save your changes and return to the Settings screen, press **OK**.

6.3 Configuring Ultrasonic Cleaner

If Included in the system, determine how often you want to use the Ultrasonic Cleaner and set the appropriate parameters. If not included, ignore these settings.

➲ To configure the Ultrasonic Cleaner:

Step 1. Access the **Control Module's** Settings screen as detailed above.

Step 2. For **Cleaner**, press ▶. The Cleaner Setting screen appears.

Step 3. For **Every**, using the ▲▼ arrows, set how often - in hours - the cleaning should run.

Step 4. To save your changes and return to the Settings screen, press ✔ **OK**.

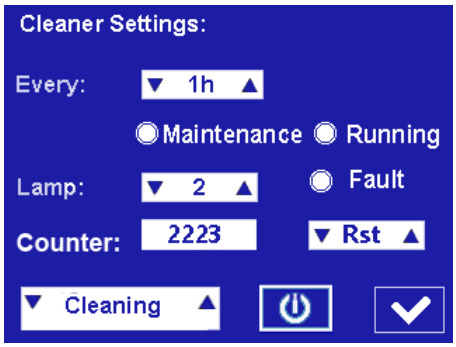


Figure 6-6: Cleaning Settings Screen

6.4 More System Settings

Access the following settings through the More Settings screen:

- Set the maximum temperature for the Atlantium unit.
- Units of Measurement - the default setting is metric; this allows you to change to American units
- Enable/Disable Cleaning- to manually disable/enable the Cleaning system
- General Alarm - to configure a facility general alarm to be triggered when there is a system alarm or if you opt for a minimum dose alarm
- External On/Off - to configure facility control of the Atlantium system

➲ To configure more system settings:

Step 1. Access the **Control Module's** Settings screen as detailed above.

Step 2. For **More System Settings**, press ▶. The System Settings screen appears.

Step 3. To set the maximum temperature threshold for the Atlantium unit, for **Max Temperature,** using the ▲▼ arrows, set the temperature.

Step 4. To change the units of measurement from Metric units, press the **American Unit** check-box until a check mark appears.

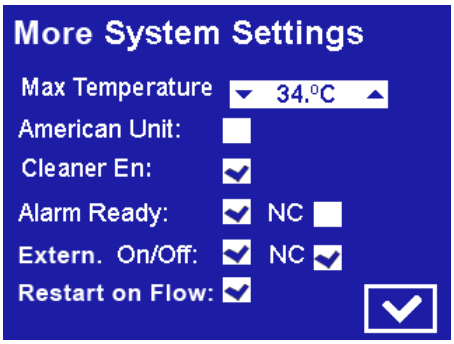


Figure 6-7: More System Settings Screen

Step 5. To manually disable the Cleaning system, press the **Cleaner En** check-box until the check-mark is removed. To re-enable the Cleaner, press the check-box until a mark appears.

Step 6. If your system configuration includes a facility general alarm, or system ready configuration, for **Alarm** mark the left check-box. For **Ready**, leave the left check-box unmarked. For electronic polarity **Normally Closed** mark the **NC** check-box or for **Normally Open** leave the **NC** check-box blank.

Step 7. If your system configuration includes a facility external control, mark the first check-box.

- For electronic polarity **Normally Closed** mark the **NC** check-box or for **Normally Open** leave the **NC** check-box blank.
- For regular wiring, set according to the table in ***External On/Off - Additional Notes*** on page 66

Step 8. To set the system to restart at the configured settings when the flow returns to normal rate, press the **Restart On Flow** check-box until a mark appears. When the check-mark is marked and the system shuts down because of low flow the main power remains on and a flow label on the main screen flashes indicating the system is waiting for a normal flow rate to resume). When normal flow rate returns, the system restarts and resumes operation according to the configured settings automatically.

Step 9. To save your changes and return to the Settings screen, press ☑ **OK**.

6.5 Configuring the Modbus

The Modbus screen displays the software version and other system version information.

Configure the Modbus node address assigned to this system in the **Control Module**. The Modbus enables communication with the rest of the facility and facilitates integration of the Atlantium Unit into the overall process.

➲ To configure the Modbus:

Step 1. Access the **Control Module's** Settings screen as detailed above.

Step 2. For **Modbus**, press ▶. The Modbus screen appears.

Step 3. For **Address**, using the ▲▼ arrows, set the Modbus address node to a number as per the serial address setting in your facility.

Step 4. For **BaudRate**, using the ▲▼ arrows, set the appropriate baud rate for this system. The options are:

- 115200
- 19200
- 57600
- 9600
- 38400

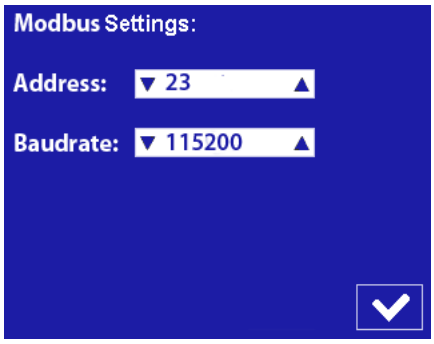


Figure 6-8: Modbus Settings Screen

Step 5. To save your changes and return to the Settings screen, press ☑ **OK**.

The software version and other system version data are displayed on the About screen.

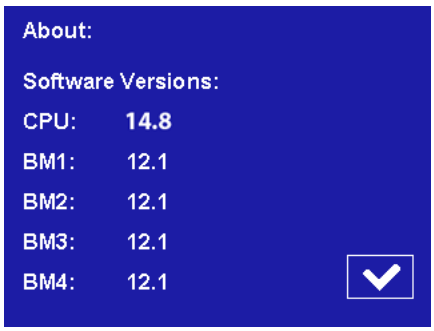


Figure 6-9: About Screen

6.6 Configuring the Technician Settings

As a precautionary measure, the Technician settings in the **Control Module** should be verified.

To configure the Technician Settings:

Step 1. From the **Main Operations** screen, press . The **Password** screen appears.

Step 2. Using the numbered keys, enter the password (default password is **8888**) for the Technician Screens and press **OK**. The **Technician 1** screen appears.

Step 3. For **System**, verify that the correct Atlantium system is defined.

Step 4. For **Auto On**, to set the Atlantium system to restart automatically when power is restored after a facility-wide power failure and reactivate according to the most recent settings, press the check-box until a mark appears.

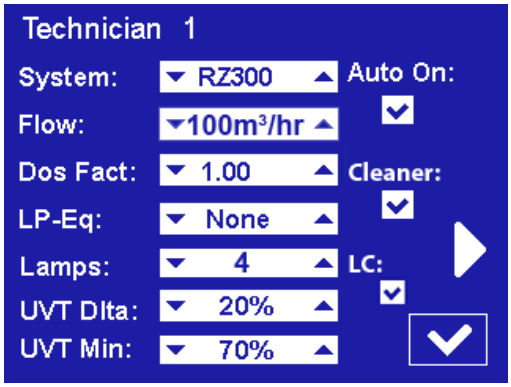


Figure 6-10: Generic Technician 1 Screen

Step 5. If the system includes a cleaning system, press the **Cleaner** check-box until a check mark appears.

Step 6. If the system includes a **flow meter**, for **Flow**, set the maximum flow rate using the ▲▼ arrows. Otherwise, ignore this parameter.

Step 7. If the system uses a **flow switch** instead of a **flow meter**, in the **Flow** field, set the constant maximum flow rate using the ▲▼ arrows.

Be Careful when using this screen as changing the unit time or other errors can substantially impact your system!

Step 8. If your system contains PG quartz, the Dose Factor must be configured. For **Dos Fact**, set the Dose Factor using the ▲▼ arrows.

Step 9. For facilities that require Dose monitoring in Low Pressure Equivalence terms, for **LP-Eq**, use the ▲▼ arrows to set the Low Pressure Equivalence figure. The Dose is expressed as a Low Pressure Equivalent PDM system.

Step 10. Verify that the system is configured correctly for the number of lamps. Use the ▲▼ arrows to set the correct number.

Step 11. If the system is equipped with a special 10m long cable (applicable to RS104 only) check the **LC** check box

Step 12. To save your changes and return to the Settings screen, press ☑.

Step 13. To access the **Technician 2** screen, press the ▶ arrow on the right of the **Technician 1** screen.

Step 14. Verify that for each lamp, the parameters listed for UVT and UVI appear in the normal range.

Step 15. Ignore the **Power** option. It is used for testing only.

Step 16. Verify that for each lamp the column on the right is set to **ON** (enabling the lamps).

Step 17. To save your changes and return to the Settings screen, press ☑.

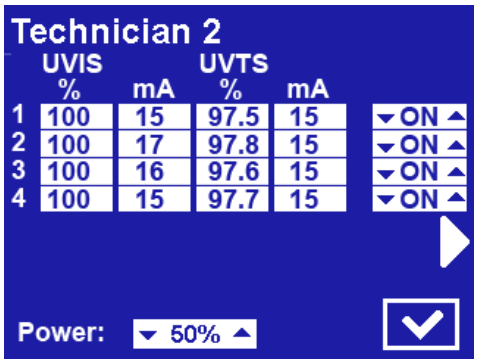


Figure 6-11: Technician 2 Screen*

Step 18. To access the **Technician 3** screen, press the ▶ arrow on the right of the **Technician 2** screen.

Step 19. Verify that the temperature displayed is within the norm. The temperature for each installed and operating lamp is displayed. An **N/A** appears for a lamp that is not operating or not installed.

Step 20. Ignore the **Power** option. It is used for testing only.

Step 21. Verify that for each lamp the column on the right is set to **ON** (enabling the lamps).

Step 22. If a lamp fails, the Volt column displays the failure code as received from the ballast PS (see following paragraph).

Step 23. To save your changes and return to the Settings screen, press ☑.

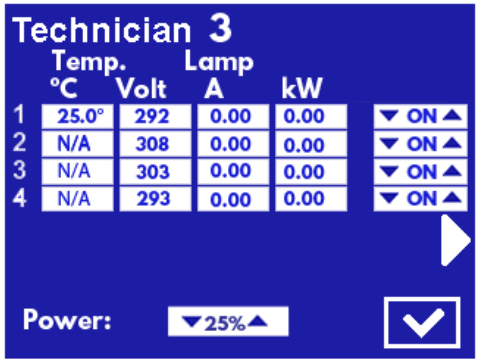


Figure 6-12: Technician 3 Screen**

*For a system with less than four lamps, only the relevant rows are displayed.

**For Temperature, for less than four lamps, only the fields for the installed lamps display the temperature. The other fields display N/A.

6.6.1 Lamp Failure codes in Technician screen 3

The table below provides the various “lamp off” or failure to ignite as identified by the ballast/s. Turn off reason can be identified in two ways:

1. By a pop-up message
2. By voltage value that acts as a **Failure code** and remains displayed in the voltage column in **Technician screen #3** after the lamp has gone off

Table 6-2: Table of Failure codes displayed in Technician screen #3

#Failure code	Reason for lamp turn off (appears in the pop-up message)
0 (+24)	None
50 (±24)	Lamp nominal voltage too low*
100 (±24)	Lamp nominal voltage too High**
150 (±24)	Device cooling insufficient
200 (±24)	Internal failure / One phase lost
250 (±24)	Lamp earth short circuit
300 (±24)	Lamp not ignited / gone out
350 (±24)	Lamp short circuit
400 (±24)	Phase failure mains voltage
450 (±24)	Mains over-voltage
500 (±24)	Manual turn off

7 Initializing the Atlantium System

The Atlantium system starts up with the initialization of the **Control Module.**

➲ To start up the Atlantium system:

Step 1. Check that the Ballast Module power cable is connected to the main power source.

The system powers up and the **Control Module's** touch-screen monitor is activated, displaying the **Main Operation** screen. The **Lamp Power** button on the bottom Left is on the **Off** position.

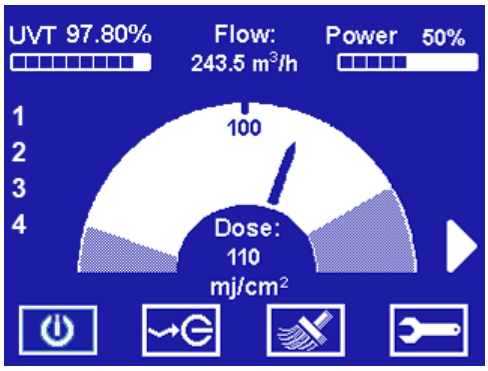


Figure 7-1: Initial Main Operation Screen

Step 2. Press the Lamp Power button to ignite the lamps.

8 First Time System Activation

Following **Control Module** configuration, completing the system commissioning includes the following major steps:

- Cleaning the Atlantium Unit (See below)
- Filling unit with water (See below)
- Igniting the UV Lamps (See below)
- Initiating the Water Flow (See page **102**)
- Adjusting the Auxiliary Equipment (See page **102**)
- System Tuning (See page **102**)
- Installation QA (See page **102**)

8.1 Cleaning the Atlantium Unit

The inner surfaces of the Atlantium Unit are to be cleaned via the CIP process at installation, before starting up and proceeding with the commissioning phase. Follow the directions in ***Cleaning In Place (CIP)*** on page **106**.

8.2 Filling unit with water

The unit must be filled with water before full activation.

Water hammer can cause damage to any hydraulic system, including the Atlantium Unit. **Make sure that water hammer is NOT present in your facility's water line.**

To prevent water hammer, consult your system engineer and check your facility system procedures, including:

- When starting the water flow when the lines are empty, keep valves open so that the air in the pipes has a simple release pathway.
- Sudden valve closures can cause water hammer. Make sure that valves close gradually enough to avoid this problem.
- Ensure that all pumps used with the Atlantium system employ soft start-up procedures.

➲ To introduce water:

- Open the **Inlet valve** and, using the **Viewport** on the Unit, check that the quartz chamber fills with water completely.

8.3 Igniting the Lamps

The lamps of the Atlantium Unit are ignited to commence live system operation.

- **Temporarily, the caps on the lamps must be left open to allow any condensation water to evaporate. During this time, set up Caution floor signs at a distance of 1m/40inch from the Atlantium Unit to warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Where appropriate eye protection, such as containing polycarbonate lens that meet EN166, CAN/CSA-Z94.3-02 and/or Z87.1 standards.
- **DO NOT look directly into openings that emit UV light.**

➲ To ignite the lamps:

Step 1. To switch on the lamps, log in to the **Control Module** and press ▶until the Lamps Power screen appears (**Figure 8-1**).

Step 2. Temporarily place Caution floor signs at a distance of 1m/40inch from the Atlantium Unit to warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.

Step 3. Open each of the lamp's caps (1 on either end of each lamp). For the location of the lamps' caps, see Figure 10-9 , on page 118.

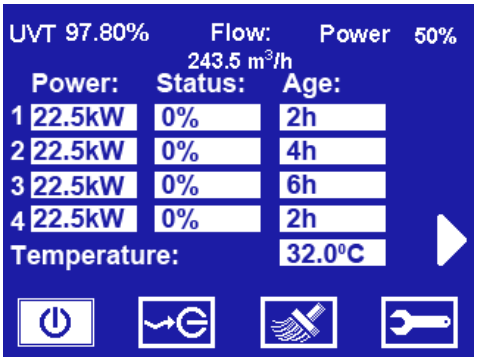


Figure 8-1: Lamp power screen

Step 4. Press the power button ⏻ and verify that power readings are as required.

Step 5. Press ▶ The System Settings screen appears.

Step 6. Configure system settings as required for your particular system. See ***Configuring the Control Module*** on page **6-88.**

Step 7. Let the system operate for at least one hour.

Step 8. Continue with Section **8.4** and Section **8.5** below while the lamps complete the first hour of operation.

Step 9. Close all of the lamps' caps and remove the Caution floor signs.

8.4 Initiating the Water Flow

When initiating the water flow, the water may be set to drain out or collect in a tank until the system is fully operational. Once the entire installation and QA testing is complete, the water is set to flow normally as part of the facility operation.

➲ To initiate the water flow:

Step 1. Open the relevant valve that allows water to flow through the Atlantium unit. This may be the draining valve or bypass valve.

Step 2. Open the viewport and verify that the Unit is completely filled with water and that no air gaps and no air bubbles exist.

Step 3. From the **Main Operations** screen, press . The **Password** screen appears.

Step 4. Using the numbered keys, enter the password (default password is **1357**) and press **OK**. The Settings screen appears.

Step 5. For **Flow**, press ▶. The Flow Settings screen appears.

Step 6. Verify that the Flow reading on the right is within normal range. If it is not, check the connections between the Atlantium system and the Flow meter. See ***Cabling Components*** on page **5-74.**

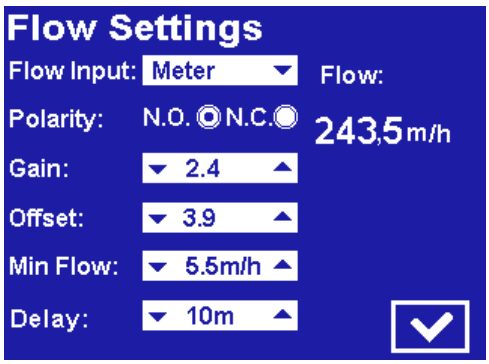


Figure 8-2: Flow Settings Screen

8.5 Adjusting the Auxiliary Equipment

All relevant auxiliary equipment , i.e., flow meter, automatic valves, PLC, etc., must be adjusted at this time. Refer to the manufacturers' user documentation of the relevant equipment.

If your configuration includes a flow meter, check that the value of the flow displayed on the **Control Module** matches the reading on the flow meter. See page **102**

8.6 System Tuning

The Atlantium system must be tuned via the **Control Module**. Follow the instructions in ***Replacing/Checking a UVT Analyzer or UVIS Sensor*** on page **124**.

8.7 Installation QA

Atlantium personnel performs a quality assurance procedure following which the Atlantium system is ready for regular facility operation.

9 Operating the Atlantium System

This section details how to operate the system and handle basic and periodic routines to be carried out by Atlantium- trained the local staff.

Cleaning, disregard the irrelevant sections.

WARNING!

- Unauthorized servicing or modification of this system in a manner not specified in this manual could expose personnel to potential electrical or other hazards.
- Improper use or adjustment of this system may invalidate the service warranty agreement.

9.1 Basic Operational Tasks

- Starting up the Atlantium system - See page **103**
- Viewing system status - See page **105**
- Taking microbial samples - See page **108**
- Cleaning System (if included) - See page **108**

9.1.1 Shutting Down the Atlantium System

In a non-emergency situation, you can shut down the Atlantium system from the **Control Module.** Be sure to enable the **Auto Power On** function in the **Control Module** if you want the system to save your settings. (See ***Igniting the Lamps*** on page **101.**)

Step 1. To shut down the Atlantium system from the Control Module: On the Main screen of the **Controller**, press . The password screen appears.

Step 2. Using the number keys, enter the password **2468**. and press **OK.** The Atlantium system is turned off.

9.1.2 Starting up the Atlantium System

You can start up the Unit via the **Control Module** and then check the water and system parameters.See page **99**

➲ To start up the Atlantium system:

Step 1. Check that the power cable of the Ballast Modules is connected to the main power source.

Step 2. Connect the power cable of the **Control Module** to the main power source.

Step 3. On the Main screen of the **Control Module**, tap ⏻

Step 4. Wait 10 minutes for lamps to turn on and the system to stabilize.

Step 5. Open the viewport and check that there are no air bubbles in the water flow. If necessary, release all air bubbles from within the system.

Step 6. Check the main screen of the **Control Module** and verify that the Atlantium system parameters for **Power**, **Flow**, and **UVT** are within wanted acceptable range (see the table below to enter your expected values).

Notify the executive maintenance engineer in case the values are out of range.

Table 9-1: Abbreviation of Table in **Appendix B**. ***Checking the System Parameters***

	Parameter	Expected Value	
		Minimum	Maximum
1	Power		
2	Flow Rate		
3	UVT		
4	UV Dose		
5	Lamp Status		

For periodic checks on Atlantium system parameters, a convenient check list is supplied in **Appendix B**

9.2 Viewing System Status

You can view the real-time status of various System parameters.

➲ To view the system's status:

Step 1. On right side of the **Controller** screen, press ▶. The System Status screen appears displaying the real-time status of the following parameters:

- **UVT %**
- **Flow**
- **Power %**
- **Per Lamp: Power**, **Status**, and **Age**
- **Temperature**

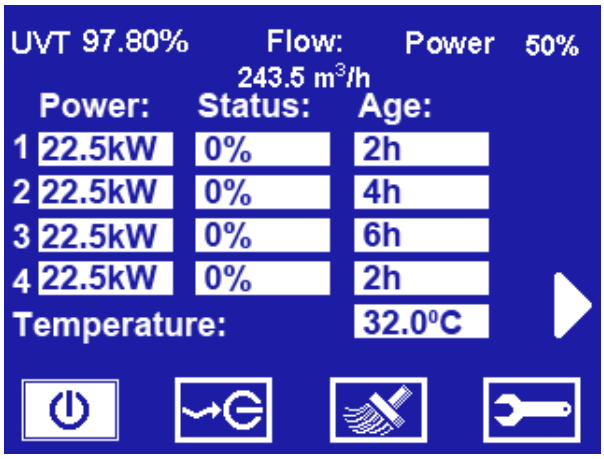


Figure 9-1: Controller Status Screen

Every defined event is written to the event log of **TRACS** (if included). You can view the data log of all events in the **Events** tab of **TRACS**.

.

9.3 Taking Microbial Samples

The designated aseptic sampling valves (i.e., EGMO ESV valves) are the **Inlet** and **Outlet Sampling Points** located on the sides of the Atlantium Unit. Before taking a microbial sample, a valve is sterilized internally with Ethanol 70%, as well as with flame.

Required Equipment

- Sampling containers that are clean, laboratory-grade, sampling bottles with a volume no less than 100ml (3.38oz).
- Ethanol 100-70%
- Bunsen Burner and lighter
- Permanent marker
- Sterile gloves
- Proper Sampler clothing, according to local laboratory protocol

CAUTION!

- Use gloves during sampling and treat them with Ethanol 70%.
- Use **sterile disposable** sampling containers.
- Prevent contact between the sampling containers and the valves' surfaces.
- Be sure that no foreign matter or liquids fall into the sampling containers.
- Perform the sample collection as fast as possible to minimize environmental contamination of the sample.
- **Use open flame to treat the surrounding air while sampling.**

➲ To take a microbial sampling:

Step 1. Open the Sampling valve and allow water to flow for 2-3 minutes. Then close the valve.

Step 2. If the sampling valve contains a cover at the top, open its cover and place it in a clean place.

Step 3. Spray all over the Sampling valve and inside it thoroughly with Ethanol 70%.

Step 4. Using a Bunsen burner, apply the flame comprehensively over the entire Sampling valve surface from its connection to the main pipe to its spout.

Step 5. Spray the cap thoroughly with Ethanol 70% and place it on the Sampling valve.

Step 6. Open the valve and allow the water to flow for 2 minutes. If you have applied the flame properly, vapor emerges from the spout. (If there is no vapor, application of the flame was insufficient. Repeat the previous step.)

Figure 9-2: Cleaning the Sampling Valve with Ethanol

ATTENTION!

Make sure never to touch the opening at the Sampling container's neck.

While the Sampling container is open, do not turn the cap upside down.

Step 7. On the Sampling container, with one hand, remove its cap carefully, avoiding touching the inside of the cap with fingers or any other object. **Do not turn the cap upside down.**

Step 8. With your other hand position the Sampling valve under the water stream and take a water sample. Make sure never to touch the valve's opening or the opening at the container's neck. Fill the sampling container, leaving ample air space in bottle (at least 2.5 cm).

Step 9. Carefully replace the container's cap and tighten it to avoid cross contamination.

Figure 9-3: Taking a Sample

Step 10. Label the Sampling container with the following data:

- Sampling site
- Sampling point
- Sampler name
- Date the sample is taken

Step 11. Close the Sampling valve.

Step 12. Spray the sampling container thoroughly with Ethanol 70%.

Step 13. Store the water samples in a dark, refrigeration unit at a temperature of 2-8°C (35.6-46.4°F) until they are examined.

9.3.1 Microbial Analysis Guidelines

The microbial analysis is to be conducted according to the company guidelines which are used in the local microbiology laboratory.

- Atlantium's recommendation is to use the microbial filtration method for the testing.
- If a number of consecutive anomalous microbiological results are obtained in one of the sampling points, take the samples only after performing CIP with a disinfection chemical (peracetic acid is recommended). It is important to let the chemicals clean and sanitize the aseptic valves, as well by letting the cleaning solution flow through them. See ***Cleaning In Place (CIP)*** on page **106.**
- In the event that system outlet counts are higher than the inlet counts, the results are to be considered as a sampling error.
- In the event that the microbial counts are not proportional in different dilutions, the results are to be considered as a laboratory error.
- For further guidance regarding microbial sampling and analysis, contact the Atlantium Application department.

9.4 Ultrasonic Cleaning (Optional)

For systems with the optional Ultrasonic Cleaning System, for whatever reason, you may want to manually disable the Ultrasonic Cleaner. Using the manual control temporarily overrides the configured settings. Manually reenabling resumes the ongoing automatic Ultrasonic Cleaning cycles as pre-configured in the settings.

➲ To run the Ultrasonic Cleaner from the Control Module:

- On the **Control Module** main screen, press . The Ultrasonic Cleaner is activated/ deactivated. Activation is for the duration set for this function.

10 System Maintenance

To keep the Atlantium system in peak form, preventive and periodic routine maintenance is essential. During commissioning, Atlantium trains your local staff how to do the tasks on a prescribed schedule or as needed.

WARNING!

- Unauthorized servicing or modification of this system in a manner not specified in this manual could expose personnel to potential electrical or other hazards.
- Improper use or adjustment of this system may invalidate the service warranty agreement.

Before you begin a procedure, carefully read through it so that you can anticipate the steps efficiently.

The periodic routine maintenance schedule includes:

- Weekly Maintenance Tasks:
  - Walk-Around Inspection Tour - see below
- Monthly Maintenance Tasks:
  - Checking the Ballast Module Cooling Fans (if relevant) (See page 121)
- Yearly Maintenance Tasks:
  - Cleaning the Quartz Sleeve (or Replacing as needed) and Replacing the Quartz Sleeve O-rings (See page 133)
  - Anti-Corrosion Maintenance (See page 145)
- Maintenance Tasks Performed as Needed:
  - Cleaning In Place (CIP) (See page 111)
  - Resetting the Cleaner Counter (See page 121)
  - Replacing a Lamp (See page 123)
  - Checking/Replacing a Sensor (See page 137)
  - Tuning the Atlantium System (See page 143)
  - Checking/Replacing the Quartz Sleeve (See page 133)

- Cleaning/Replacing a UVIS Wave Guide (See page 131)
- Temperature Sensor Maintenance
  - Replacing the Temperature Sensor (See page 147)

10.1 Walk-Around Inspection Tour

Walk-Around and inspect the Atlantium system once a month to check for water leaks, visible damage to the cables and harnesses, or any other part of the system. Report any problem you find to an authorized technician.

➲ To perform a Walk-Around inspections:

Step 1. Walk around the Atlantium unit and inspect it. Look for leakage at the incoming and outgoing pipes.

Step 2. Inspect the **Cable Harness** and cables running between the Unit and the Ballast Module(s) and **Control Module** for signs of fraying and any other visible damage.

Step 3. Inspect the Ballast Module(s) and **Control Module** for any signs of damage.

Step 4. Report any leakage or damage to an authorized technician immediately.

10.2 Cleaning In Place (CIP)

Performed as needed

The CIP procedure consists of two parts:

- Cleaning to remove the accumulated deposit (particles) that collect on the quartz sleeve over time
- Disinfectant cleaning to sterilize the quartz sleeve (not applicable to some Municipal installations)

Clean the Atlantium Unit's inner surfaces and quartz sleeve generally on a periodic basis that is determined by observing that the performance is on target and the UVT Sensor readings are stable. You can clean using this process less frequently, depending on the specific chemistry of your water.

A CIP recirculation system is available from Atlantium as an optional CIP Kit. Connect the recirculation system to the CIP ports to create a closed-loop flow line that pumps a diluted solution through the Unit.

- In facilities where CIP is performed on the plant level, circulating though all piping, the CIP process is the same for the Atlantium System. The inlet and outlet valves referred to in these sections are irrelevant.
- If the Atlantium Unit(s) is/are integrated into the facility's CIP loop, ensure that the water flow rate through the Unit is at least 60 m3 per hour.
- In facilities with plant-wide CIP, perform CIP with the Atlantium System running.
- In facilities that utilized sea water, sea water may be used in the CIP process provided that the chemical manufacture of the cleaning/disinfectant solution approves.

The estimated time for a CIP process is 1.5 hours.

Prepare the following:

- Protective chemical-safe goggles, clothing and gloves
- An external pump-driven recirculation system with a built-in 100 liter reservoir (26.5 US gallons) or Atlantium's CIP Kit including its accessories (See ***Sampling Valve Components and Kits*** on page **3-35.**)
- Scalant cleaning solution
- Disinfectant cleaning solution (if disinfection is planned)

10.2.1 Selecting the Correct Chemicals for CIP

It is **critically important** to select the correct cleaning and disinfecting chemical for the CIP process. It must be powerful enough to clean the inner surfaces of the Atlantium unit, while at the same time:

- Does not harm the components of the Unit that it comes into contact with (stainless steel 316, quartz and Viton® o-rings).
- Does not contaminate the product for which the water being treated by the system is used.

WARNING!

- The chemicals selected and the concentrations used must be approved for use by the facility's quality assurance and safety officials, and must comply with the written protocols and procedures set in place by those departments.
- Use and handle in strict accordance with their manufacturer's instructions, product information sheets, and material safety data sheets (MSDS).
- Use of NaOH (Caustic Soda) may cause irreversible damage to the quartz. Use of this material requires an analysis of temperatures, concentration, frequency and duration and written approval from Atlantium's application engineer.

- All chemicals must be obtained locally.
- Atlantium does not supply chemicals - get them locally from commercial chemical suppliers.

Recommended Scalant Cleaning Solutions

The following chemicals have been tested and found to be very effective for cleaning and are recommended for use when cleaning the Atlantium system:

- Phosphoric acid (H_3PO_4) 2% diluted solution
- Nitric acid (HNO_3) 2% diluted solution
- HCl 2% diluted solution - recommended for metal based (rust-colored) deposits
- Sulfamic acid (H_3NSO_3) 1.5% diluted solution - recommended for calcium based deposits

Recommended Disinfectant Solutions (not applicable to some Municipal installations)

(The Disinfectant CIP process is not applicable to some Municipal installations)

The following have been tested and found to be very effective for disinfection and are recommended for use when disinfecting the Atlantium system:

- Peracetic acid-based chemicals - 30 min. contact time, for example:
  - P-3 Oxonia Active 150 - 0.5% diluted solution according to manufacturer's recommendation

- Divosan Forte - 0.5% diluted solution according to manufacturer's recommendation

- Hot water at 85°C (185°F) for 30 min.

WARNING!

- **It is the responsibility of the Atlantium system's owner/operator to use and apply the correct chemicals.**
- **The use of any other cleaning and disinfection chemicals requires Atlantium's approval.**

10.2.2 Preparing the Chemical Solution

For both cleaning and disinfecting, the liquid in the reservoir must contain the desired percentage of the chemical additive (see above). Therefore, the purchased chemical solution must be diluted and added to the reservoir. Calculate the correct amount of chemical solution to be used for the CIP process.

Protect yourself from the ill-effects of chemicals with protective chemical-safe goggles, clothing and gloves.

- The concentrations refer to room temperature.
- The diluted solution values denote the required dilution of the chemical circulating through the Atlantium Unit.
- While calculating the diluted volumes, consider all volumes including the Unit, pipes and the recirculation reservoir.

➲ To calculate the correct amount of chemical solution:

Step 1. Calculate the total volume you need for the CIP* by adding the water volumes in:

- The Atlantium Unit volume (measurements in liters/gallons) - Refer to the ***Physical Specifications*** Table in ***Atlantium System Footprint*** on page **15**
- The pipes between the Atlantium Unit and the **Inlet** and **Outlet Valves**
- The CIP recirculation system
- The hoses between the Unit and the recirculation system

Step 2. Calculate the amount of chemical additive to be added to the reservoir to achieve the desired dilution. Use this formula:

$$C = \frac{A * V_t}{B}$$

A - The target concentration for the liquid (chemical substance + water) in %
B - The concentration of the chemical substance to be added in %
C - Volume of the chemical substance to be added in liters/gallons
V_t - Volume of the water in the Unit + piping + CIP Reservoir (as described above*) in liters/gallons

OR:

Assume that you need a 2% diluted solution in the CIP process and assume that your purchased solution is of a 35% concentration in a 4-liter container.

Calculate according to this calculation: [200 liter X 2%] / [35% X 4 liter] = [400] / [140] = 2.9 = 3 containers.

CAUTION!

* If you intend to collect the solution into the CIP reservoir or other container, the total volume of the cleaning
solution must not exceed the volume of the CIP reservoir or container.

Be sure to obtain from an engineer the correct calculation for the amount of chemicals to be used for the process you intend to use.

10.2.3 Performing CIP

CAUTION!

The anti-scaling procedure is be performed only if necessary. Atlantium recommends that anti-scaling always be followed by disinfection.

The CIP process is performed utilizing fresh water. For systems with sea water, drain the system and rinse before preforming CIP.

- In facilities where CIP is performed on the plant level, circulating though all piping, the CIP process is the same for the Atlantium System. The inlet and outlet valves referred to in these sections are irrelevant.
- If the Atlantium Unit(s) is/are integrated into the facility's CIP loop, ensure that the water flow rate through the Unit is at least 60 m^3 per hour.
- In facilities with plant-wide CIP, perform CIP with the Atlantium System running.

This section details the CIP process with recommended steps for carrying it out. The procedure is to be adapted according to the specific circumstances of your installation.

The CIP process is to follow these major steps:

(Refer to the next section for the detailed description on the CIP process.)

Step 1. Close the inlet and outlet valves and immediately shut down the Atlantium system

Step 2. For systems with sea water, if needed, drain the system

Step 3. Remove the caps covering the CIP ports

Step 4. Connect the supply hose to the bottom CIP port and connect the return hose to the top CIP port

Step 5. Fill the CIP reservoir with fresh water or sea water

Step 6. Starting the circulation

Step 7. Adding the cleaning solution (See ***Selecting the Correct Chemicals for CIP*** on page **10-111.**)

Step 8. Running the CIP circulation for at least 30 minutes or as specified according to the chemical manufacturer

Step 9. Rinsing the system and the CIP cart

Step 10. Performing CIP Disinfection with the disinfection solution

Step 11. Ending CIP

The CIP diagram below is followed by the full recommend CIP procedure.

CIP Setup

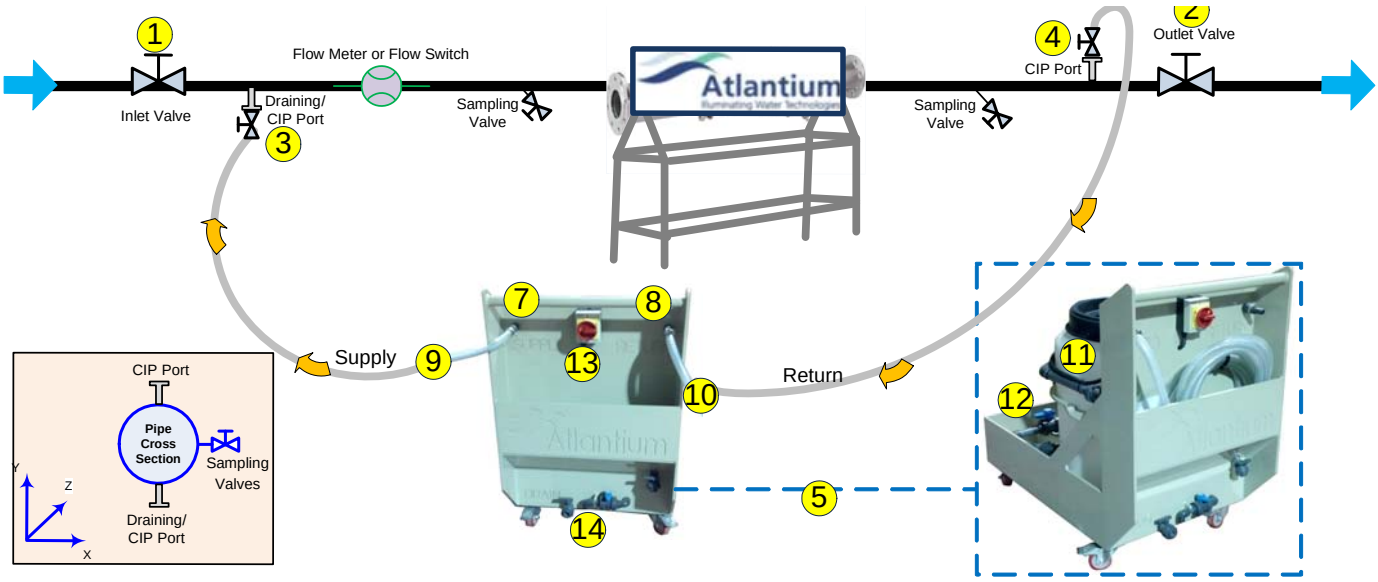


Figure 10-1: Recommended CIP Setup

If the installation is vertical, the water flow is from bottom to top. The Draining/CIP port **(3)** is on the bottom, under the Atlantium Unit and the CIP port **(4)** is above the Atlantium Unit.

➲ To Perform CIP:

ATTENTION!

If you are only performing disinfection: Follow Steps 1 through 20. Then resume with Step 22.

If you are performing both procedures: Follow all of the steps below.

Shutting Down the System

Step 1. Close the Unit's inlet isolating valve **(1)** and outlet isolating valve **(2)**. Immediately proceed to the next step.

Step 2. Shut down the Atlantium system, on the **Control Module**, by pressing the **Main Unit Operation Button Operation** button in the upper-right corner.

Step 3. On each CIP port, remove the cap's clamp and remove the caps covering the CIP ports **(3 & 4)**. Place the caps in a safe place for reuse later. Verify that the gasket is still inserted into the ferrule.

ATTENTION!

Make sure that the hose fittings are secure and tight.

Water must fill the entire path through the piping on both sides of the Atlantium Unit and the Unit itself, as well as the Supply and Return hoses and the CIP Kit's reservoir.

Connecting the CIP Hoses

Step 4. Remove the hoses from their pocket on the Cart.

Step 5. Using the CIP cap's clamp, connect the CIP Kit's Supply hose to the bottom CIP port **(3)**. Verify that the gasket is present.

Step 6. Using the CIP cap's clamp, connect the CIP Kit's Return hose **(10)** to the top CIP port **(4)**. Verify that the gasket is present.

Filling the CIP Reservoir

Step 7. To fill the CIP reservoir **(A)** with water:

- **Filling option 1:**
  - Slightly open the Unit's inlet isolating valve **(1)** and open the top CIP port on the outlet side **(4)** and allow the facility's water to fill the CIP Kit's reservoir **(11)**.
  - There is a full line indicator on the reservoir **(11)**. Make sure the water in the reservoir is up to the 100 liter (26.5 US gallons) level.
    - Close the Unit's inlet isolating valve **(1)**.
    - Close the CIP port **(4)** valve.
- **Filling option 2:**
  - Open the top cover of the CIP Kit's reservoir **(11).**
  - Using a hose of an external water source, fill the reservoir with 100 liter reservoir (26.5 US gallons of room-temperature tap water (±25°C/75°F).
  - There is a full line indicator of the reservoir. Make sure the water in the CIP Kit's reservoir **(11)** is at the 100 liter reservoir (26.5 US gallons level.
  - Close the cover.

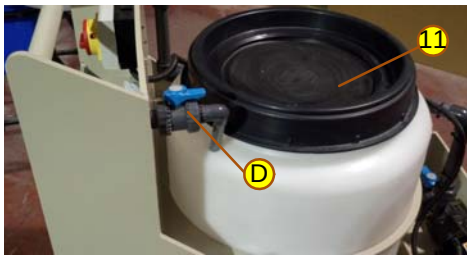


Figure 10-2: CIP Kit's Reservoir

Figure 10-3: CIP Cart's Inlet and Drain Port

WARNING!

Observe chemical safety rules! Follow all safety recommendations in the chemical manufacturer's product information and MSDS publications!

Adding the Chemical Solution

Protect yourself from the ill-effects of chemicals with protective chemical-safe goggles, clothing and gloves.

Step 8. Open the top cover of the CIP Kit's reservoir **(11)** and add the prepared chemical solution to the water in the reservoir. (See ***Selecting the Correct Chemicals for CIP*** on page **10-111.**)

Starting the CIP Circulation Process

ATTENTION!

The Atlantium Unit must be absolutely full for the solution to reach all of the inner surfaces.

Step 9. If the bottom and top CIP ports **(3 & 4)** are valves, open them.

Step 10. To start the CIP recirculation pump, on the CIP cart, turn the switch **(13)** to the **On** position and allow the solution to circulate through the Unit for **the period of time specified in the chemical's Use instructions, but not less than 30 minutes**.

Step 11. Verify that the Atlantium Unit is full of water and that the water is circulating correctly.

Step 12. During the circulation period open and close the sampling valve several times to assure chemical contact with the CIP chemicals.

If the draining valve is separate from the CIP ports, open and close the draining valve.

Rinsing the System and CIP Kit

After the chemical solution has circulated through the Unit for a sufficient duration, the chemicals solution must be rinsed from the system.

Step 13. Connect the CIP recirculation system's drain valve **(B)** to the facility's sewage system. See ***Alternatives for Draining the Water with Cleaning Solution*** on page **120**.

Step 14. Turn on the Atlantium system, on the **Control Module**, by pressing the **Main ON/OFF Operation** button in the upper-right corner.

ATTENTION!

Before performing the next step, verify that the lamp(s) are on and the system has stabilized and indicates it is ready. If not, restart the lamp(s) again and wait until the system has stabilized and indicates it is ready.

Step 15. Close the inlet CIP port **(3)**, stop the recirculation pump **(12)** and open the CIP recirculation system's drain valve **(C)**. Immediately slightly open the inlet valve **(1)** to rinse the Atlantium system thoroughly.

Allow this rinsing process to operate until at least three cycles of water volume have gone through the Atlantium Unit to remove all of the scalant cleaning/disinfectant solution.

Step 16. To assure that all the chemicals are flushed out, use a measurable indicators, such as measuring the pH, or other indicators depending on the chemical being used (consult your chemical supplier).

Step 17. Slightly open the outlet valve **(2)** and close the CIP port on the outlet side (4). To avoid hammer effect if there is air trapped in the system, carefully open the inlet valve **(1)** completely. Resume normal operation.

Rinsing the CIP Reservoir

After the chemical solution has circulated through the Unit for a sufficient duration as specified by the chemical supplier (typically 30 minutes), the chemicals solution must be rinsed from the system by the steps below.

Step 18. Open the CIP cart's drain valve **(C)** and completely drain the CIP reservoir **(11)**.

Step 19. Using an external water source, rinse the CIP cart's reservoir, letting the rinse water drain out of the CIP cart's drain valve **(C)**.

Step 20. Close the CIP cart's drain valve **(C)**.

Performing CIP Disinfection

Step 21. If you are planning to perform an additional CIP process using the disinfectant solution, refer to ***Preparing the Chemical Solution*** on page 113 and follow the CIP process again from ***Filling the CIP Reservoir*** in **Step 7.** above.

Or,

To return to normal operating status, proceed to the next step.

Ending CIP

Step 22. To drain out any water inadvertently collected in the CIP cart's storage pocket or base, use the CIP cart's drain valves **(A** and **B)**.

Step 23. Roll up the CIP Kit's hoses and return them to their storage pocket **(F)**.

Refilling the System

ATTENTION!

Make sure that the Atlantium Unit is filled with water before powering it up and turning on the lamps.

Step 24. Turn on the Atlantium system at 100% power and wait until the system has stabilized and indicates that it is ready.

Returning Power to the Lamps

Step 25. On the Connection Box next to each lamp, turn the switch to the **On** position.

Step 26. Turn on the Atlantium system, on the **Control Module**, by pressing the **Main ON/OFF Operation** button in the upper-right corner.

Alternatives for Draining the Water with Cleaning Solution

- **If local regulations permit, and a physical drain setup exists**, drain the cleaning solution to the facility's sewage system:

  a Connect the CIP recirculation system's drain valve **(B)** to the facility's sewage system.

  b With the Atlantium System operating, sightly open the Inlet valve.

  c Close the inlet bottom CIP port **(3).**

  d To stop the CIP recirculation, on the CIP cart, turn the switch **(13)** to the **Off** position.

  e Open the CIP cart's drain valve **(C)**. The water with the chemical solution begins to drain out.

  f Connect the CIP cart's drain valve **(C)** to the facility's sewage system.

Or,

- **According to local regulations, use the Draining/CIP port to drain the solution in the system into a external container.**

Or,

- **Use the CIP reservoir (11) to collect the chemical solution**.

  a Connect the CIP recirculation system's drain valve **(B)** to the facility's sewage system.

  b With the Atlantium System operating, close the inlet bottom CIP port **(3).**

  c To stop the CIP recirculation pump, on the CIP cart, turn the switch **(13)** to the **Off** position.

  d Allow the chemical solution to drain entirely into the CIP reservoir **(11)**.

  e Dispose of the contents according to regulations. Connect a drain pipe to the CIP cart's drain pipe and open the CIP cart's drain valve **(C)**. The water with the chemical solution begins to drain out.

  f Disconnect and close the CIP cart's drain valve **(B)**.

10.3 Resetting the Cleaner Counter

Performed as needed

You can reset the counter that keeps track of the number of time the cleaner has performed a cleaning.

➲ To resent the counter:

Step 1. From the **Main Operations** screen, tap . The **Password** screen appears.

Step 2. Using the numbered keys, enter the password (default password is **1357**) and tap **OK**. The Settings screen appears.

Step 3. For **Cleaner**, tap ▶. The Cleaner Setting screen appears.

Step 4. For Lamp, using the ▲▼ arrows, select the number of the lamp for which you want to reset the counter.

Step 5. Tap **Rst**. A number pad appears displaying **0**. Tap . The counter is reset to **0**.

10.4 Checking the Cooling Fans

Perform Monthly

For Ballast Module B, once a month the cooling fans of the Ballast Modules must be checked to verify that the air is flowing out.

➲ To check the cooling fans:

- Check that air is being blown out of the fan on top of the Ballast Module.

10.5 Checking the Temperature Sensors' Readings

Perform Monthly

Once a month both temperature sensors must be checked to verify that they are working properly.

Step 1. **To check the temperature sensors' readings:** On the **Control Module**, to access the Technician mode, tap . The Password dialog box appears. Enter the password (default password is **8888**) and tap **OK**. The Technician Settings 1 screen appears.

Step 2. Tap twice on the ▶at the right of the screen to access Technician 3 screen (Figure 6-12 , on page 97). The temperature for each installed and operating lamp is displayed. Verify that the temperatures displayed are within the norm. An **N/A** appears for a lamp that is not operating or not installed.

DO NOT leave the Control Module unattended while in **Technician mode** even for a short time. Unauthorized changes to the **Technician mode** settings could damage the operation of the Atlantium system.

10.5.1 Lamp Failure codes in Technician screen 3

If one of the UV lamps fails, a pop-up message is displayed and the Volt column in Technician screen 3 (see ***Configuring the Technician Settings*** on page 95) displays the failure code as received from the ballast PS.

Step 1. To access the **Technician 3** screen, On the Control Module, to access the Technician mode, tap . The Password dialog box appears.

Step 2. Enter the password (default is 8888) and tap **OK**. The Technician Settings 1 screen appears.

Step 3. Tap twice on the ▶at the right of the screen to access Technician 3 screen.

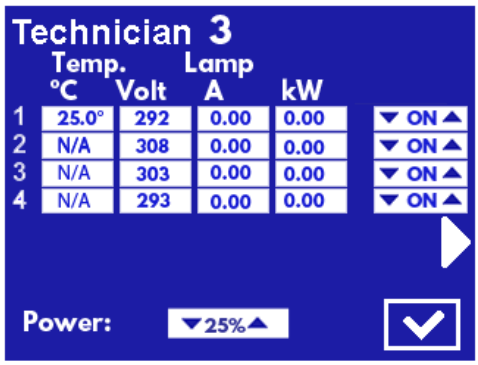


Figure 10-4: Technician 3 Screen

The table below provides the various "lamp off" or failure to ignite as identified by the ballast/s. Turn off reason can be identified in two ways:

1. By a pop-up message
2. By voltage value that acts as a **Failure code** and remains displayed in the voltage column in **Technician screen #3** after the lamp has gone off and the corresponding pop-up message has been acknowledged.

Table 10-1: Table of Failure codes displayed in Technician screen #3

#Failure code	Reason for lamp turn off (appears in the pop-up message)
0 (+24)	None
50 (±24)	Lamp nominal voltage too low*
100 (±24)	Lamp nominal voltage too High**
150 (±24)	Device cooling insufficient
200 (±24)	Internal failure / One phase lost

#Failure code	Reason for lamp turn off (appears in the pop-up message)
250 (±24)	Lamp earth short circuit
300 (±24)	Lamp not ignited / gone out
350 (±24)	Lamp short circuit
400 (±24)	Phase failure mains voltage
450 (±24)	Mains over-voltage
500 (±24)	Manual turn off

10.1 General Lamps, Sleeves Maintenance Tips

The following are some tips that are meant to help you understand maintenance requirements:

- Lamps and quartz sleeves traverse the width of the Unit and are held in place by holders and connectors at both sides. For maintenance, you can access the lamps and quartz sleeves from either side. Depending on the constraints of the installation (sometimes one side is less accessible) you can usually decide from which side to access the them.
- To access the quartz sleeve, you must remove the lamp housed within it.
- On rare occasions, quartz sleeves and their o-rings may need to be replaced because of a maintenance issue that arises.

10.2 Replacing a UV Lamp

The UV lamp can be used as long as it provides the required dose and efficient service. (Generally, a lamp service status of 80% or more is considered efficient service.) A lamp should be replaced ONLY when necessary.

Each lamp's performance is measured directly by a dedicated UV Intensity Sensor. Replace a lamp when:

- Its status or performance declines and the lamp no longer provides the required performance at the appropriate electrical cost
- Once its rated hours expire, be alert to any changes in efficiency that could indicate the need to change the lamp

In the **Control Module**, you can set a margin to assure notification as status declines and a warning is sent to the operator at both the user setting level and the internally coded minimum.

For systems with multiple lamps, the system may continue to operate while any one of the lamps is being replaced, however, precautions must be carried out to protect you from the UV.

Before commencing this procedure:

CAUTION!

- Allow the old lamp at least 10 minutes to cool down before starting.
- Read carefully the instructions provided in the new Lamp package.
- Wear white cotton gloves.
- Before installing a new lamp, verify that it is clean and free of grease.
  If not, rub it gently with the provided lens tissue. Never touch the lamp with your bare hands - hold it only by the end-connectors. Oily residues from fingerprints can damage the lamp.

WARNING!

- The UV lamps are designed with high internal positive pressure. Wear protective eye-wear while replacing a lamp. Pay strict attention to the safety warnings and precautions in the ***Safety Overview*** on page **40** in the front of this manual.
- In case of lamp breakage, refer to the Safety instructions in ***Safety Overview*** on page **40**.

When replacing a lamp while the rest of the system is still in operation, you must follow these safety requirements:

- Shutdown the electricity supplied to the lamp to be replaced.
- **During the time when the lamp's caps are removed, use Caution floor signs at a distance of 1m/ 40inch from the Atlantium Unit that warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Wear appropriate eye protection.
- **Do not look directly into the lamp enclosure or into any openings that emit UV light during the entire lamp replacement procedure.**

CAUTION!

While carrying the steps of this procedure, replace all of the O-rings on the components, such as the lamp housings and holders.

10.2.1 Lamp Replacement Options

- Replacing a Lamp after System Shutdown (See page **125**)
- Replacing a Lamp while System is Operating ()

Generally, a lamp is replaced after shutting down the entire Atlantium system. However, in a facility where there is no redundant UV purification, and you require minimizing system downtime, you can replace a lamp while other lamps continue operation provided you follow these safety requirements:

Replacing a Lamp after System Shutdown

Prepare the following:

- Replacement lamp
- Lamp cleaning cloth (comes with the lamp)
- White cotton gloves
- Eye protection
- #3 Allen wrench
- #4 Allen wrench
- Caution signs

10.2.2 Shutting Down the Entire System

Generally, a lamp is replaced after shutting down the entire Atlantium system.

➲ To shut down the entire system:

Step 1. On the Main screen of the **Control Module**, tap ⏻.

Step 2. Enter the password (the default password is **2468**). The Atlantium system is turned off.

Step 1. Using the viewport on the Atlantium Unit, check that the lamps are indeed turned off.

Step 2. Check the main screen of the **Control Module** and verify that the system parameter for **Power** is **0%**.

A severe danger of electrocution exists if the lamp is not turned off during replacement. Be absolutely sure you have turned off the correct lamp's switch before you proceed to replace the lamp.

Step 3. For the lamp that is to be replaced, on the connection box, turn the lamp switch to the **Off** position.

Step 4. Continue with the procedure, ***Replacing the Lamp*** on page 126.

10.2.3 Replacing a Lamp while System is Operating

If you are replacing a lamp while other lamps are still in operations, follow these safety requirements:

- Shutdown the electricity supplied to the lamp to be replaced.
- **During the time when the lamp's caps are removed, use Caution floor signs at a distance of 1m/ 40inch from the Atlantium Unit that warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Where appropriate eye protection, such as containing polycarbonate lens that meet EN166, CAN/ CSA-Z94.3-02 and/or Z87.1 standards. The UV lamps are designed with high internal positive pressure.
- **Do not look directly into the lamp enclosure or into any openings that emit UV light during the entire lamp replacement procedure.**

➲ To shut down electricity to a single lamp:

Step 1. Deactivate the lamp to be replaced:

a To access the **Control Module's** Settings screen, from the **Main Operations** screen, tap . The **Password** screen appears.

b Using the numbered keys, enter the password (default password is **1357**) and tap . The Settings screen appears.

c For **Lamps**, tap ▶. The Lamps Settings screen appears.

d To disable a lamp, tap its **Enable** box to remove the checkmark.

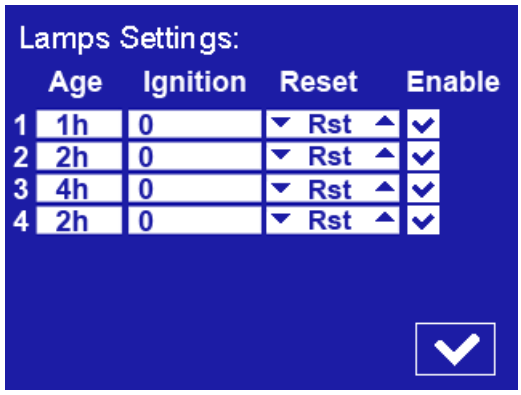


Figure 10-5: Lamp Setting Screen

- A severe danger of electrocution exists if the lamp is not turned off during replacement. Be absolutely sure you have turned off the correct lamp before you proceed to replace the lamp.

Step 5. Continue with the procedure, ***Replacing the Lamp***, below.

10.2.4 Replacing the Lamp

Once the electricity is turned off for the system or for the specific pair of lamps (at least one of which is to be replaced) and preparations are made (see above) you can replace the lamp.

➲ To replace a lamp:

Step 1. If you are replacing a lamp while other lamps are still in operations, follow these safety requirements:

- **During the time when the lamp's caps are removed, use Caution floor signs at a distance of 1m/ 40inch from the Atlantium Unit that warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.**
- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Where appropriate eye protection, such as containing polycarbonate lens that meet EN166, CAN/ CSA-Z94.3-02 and/or Z87.1 standards.
- **DO NOT look directly into the lamp enclosure during the entire lamp replacement procedure.**

ATTENTION!

While carrying the steps of this procedure, check that the O-rings on the components, such as the lamp housings and holders, are not damaged. If damaged, replace them before refastening the component in place.

Accessing the Lamp

ATTENTION!

For systems that contain wave guides, **be sure to access the lamp from the side that contains the sensors**. The wave guide is fragile and may be damaged easily if the lamp is inserted from the opposite site.

Step 2. Using a #3 Allen wrench, remove the four Allen screws **(1)** that secure the lamp's connector assembly **(2)** and remove the connector assembly. Each screw has a washer.

Step 3. **To prevent any damage to the lamp**, make sure you pull the lamp's connector assembly straight out and not at an angle.

Step 4. Using a #4 Allen wrench, remove the 4 screws **(3)** of the lamp's housing **(4)**. Each screw has a washer.

Step 5. Grasp the lamp's housing and rotate it clockwise and counterclockwise until it loosens. **Make sure the housing is not tilted against the quartz sleeve while rotating it**.

Figure 10-6: Lamp Replacement (Generic View 1)

(The O-Ring on the sleeve stopper between the holder and the stopper may cause the two parts to stick due to the heat. The slight rotation releases it).

To prevent any damage to the lamp and quartz sleeve, make sure you pull the lamp's housing straight out and not at an angle.

Figure 10-7: Lamp Replacement (Generic View 2)

Removing the Faulty Lamp

The lamp is hot! DO NOT touch the lamp with bare hands, particularly the quartz portion.

Wait at least ten minutes until the lamp has cooled down or use an appropriate pair of long-nose pliers to carefully pull out the lamp.

Step 6. Wearing appropriate cotton gloves and skin covering, and using a suitable long-nose pliers, pull the lamp straight out of the quartz sleeve in the Unit and set it aside on a stable surface.

Step 7. Check the inner surface of the quartz sleeve for residual moisture or high humidity. If any visible moisture exists, using a lens cleaning cloth, dry it.

Figure 10-8: Lamp Replacement (Generic View 3)

For Systems with Wave Guide

Every time the lamp is replaced, the wave guide must be cleaned or, if needed, replaced. See ***Cleaning/Replacing the UV Intensity Wave Guide*** on page **10-131.**

Cleaning/Replacing the UV Intensity Wave Guide or Mirror

Step 8. If you need to clean or replace the **UV Intensity Wave Guide** or **Mirror**, do so now before installing the new lamp.

- For wave guide, refer to ***Cleaning/Replacing the UV Intensity Wave Guide*** (on page **131**)
- For mirror, refer to ***Cleaning/Replacing the UV Intensity Sensor's Mirror*** (on page **133**)

Installing the Replacement Lamp

Step 9. Unpack the new UV lamp from its vacuumed sealed package.

Step 10. Make a note of the serial number of the new lamp.

Step 11. Clean the new lamp using the cleaning pad provided with the lamp. Verify that no visible fingerprint or dust is on the lamp.

Step 12. Insert the new lamp into the quartz sleeve in the Unit, all the way until it stops. Ensure that the male contact point **(6)** on the end of the lamp has meshed with the female contact point in the lamp's housing on the other side of the Unit.

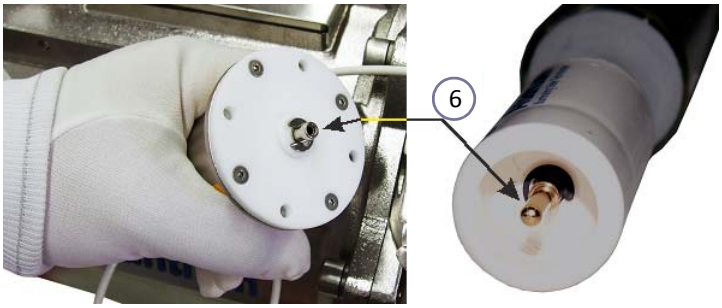


Figure 10-9: Lamp Replacement (Generic View 4)

Assembly

Step 13. Replace the lamp housing's O-ring.

Step 14. Insert the lamp's housing that you removed in above. Ensure that the male contact point on the end of the lamp has meshed with the female contact point in the lamp's housing on the other side of the Unit.

Step 15. Replace the lamp's housing and, using the # 4 Allen wrench, fasten the four Allen screws with their washers to secure it to the Unit.

Step 16. Replace the lamp's connector assembly pushing it straight in and using the # 4 Allen wrench, fasten the four Allen screws with their washers to secure it to the Unit.

Allowing the Atlantium Unit to Run with Caps Open

Step 17. Place Caution floor signs at a distance of 1m/40inch from the Atlantium Unit that warn passersby against possible UV light exposure and not to approach without proper eye and skin protection.

Step 18. Unscrew and remove the two caps **(7)** one on each side of the lamp. (See the figure to the right.)

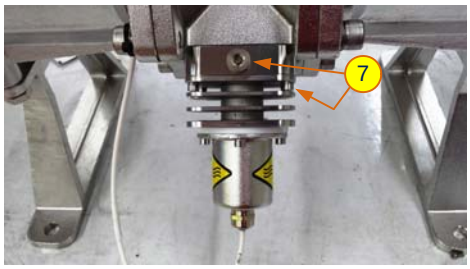


Figure 10-10: Lamp's Caps (Generic View 5)

Reactivating the Lamp

The newly replaced lamp must be tested to verify that it is in working order.

Step 19. For the lamp that was to be replaced, on the connection box, turn the lamp switch to the **On** position.

If the Entire System is Off

Step 1. Return to the Main screen of the **Control Module** and tap ⏻.

If only one lamp is off - Reactivation the Lamp

Step 2. Return to the **Control Module's** Lamps Settings screen and activate the replaced lamp by using the ▲▼ arrows, set the desired lamp to **ENB**.

Step 3. To reset the lamps' usage clock, tap .

Step 4. Tap . The settings are saved and you are returned to the Settings screen.

Step 5. Operate the system without the lamp's caps for one hour.

Step 6. Return the lamp's caps and remove the Caution floor signs.

Step 7. Perform system tuning according the instructions on page **142**.

Step 8.

Reactivation the Lamp

Step 9. For the lamp that is replaced, on the connection box, turn the lamp switch to the **On** position. Check that the lamp ignites properly.**Figure 10.2**

Step 10. Return to the **Control Module's** Lamps Settings screen and activate the replaced lamp by using the ▲▼ arrows, set the desired lamp to **ENB**.

Step 11. To reset the lamps' usage clock, tap .

Step 12. Tap . The settings are saved and you are returned to the Settings screen.

For All

Step 13. Check that the lamp ignites properly.

Step 14. Operate the system without the lamp's caps for one hour.

Step 15. Return the lamp's caps **(G)**.

System Tuning

Step 16. Perform system tuning according the instructions on page **143**.

Observing Lamp Efficiency

Step 17. Observe the system's efficiency for **100 hours**. If after this period there is a performance degradation of more than 5%, remove the UV intensity sensor and check for humidity. Clean if necessary.

Step 18. Perform **System Tuning** according the instructions on page **143**.

10.3 Cleaning / Replacing the UV Intensity Wave Guide

Performed as needed

If your system includes wave guides:

For each UVIS sensor in the Atlantium Unit, there is an internal wave guide for directing UV light into the sensor. The UV Intensity Sensor's wave guide must be cleaned every time the lamps are replaced.

Prepare the following:

- Replacement UV Intensity Sensor's Wave Guide
- Wave Guide tool
- White cotton gloves
- Alcohol/isopropanol and lint-free lens cleaning paper
- Appropriate Phillips screwdrivers

- Allow the old lamp at least 10 minutes to cool down before starting.

- Be very careful when handling the wave guide. The wave guide is made of quartz and is fragile and easily broken.

➲ To clean / replace the wave guide:

Accessing the Wave Guide to be Replaced

Step 1. On the side of the Unit where the UVIS sensors are located, identify the lamp whose wave guide is to be cleaned/replaced.

Step 2. Follow the directions for removing a lamp. See ***Replacing a UV Lamp*** on page **10-123.**

Removing the Wave Guide

Step 3. Screw the Wave Guide tool **(L)** onto the wave guide socket **(J)**.

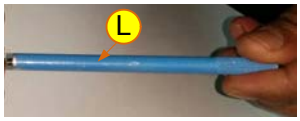


Figure 10-11: Wave Guide Tool

Step 4. While holding the Wave Guide tool **(L)** now attached to the wave guide **(G)**, using an appropriate Phillips screwdriver, **(M)** unfasten each of the screws **(K)**.

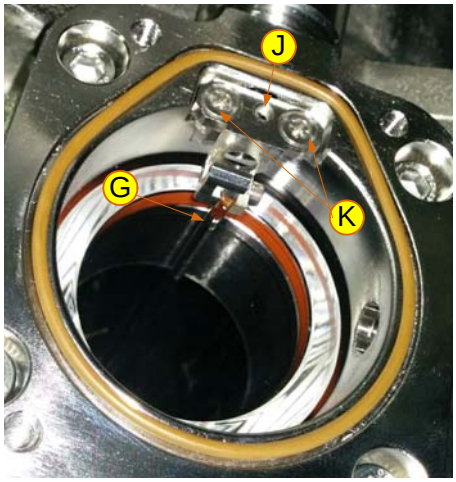


Figure 10-12: Wave Guide Elements (Generic View)

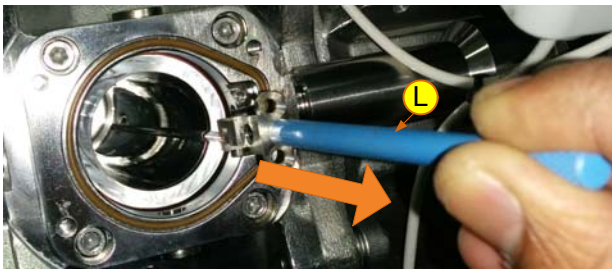


Figure 10-13: Removing the Wave Guide (Generic View)

Step 5. With the Wave Guide tool **(L)**, carefully pull out the wave guide.

Step 6. Unscrew the Wave Guide tool **(L)** and detach it from the wave guide **(G)**.

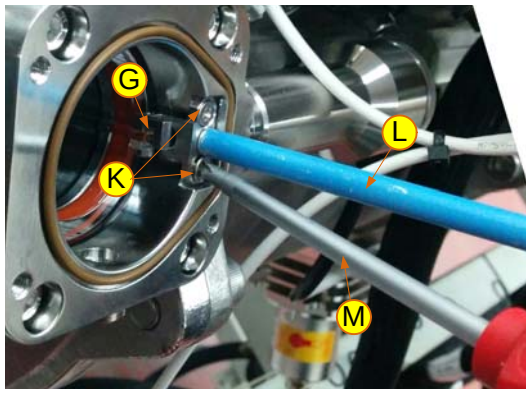


Figure 10-14: Unfastening the Wave Guide (Generic View)

Figure 10-15: Wave Guide Tool and Wave Guide

Cleaning the Wave Guide

Step 7. While wearing white cotton gloves, carefully clean the wave guide with isopropanol and lint-free lens cleaning paper.

Step 8. Ensure that the wave guide is completely dry before returning it to its place.

Step 9. Examine the wave guide for marks and damage. Fingerprints or other stains may cause erroneous UV intensity sensor readings. If the wave guide's surface is damaged, it must be replaced.

Preparing the Replacement Wave Guide

Step 10. Screw the Wave Guide tool **(L)** onto the wave guide socket **(J)**.

Step 11. With the Wave Guide tool **(L)**, carefully insert the wave guide in place.

Step 12. While holding the Wave Guide tool **(L)** now attached to the wave guide **(G)**, using an appropriate Phillips screwdriver, **(M)** fasten each of the screws **(K)**.

Step 13. Check that the wave guide is parallel to the side of the quartz sleeve and not pointing at an angle. If it is at an angle, the quarts sleeve is not installed straight. Remove the wave guide and remove and reinstall the quartz sleeve following the directions in ***Cleaning/Replacing a Lamp's Quartz Sleeve and O-Rings*** on page **133**. When complete, install the wave guide again.

The correct position of the wave guide is parallel to the side of the quartz sleeve and a distance of 0.75mm from the quartz sleeve.

Step 14. Follow the directions to reinsert the lamp and refasten the lamp's housing and connector See ***Replacing a UV Lamp*** on page **10-123.**

10.4 Cleaning / Replacing a Lamp's Quartz Sleeve and O-Rings

The quartz sleeve is cleaned as needed according to the UVT Sensor readings and when deposit collects on the quartz sleeve. **The quartz sleeve's rubber O-rings must be replaced when dismantling the quartz sleeve**.

Depending on the quality of the water being treated by the Atlantium Unit, the exterior surface of the quartz sleeve that houses a lamp may require periodic cleaning or replacement if after the optional Cleaning system and the CIP process the quartz sleeve is still not sufficiently clean. This is to be determined by the quality control officer at the facility.

Under normal environmental conditions, the quartz sleeve's rubber O-rings (as shown in **Figure 10-16** below) must be replaced once a year even if they pass visual inspection. For environments where the water temperature is above 70°C (160°F), the quartz sleeve's O-rings must be replaced yearly.

Prepare the following:

- Replacement quartz sleeve (if replacing)
- Two O-rings (See view **B** in **Figure 10-16** below)
- #3 Allen wrench
- #4 Allen wrench
- Protective gloves
- Appropriate cleaning cloth. A mild, household cleaning solution
- Alcohol

ATTENTION!

While carrying the steps of this procedure, check that the O-rings on the components, such as the lamp housings and holders, are not damaged. If damaged, replace them before refastening the component in place.

➲ To clean or replace the quartz sleeve:

Powering Down the System

Step 1. On the Main screen of the **Control Module**, tap ⏻.

Step 2. Enter the password (the default password is **2468**). The Atlantium system is turned off.

Step 3.

A severe danger of electrocution exists if the lamp is not turned off during replacement. Be absolutely sure you have turned off the correct lamp's before you proceed to replace the lamp.

Step 4. Using the viewport on the Atlantium Unit, check that the lamps are indeed turned off.

Step 5. Check the main screen of the **Control Module** and verify that the system parameter for **Power** is **0%**.

Isolating the Unit

Step 6. To isolate the Unit, close the Inlet and Outlet valves.

Step 7. Using the Drain valve, drain the water from the Unit.

Accessing the Quartz Sleeve

Quartz sleeve removal requires that the lamp connector assemblies, lamp housings (on both ends of the lamp) and the lamp be removed, as well as the quartz sleeve holder.

Step 8. Quartz sleeve removal requires that the lamp connector assemblies, lamp housings (on both ends of the lamp) for systems that have them, and the lamp be removed. In the section, 10.2.4. Replacing the Lamp, follow from **Step 13.** on page **123** and remove the lamp connector assembly and lamp housing **from both sides of the Unit,** as well as the lamp, placing them in a secure location until they need to be reinstalled.

While pulling the lamp housing away, DO NOT tilt it and DO NOT pull it away from the lamp/ sleeve axis at an angle. For systems with wave guide, the wave guide is fragile. Take care not to damage it when removing the lamp housing.

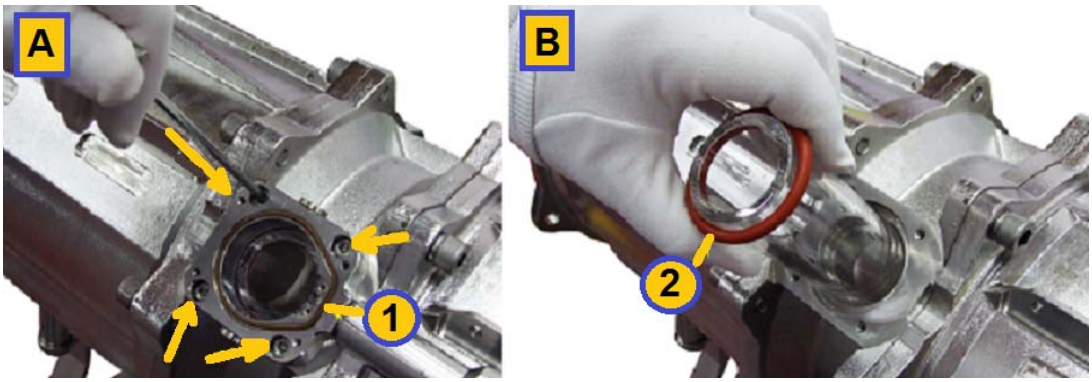


Figure 10-16: Replacing the Quartz Sleeve (Generic View 1)

Step 9. On one side of the quarts sleeve, using a #4 Allen wrench, remove the four Allen screws, opening them in a diagonal pattern (i.e. upper left, lower right, upper right, lower left) (arrow indicate the screws in view **A**) that secure the sleeve holder to the Unit. Loosen one screw gradually (2 turns) on one side and do the same on the screw at the diagonal opposite side of the sleeve (2 turns).

Step 10. Continue to alternate until both sleeve holders are loose, and then remove the screws completely.

Step 11. On the other side of the quarts sleeve, repeat **Step 8.** to **Step 10.**

Removing the Quartz Sleeve

Wear protective gloves. To protect the quartz sleeve, do not touch the internal surface with your bare hands.

Step 12. On one side of the quartz sleeve, place the gloved palm of your hand over the quartz sleeve and gently push in. The quartz sleeve shifts inward. The o-ring on the side where you pushed slips out (if not, take it off).

Step 13. On the other side of the quartz sleeve, the sleeve sticks out slightly. **To prevent any damage to the quartz sleeve, make sure you pull the lamp's connector assembly straight out.** Carefully pull out the quartz sleeve and remove the O-ring (shown in view **B** above).

Cleaning the Quartz Sleeve

While pulling the connector assembly away, DO NOT tilt it and DO NOT pull it away at an angle.

Step 14. If there is a deposit buildup on the sleeve's exterior surface, clean it with a mild, household cleaning solution.

Step 15. Clean the exterior and interior surfaces with alcohol. If the sleeve cannot be cleaned to a pristine condition, replace it with a new one.

Step 16. Ensure that there is no residual moisture or high humidity on the inner surface of the quartz sleeve. If any visible moisture exists, dry it before inserting it in place.

Installing the Quartz Sleeve

Step 17. Mount the new O-ring onto the quartz sleeve (as shown in view **B** in **Figure** above) and insert the sleeve into the Unit. Make sure it is centered.

Step 18. Add the second new O-ring on the other side of the quartz sleeve.

Step 19. Install the quartz sleeve holder on the side without the UVIS mirror and close it loosely with 4 screws in a diagonal pattern (upper left, lower right, upper right, lower left). **Do not tighten the screws!!**

Figure 10-17: Replacing the Lamp's quartz sleeve (Generic View 2)

Assembly

Step 20. Install the quartz sleeve holder on the opposite side – the one with the UVIS mirror or wave guide.

For systems with wave guide, while reinstalling the sleeve holder, DO NOT tilt it and DO NOT insert it at an angle. the wave guide is fragile. Take care not to damage it when inserting the lamp housing.

Step 21. Install the quartz sleeve holder on the lower side and close it loosely with 4 screws in a diagonal pattern (upper left, lower right, upper right, lower left). **Do not tighten the screws!!**

Step 22. Install the quartz sleeve holder on the upper side.

Step 23. Using a #4 Allen wrench, close the screws in diagonal pattern and tighten them.

Step 24. To reinstall the lamp, perform from **Step 9.** on page **128** in the section,10.2. Replacing a UV Lamp.

Step 25. Reinstall the lamps' connector assemblies pushing each straight in and using the # 4 Allen wrench, fasten the four Allen screws with their washers to secure them to the Unit.

Step 26. Connect the UVIS sensors' cables.

Step 27. On the corresponding Connection Box to where the sleeves/lamps are located, turn the switch to the **On** position.

Reintroducing the Water

Step 28. Carefully open the Inlet and Outlet valves (Be sure to avoid water hammer). Follow the directions in the section ***Filling unit with water*** on page **100**.

Powering Up the System

Step 29. On the Main screen of the **Control Module**, tap ⏻.

Step 30. Enter the password (the default password is **2468**). The Atlantium system is turned off.

Verifying System Performance

Step 31. Wait 10 minutes for lamps to turn on and the system to stabilize.

Step 32. Open the viewport and check that there are no air bubbles in the water flow. If necessary, release all air bubbles from within the system.

Step 33. Run the system with the caps of the lamp off as described in the section, 10.2.4. Replacing the Lamp, see from **Step 17.** to the end of the procedure, starting on page **129**.

WARNING!

While lamp caps are off, be sure to follow safety requirement:

- Wear white cotton gloves.
- Wear long sleeves rolled down to your wrist.
- Wear appropriate eye protection.
- **Do not look directly into the lamp enclosure or into any openings that emit UV light during the entire lamp replacement procedure.**

Step 34. Check the main screen of the **Control Module** and verify that the Atlantium system parameters for **Power**, **Flow**, and **UVT** are within wanted acceptable range and record is using the Table in **Appendix B**. ***Checking the System Parameters*** on page **165**.

If the values are out of range, notify the executive maintenance engineer.

10.5 Checking / Replacing Sensors

From time to time, sensors may need to be checked (most frequently for condensation). During the Calibration process, the UVIS sensor also requires to be removed and temporarily replaced with a Reference Master sensor during. At times a sensor needs to be replaced. There are two types of UVIS sensors. The sensor itself is the same in both types. The difference is the sensor casing and how it is fastened to the Atlantium Unit. The two types are:

- **Adjustable UVIS sensor** - The Sensor's position is adjustable and its casing is fastened with nuts. See below.

- **Fixed UVIS sensor** - The sensor's casing screws into a fixed position. This type of sensor may contain a pinhole. See ***Replacing/Checking a UVT Analyzer or Fixed UVIS Sensor*** on page **142**.

10.5.1 Checking an Adjustable UVIS Sensor

Performed as needed

Before removing the UVIS sensor to check it for moisture, do one of the following:

If you do not have a caliper measurement tool: You must mark the placement of the sensor so that you can replace it in the correct position.

If you have a caliper: You must measure the position of the sensor. (See on page **10-122.**) or by using a marker to mark the position before removal.

Prepare the following:

- Wrenches # 24 and # 30
- Caliper measuring tool or thin marker

➲ To check an adjustable sensor (using a marker):

Step 1. In the **Control Module**, note the value for UVIS for the lamp on which the sensor to be replaced. For details on finding the **Control Module** screen with this information, see the section, ***System Tuning*** on page **143**.

Step 2. Mark the placement of the sensor so that you can replace it in the correct position. Use the thin marker to mark the position before removal.

Mark the sensor in two places, a line along the nut and across the nut and Perpendicular to the first line. as shown in **(1)**.

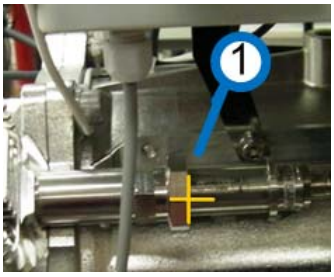


Figure 10-18: Marking the Sensor

Step 3. Check the sensor's lens for condensation. If present, using a dry lens cloth, dry the lens thoroughly.

Be sure that the lens is completely dry before it is returned to its place.

Step 4. When you return the sensor to its place, make sure the thin lines line up to the nut and align together.

Step 5. In the **Control Module**, check that the value for UVIS for the lamp on which the sensor was replaced is identical to the value before the sensor was removed. For details on finding the **Control Module** screen with this information, see the section, ***System Tuning*** on page **143**.

➲ To check an adjustable sensor (using a caliper):

Step 1. Place the measurement tool on the UVIS sensor as show in **Figure 10-19** below. Use the screw to adjust the placement of the caliper measurement tool.

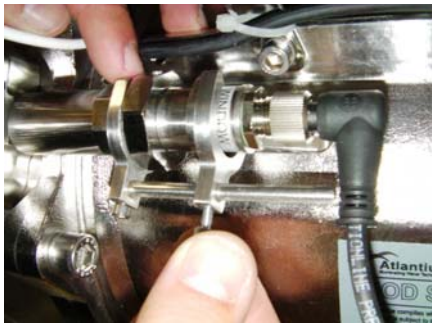


Figure 10-19: Measuring the UVIS sensor

Figure 10-20: Caliper Measurement Tool

Step 2. Carefully remove the caliper, taking care not to accidentally change its position, and set it aside.

Step 3. Remove the sensor according to the relevant steps in ***Replacing an Adjustable UVIS Sensor*** on page 140.

Step 4. If you are in the midst of a Calibration process for the UVIS sensor, replace the original UVIS sensor with the Reference sensor.

Step 5. Check the sensor's lens for condensation. If present, using a dry lens cloth, dry the lens thoroughly.

Be sure that the lens is completely dry before it is returned to its place.

Figure 10-21: Adjusting the Reference sensor

Step 6. When you insert a Reference Master sensor or return a sensor to its place, using the set caliper (calibrated to the original UVIS sensor's position) adjust the sensor so that it is set to the position of the original. Refer to **Figure 10-21** to the left.

10.5.2 Replacing an Adjustable UVIS Sensor

Performed as needed

Before removing the UVIS sensor when calibrating it against the Reference Master sensor or checking for moisture, you must mark the placement of the sensor so that you can replace it in the correct position. This is accomplished using the caliper measurement tool (See on page **10-122.**) or by using a marker to mark the position before removal.

Replacing the UVIS sensor involve the loosening of the nut holding it in place.

Prepare the following:

- Replacement UVIS sensor or appropriate type
- Wrenches # 24 and # 30
- Caliper measuring tool or thin marker

➲ To replace an adjustable UVIS sensor:

Step 1. To disconnect the harness connector (shown in view **A** in **Figure 10-23**), unscrew the sensor's threaded locking nut.

Step 2. Grip the Sensor casing with wrenches **# 24** and **# 30**, as shown in view **B** below. The inset shows the wrench **#24** in the left hand gripping the casing's left band and wrench **#30** in the right hand gripping the casing's large nut on the right.

Step 3. Holding the left wrench **#24** steady, twist the right wrench **#30** to loosen the nut.

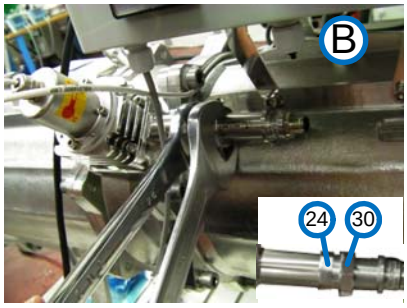


Figure 10-22: Replacing the Sensors (Generic View 2)

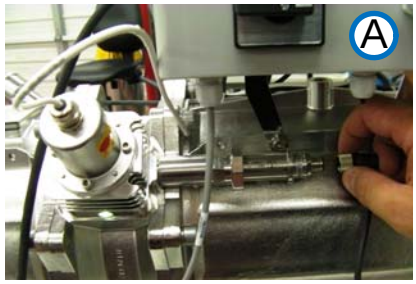


Figure 10-23: Replacing the Sensors (Generic View 1)

Step 4. Carefully pull the sensor out of the housing **(C)**. The Large nut comes off together with the sensor.

Step 5. To enable the system to track relevant data about the sensor, make a note of the serial number of the replacement sensor (for later use).

Step 6. Verify that the sensor lens **(1)** is clean. If needed, using a lens cloth, clean it. (See below.)

Be sure that the lens is completely dry before it is returned to its place.

Step 7. When replacing a sensor, remove the nut **(2)** and the washer **(3)** shown in **Figure 10-25** below.

Figure 10-24: Replacing the Sensors (Generic View 3)

Step 8. Position the nut **(2)** on the sensor as shown above. Then position the washer **(3)** on the sensor as shown above.

Step 9. Insert the sensor into the housing.

- If you are replacing the sensor, positioning it about 2/3 the way in. Fasten the nut onto the housing.
- If you are returning a sensor that was removed for checking or calibration, see ***To check a sensor or remove it for calibration*** on page **138**.

Step 10. Grip the Sensor casing with wrenches **# 24** and **# 30**, as shown in view **B** above. The inset shows the wrench **#24** in the left hand gripping the casing's left band and wrench **#30** in the right hand gripping the casing's large nut on the right.

Step 11. Holding the left wrench **#24** steady, twist the right wrench **#30** to tighten the nut.

Step 12. Reconnect the harness's threaded cap to the new sensor.

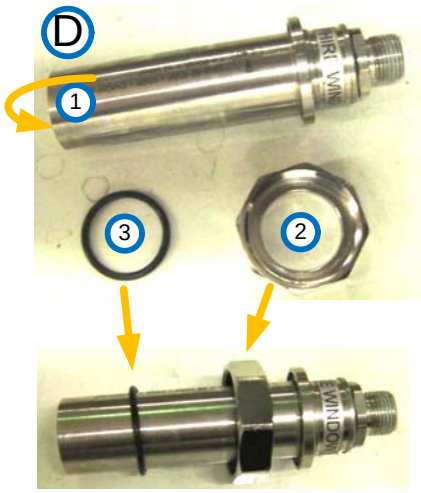


Figure 10-25: Sensor, washer and nut

Step 13. For a replacement sensor, in the **Control Module**, perform **System Tuning**, See the section, ***System Tuning*** on page **143**.

10.5.3 Replacing / Checking a UVT Analyzer or Fixed UVIS Sensor

Performed as needed

For system that may contain a fixed UVIS sensors, this sensor-replacement procedure is required. If your system contains the adjustable UVIS sensor, see ***Checking an Adjustable UVIS Sensor*** on page **138**.

For some of the sensors, there is an pinhole located on the sensor's window. This pinhole is calibrated by Atlantium for the specific measurement point. Do not switch disks between ports. Before inserting the sensor, be sure to include the pinhole if present.

Prepare the following:

- Replacement sensor

➲ To replace a sensor:

Step 1. To disconnect the harness connector (shown by the arrow in view **A** below), unscrew the sensor's threaded locking nut.

Step 2. Unscrew the sensor's threaded locking nut from the bottom of the sensor housing, as shown in view **B**.

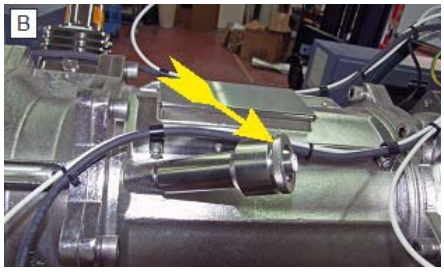


Figure 10-26: Replacing the Sensors 2 (Generic View 2)

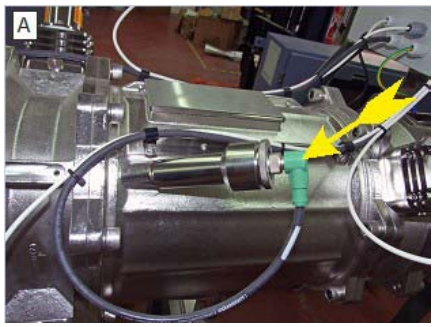


Figure 10-27: Replacing the Sensors 2 (Generic View 1)

Step 3. Carefully pull the sensor out of the housing **(C)**.

Step 4. If an pinhole **(E)** is present it slides out.

Step 5. Unpack the new sensor **(D)** and (if it is supplied) the sensor pinhole.

Figure 10-28: Replacing the Sensors 2 (Generic View 4)

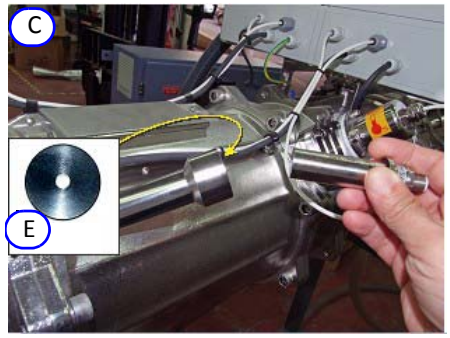


Figure 10-29: Replacing the Sensors 2 (Generic View 3)

Step 6. To enables the system to track relevant data about the sensor, make a note of its serial number.

Step 7. Verify that the sensor lens **(F)** is clean. If needed, using a lens cloth, clean it.

Step 8. If an pinhole **(E)** is present, insert it.

Step 9. Insert the sensor into the housing, and screw the threaded cap back onto the housing, securing the sensor.

Step 10. Reconnect the harness's threaded cap to the new sensor.

Step 11. In the **Control Module**, perform **System Tuning**, see the section, ***System Tuning***, below.

10.6 System Tuning

Performed as needed

System tuning is performed on these occasions:

- As the last step in system installation
- As the last step when lamps are changed and after 100 operating hours (as needed)
- Upon advice of an Atlantium Service Representative when the system is reset

➲ To tune the Atlantium system:

Step 1. Just prior to tuning, obtain a water sample from the sampling valve on the inlet side of the Atlantium Unit. Test the water sample in an external spectrophotometer. Write down the results. If a spectrophotometer in not available, record the current UVT value shown on the system's display.

Step 2.

Step 3. To shutdown the Atlantium system, on the **Control Module** main screen, tap ⏻ and wait approximately three minutes for the **Ready Lamps Off** signal to transmit.

Step 4. On the Atlantium Unit, locate each UVIS and UVT Intensity Sensor.

Step 5. On each sensor (refer to the figures in the section, ***To replace an adjustable UVIS sensor:*** on page **140**):

a Unscrew the locking nut and gently pull out the sensor.

b With a dry and clean lint free cloth, wipe the UVIS sensor's quartz window.

c Reinsert the sensor and replace the locking nut.

Step 4. To turn on the Atlantium system, on the main screen, tap ⏻ and wait approximately three minutes for the **Ready Lamps On** signal to transmit.

Step 5. To access the **Control Module's** Settings screen, from the **Main Operations** screen, tap . The **Password** screen appears.

Step 6. Using the numbered keys, enter the password (default password is **1357**) and tap . The Settings screen appears.

Step 7. To verify that the lamps are already on, for **Lamps**, tap ▶. The Lamps Settings screen appears. (Figure **10-5** , on page **126**.) Note that the lamps are set to **On**.

Step 8. Tap . The settings are saved and you are returned to the Settings screen.

Step 9. For **Mode**, tap ▶. The Mode Settings screen appears.

Step 10. For **Power**, using the ▲▼ arrows, turn each lamp up to **100%** power.

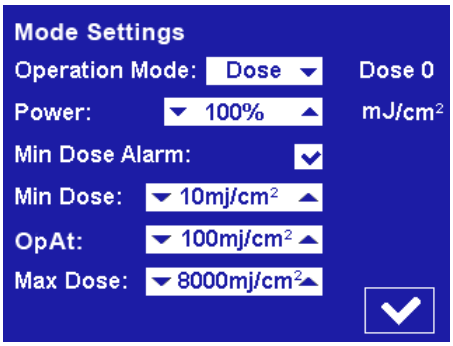


Figure 10-30: Mode Settings Screen

Step 11. Tap . The settings are saved and you are returned to the Settings screen.

Step 12. For **Tuning**, tap ▶. The System Tuning screen appears.

Step 13. For **UVT**, use the ▲▼ arrows to enter the water test results.

Step 14. For **UVIS 1** and **2** (if your system has two lamps):

- For tuning when replacing a lamp, use the ▲▼ arrows, to enter 100%.
- For tuning with existing lamps, leave the settings as is.

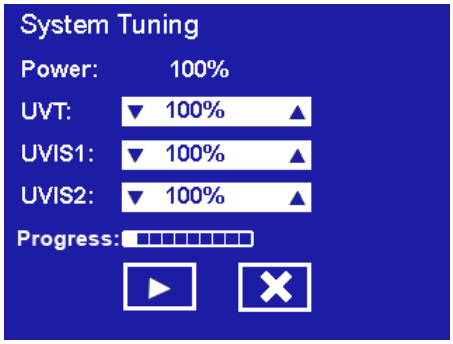


Figure 10-31: System Settings Screen

Step 15. Tap . The systems tunes and the **Tune** bar displays the progress.

Step 16. When tuning is complete, the activate arrow turns to . Tap it. The settings are saved and you are returned to the Settings screen.

10.7 Anti-Corrosion Maintenance

The Atlantium Unit has been coated with the OKS 2101 – Protective Film for Metals to protect the Unit against corrosion. This protective film must be reapplied yearly to make sure the Atlantium Unit is constantly protected from the corrosion that can be affected in a seawater environment.

Prepare the following:

- OKS 2101 Spray
- Safety glasses with side-shields conforming to EN166
- Protective gloves that satisfy the specifications of EU Directive 89/686/EEC and the standard EN 374 derived from it
- Respiratory protection mask with filter type A-P

WARNING!

Refer to the OKS 2101 Safety Data Sheet according to Regulation (EC) No. 1907/2006 - GB.

Follow these precautions for safe handling

Do not use in areas without adequate ventilation.

Do not breathe vapors or spray mist.

In case of insufficient ventilation, wear suitable respiratory equipment.

Avoid contact with skin and eyes.

Keep away from fire, sparks and heated surfaces.

Persons with a history of skin sensitization problems or asthma, allergies, chronic or recurrent respiratory disease should not be employed in any process in which this mixture is being used.

Smoking, eating and drinking should be prohibited in the application area.

Wash hands and face before breaks and immediately after handling the product.

Do not get in eyes or mouth or on skin.

Do not get on skin or clothing.

Do not ingest.

Do not use sparking tools.

These safety instructions also apply to empty packaging which may still contain product residues.

Pressurized container: protect from sunlight and do not expose to temperatures exceeding 50 °C. Do not pierce or burn, even after use.

➲ To apply the protective film:

Step 1. Wear the protective glasses, gloves and filter mask as specified above.

Step 2. For optimum effect, clean the surfaces. Best way is to clean mechanically first and then with OKS 2610/OKS 2611 universal cleaner. The surfaces must be bright metal and dry. Stir OKS 2100 thoroughly before use.

Step 3. Shake the OKS 2101 can.

Step 4. Spray on a thin protective film in an evenly manner to the prepared surfaces Move the attached cables aside as you spay. Make sure to cover the entire body, including under the cables. Avoid local excesses. The optimal layer thickness is 50µm.

Step 5. At an environmental temperature of 20°C, allow the Unit to dry for 30 minutes. In colder environments, allow additional time.

10.8 Advance Maintenance

The advanced maintenance procedures require opening the Connection Box. **Local regulations may require a certified electrician to perform any of these procedures.**

This section contains:

- ***Replacing the Temperature Sensor*** on page **147**
- ***Replacing the Temperature Converter Card*** on page **148**

10.8.1 Replacing the Temperature Sensor

In the event that the Temperature sensor requires replacement, this section details the procedure.

Follow your local regulations regarding whether a certified electrician may be required to perform this procedure.

Any person involved in handling the Connection Box card or its components is required to wear an electrostatic discharge (ESD) Wrist Strap.

Connect the ESD Wrist Strap to any bolt on the body of the Atlantium Unit or to the metal bracket under the Connection Box using a banana plug or an alligator clip.

Prepare the following:

- Replacement Temperature sensor
- #5 Allen wrench
- #3 Flat screwdriver
- #2 Phillips screwdriver

➲ To replace the Temperature sensor:

Step 1. Turn the Mains circuit breaker to the **Off** position.

Step 2. Using the appropriate Allen wrench, open the Connection Box cover.

Step 3. Verify with volt meter that there is no power in first Connection Box card and that no LED is lit on the card.

Step 4. Locate and disconnect the Temperature Sensor connector **(A)**.

Step 5. Detach the wire from the terminal connector and remove the wire from the Connection Box via the gland.

Step 6. Using a #2 Phillips screwdriver, detach the temperature sensor **(C)** from the Atlantium Unit.

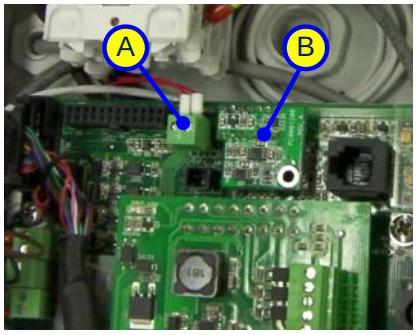


Figure 10-32: Location of the Temperature Sensor Connector (Generic)

Step 7. Attach the replacement Temperature sensor **(C)** to the Atlantium Unit.

Step 8. Thread the wire through the gland and attach it to the terminal connector.

Step 9. Insert the connector **(A)** to its socket on the terminal block.

Step 10. Return the Connection Box cover and using the Allen wrench, fasten the screws. Tighten them in a diagonal pattern.

Step 11. Turn the Mains circuit breaker to the **On** position.

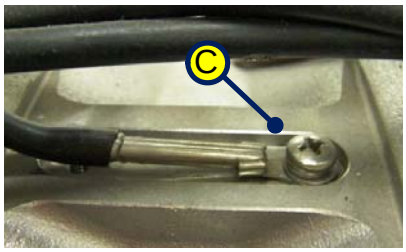


Figure 10-33: Location of the Temperature Sensor on the Atlantium Unit

10.8.2 Replacing the Temperature Converter Card

Performed as needed

In the event that the Temperature sensor card requires replacement, this section details the procedure.

Follow your local regulations regarding whether a certified electrician may be required to perform this procedure.

Any person involved in handling the Connection Box card or its components is required to wear an electrostatic discharge (ESD) Wrist Strap.

Connect the ESD Wrist Strap to any bolt on the body of the Atlantium Unit or to the metal bracket under the Connection Box using a banana plug or an alligator clip.

Prepare the following:

- Replacement Temperature Converter card
- #5 Allen wrench
- #2 Phillips screwdriver

➲ To replace the Temperature sensor card:

Step 1. Turn the Mains circuit breaker to the **Off** position.

Step 2. Using the appropriate Allen wrench, on the first Connection Box, open the Connection Box cover.

Step 3. Verify with volt meter that there is no power in first Connection Box card and that no LED is lit on the card.

Step 4. Locate and carefully pull off the Temperature Sensor card **(C)**.

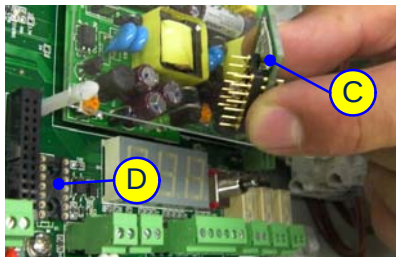


Figure 10-34: Replacing the Temperature Sensor Card (Generic)

Step 5. Aligning the pins of the replacement Temperature Sensor card **(C)** with the connector (on the board) on the Connection Box card, carefully insert the small card into the connector of the Connection Box card.

Step 6. Return the Connection Box cover and using the Allen wrench, fasten the screws. Tighten them in a diagonal pattern.

Step 7. Turn the Mains circuit breaker to the **On** position.

11 Troubleshooting with Standard Control Module

The sections here are comprised of messages headers that appear as error messages in the **Standard Control Module.**

Utilize the following safety notes when reviewing the troubleshooting issues:

The solution to the problem might involve exposure to UV light.

The system generates ultraviolet (UV) light within the Atlantium Unit, which can cause serious eye damage or blindness if you stare at it directly when it is working.

Do not look directly into the lamp enclosure during system operation, or while examining or servicing the system's internal components when the system is operating, or when a lamp's caps are open.

The solution to the problem might involve working in a high-voltage environment.

Before replacing a UV lamp or other components, and during any maintenance requiring lamps to be turned off, **make sure the switch on the ballast of the relevant lamps are turned OFF.**

Place a sign on the relevant Ballast Modules and **Control Module** alerting others **NOT** to touch the switch and touch screen during maintenance so that no one can inadvertently turn it on while maintenance is in progress.

The solution to the problem might involve working with hot surface. Do not touch the UV lamp with your bare hands. Lamps can reach a temperature of 1,000°C under operating conditions.

Wait at least 10 minutes until the lamp is cooled down before replacing it.

Use appropriate protective gloves both to protect your hands and to avoid skin oils that leave fingerprints and/or harm the UV lamp.

11.1 Voltage Error Codes

For Ballast Module version C, the Voltage parameter provides a value that functions as an additional Error code and indicates additional information about the fault.

Table 11-1: Ballast Module Version 3 Voltage Error Code Key

Lamp voltage value when lamp is off	Lamp turn off reason
0 (±24)	None
50 (±24)	Lamp nominal voltage too low*
100 (±24)	Lamp nominal voltage too High**

Table 11-1: Ballast Module Version 3 Voltage Error Code Key

150 (±24)	Device cooling insufficient
200 (±24)	Internal failure / Phase lost
250 (±24)	Lamp earth short circuit
300 (±24)	Lamp not ignited / gone out
350 (±24)	Lamp short circuit
400 (±24)	Phase failure mains voltage
450 (±24)	Mains overvoltage
500 (±24)	Manual turn off

*** Voltage of lamp is low while the current to the lamp is at the maximum allowed value – in this state, the required power cannot be delivered to the lamp.**

**** The voltage required by the lamp cannot be created by the ballast. The current going to the lamp is zero.**

The above Error Codes are displayed in the Volt column in Technician screen 3 (see ***Configuring the Technician Settings*** on page **6-95**).

Step 1. To access the **Technician 3** screen, On the **Control Module**, to access the Technician mode, tap . The Password dialog box appears.

Step 2. Enter the password (default is **8888**) and tap **OK**. The Technician Settings 1 screen appears.

Step 3. Tap twice on the ▶at the right of the screen to access Technician 3 screen.

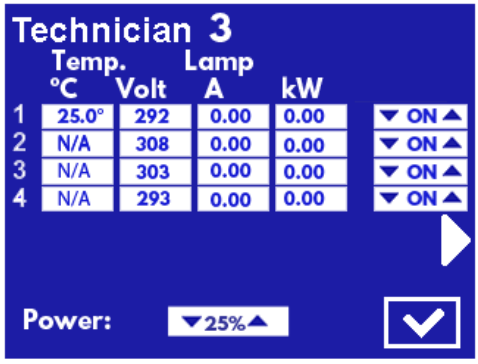


Figure 11-1: Technician 3 Screen

Troubleshooting issues

The UV dose is below the allowable threshold on page **11-152**
Unit temperature is too high on page **11-153**
Unit temperature is critically high! on page **11-154**
The rate of the water flow is too low on page **11-155**
UV lamp ignition failed on page **11-156**
Temperature sensor fault on page **11-157**
Ballast temperature is too high on page **11-157**
System type settings error on page **11-158**
Lamp mains failure on page **11-159**
Unexpected failure for an unknown reason on page **11-159**
Lost communication with ballast module on page **11-160**
Clean system deposit on page **11-161**

11.2 The UV dose is below the allowable threshold

Control Module displays this warning message
UV Dose Below Allowable Threshold

!

	Possible Cause	Solution	
1	The Water flow rate is too high.	Check the water flow on the Control Module's main screen. If it is higher than normal, decrease the rate of the water flow through the system.	
2	UVT has declined because of chemicals (chlorine, etc.) or solids (sand, silt, etc.) in the water flow.	Carry out cleaning procedures. See ***Ultrasonic Cleaning (Optional)*** on page **97.** ***Cleaning In Place (CIP)*** on page **10-111**. Try to improve the water condition. Check on the status of upstream filtering equipment and upstream chemical processes.	
3	UV lamp efficiency has dropped.		Check the UVIS sensors and clean them if necessary. Check the lamp's efficiency reading. If it is low, replace the lamp. See ***Replacing a UV Lamp*** on page **123.**
4	The UVT sensor has malfunctioned.		Switch between the UVT and UVIS sensors and see if the reading increased. If the reading is increased, switch the sensors back and replace the UVT sensor. See ***Checking/ Replacing Sensors*** on page **137.**
5	The system is operating on **Power mode** instead of **Dose mode**.	If the system settings are in Power mode, in the **Control Module**, access the Settings screen and change the **Operation Mode** from **Power** to **Dose** setting. Or, change the Power mode settings to allow higher power to respond to operating conditions.	

11.3 Unit temperature is too high

Control Module displays this warning message
Unit Temperature Is Too High

	Possible Cause	Solution
1	The Max temperature value is too low.	Check the system's **Max Temperature** settings.
2	The water flow has significantly decreased.	Check the flow reading in the **Control Module**'s main screen to see if it has deviated from normal flow. Check the flow via the flowmeter. Check the water source for a problem.
3	There are air bubbles or the system is not fully filled with water.	Verify this by looking through the Atlantium Unit's view port.
4	You did not install a designated air release valve at the highest point on the inlet side of the system.	Release the air by installing a designated air release valve at the highest point on the inlet side of the system.

11.4 Unit temperature is critically high!

Control Module displays this warning message
Unit Temperature Critically High

	Possible Cause	Solution
1	If this message appeared right after a message indicating loss of communication with the Ballast Module (and both messages refer to the same lamp) and the temperature display on the monitor is 165°C/ 365°F, then most likely the communication between the **Control Module** and the Connection box is lost.	Follow the solutions offered in ***Ballast temperature is too high***
2	The temperature reading dropped to 185°C / 365°F due to a short-circuited in the Temperature sensor, Temperature converter card or Connection Box Card	Replace the temperature sensor. See ***Replacing the Temperature Sensor*** on page **147.** If the problem persists replace the Temperature Converter card. See ***Replacing the Temperature Converter Card*** on page **148.** If the problem still persists replace the Connection Box card. Consult your Atlantium Representative.
3	The Max temperature value is too low.	Check the system's **Max Temperature** settings. See ***Configuring More Settings*** on page **68.**
4	There is no water flow.	Check the flow reading in the **Control Module**'s main screen to see if it has deviated from normal flow. Check the flow via the flow meter. Check the water source for a problem.
5	There are air bubbles or the system is not filled with water to capacity.	Verify this by looking through the Atlantium Unit's view port.
6	You did not install a designated air release valve at the highest point on the inlet side of the system.	Release the air by installing designated air release valve at the highest point on the inlet side of the system.

	Possible Cause	Solution
7	The Temperature jumper is missing.	Verify the existence of the Temperature jumper in its designated location on the Connection Box card #1. On the last Connection Box, open the top cover and locate Jumper **J32** as shown marked **A** in the figure. Figure 11-2: Jumper Location
8	The Temperature sensor is faulty.	Replace the temperature sensor. See ***Replacing the Temperature Sensor*** on page **147.**
9	The Temperature Converter card is faulty.	If the problem persists replace the Temperature Converter card. See ***Replacing the Temperature Converter Card*** on page **148.**

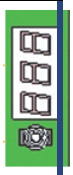
11.5 The rate of the water flow is too low

Control Module displays this warning message
Water Flow Rate: Too Low

	Possible Cause	Solution
1	There is an obstruction in the upstream system or pump malfunction.	If this is a planned decrease or temporary stop, change the **Restart on Flow** setting from **Shutdown mode** to **Restart**. This sets the system to Standby. Check pumps, filters and system feed and increase the flow rate.
2	There is no water running through the system.	If this is a planned stop, change the **Restart on Flow setting** to **Restart**. This sets the system to Standby.
3	The flow meter/flow switch is faulty or a wire has been disconnected.	Check the wires connecting the flow meter/flow switch and reconnect or replace if damaged. Replace the flow meter/flow switch is faulty.

11.6 UV lamp ignition failed

Control Module displays this warning message
UV Lamp ID Ignition Failed

	Possible Cause	Solution	
1	The system is set to the wrong model.	In the **Control Module**, access **Technician screen #2**. Verify that the system is set to the correct model, number of lamps and branches.	
2	One or more connectors are burnt out.		Turn off the system's main power source On each Connection Box, turn the switch to the Off position. Disconnect and thoroughly inspect the Ballast Module connectors. IF replacement is needed, contact Atlantium Technical Support.
3	The power supply inside the Ballast Module is faulty.	 	In a system with more than one lamp, if possible, switch between the Ballast Module connectors in order to verify the Ballast is faulty, if so replace the Ballast Module. In a system with only one lamp, replace the Ballast Module with a new one.
4	A lamp is faulty/burned out.	 	In a system with more than one lamp, switch between the lamps in order to verify the lamp is faulty, if so replace the lamp. In a system with only one lamp, replace the lamp with a new one. See ***Replacing a UV Lamp*** on page **123.**
5	The ID number of the Connection Box card does not match ID number of the Ballast Module.		On the each Connection Box, open the top cover and locate the 3-digit display (as shown on the right). Check that the set number is identical to the number marked on corresponding Ballast Module. If not identical, use the switch to change it to the correct number. 
6	One or more of the fuses in the Connection Box is burned out.		Replace the burned out fuse in the Connection Box. Consult your Atlantium representative. Contact Atlantium support for further instructions.
7	The Connection Box card is faulty.		On the Connection Box, open the top cover and check that the 3-digit display and LEDs are lit. If they are dark, replace the Connection Box card. Contact Atlantium support for further instructions.

11.7 Temperature sensor fault

Control Module displays this warning message
Temperature Sensor Fault

	Possible Cause	Solution
1	If this message appeared right after a message indicating loss of communication with the Ballast Module (and both messages refer to the same lamp) and the temperature display on the monitor is 165°C/365°F, then most likely the communication between the **Control Module** and the Connection box is lost.	Follow the solutions offered in ***Ballast temperature is too high***.
2	The temperature reading dropped to 0°C / 32°F due to a short-circuited in the Temperature sensor, Temperature converter card or Connection Box Card	Replace the temperature sensor. See ***Replacing the Temperature Sensor*** on page **147.** If the problem persists replace the Temperature Converter card. See ***Replacing the Temperature Converter Card*** on page **148.** If the problem still persists replace the Connection Box card. Contact Atlantium support for further instructions.

11.8 Ballast temperature is too high

Control Module displays this warning message
Ballast Temp Too high Lamp ID Fault

	Possible Cause	Solution
1	The Connection Box environment's temperature is above 80 degrees Celsius.	Check the environment temperature in the Connection Box area, open the connection box and look for a possible cause for the heating.

11.9 System type settings error

Control Module displays this warning message
Unset Error Fault at Lamp ID

	Possible Cause	Solution
1	**Lamp Unset:** The System is set to the wrong system model. (Example: System is **RZ163**, but the **Control Module** settings read **RZ300**.)	In the **Control Module**, access the **Technician screen #2**. Verify that the system is set to the correct system model, number of lamps and branches. If the system is already set to the correct model, you might need to refresh the selection: **Step 1.** Change the model to a different one (e.g. if your system is **RZ163**, change the selection to **RZ300**). **Step 2.** Exit the **Technician** screen and close the **Control Module** circuit breaker. **Step 3.** Open the **Control Module** circuit breaker and enter **Technician** screen. **Step 4.** Change the model back to the correct model of your system. **Step 5.** Exit the technician screen and close the **Control Module** circuit breaker. **Step 6.** Open the **Control Module** circuit breaker and turn the lamps on.
2	Your system is supposed to contain High Power lamps, but actually does not contain this type of lamp(s).	Make sure the lamps inside the system are the High Power type. Refer to ***Replacing a UV Lamp*** on page **10-123** for details on checking the lamps.

11.10 Lamp mains failure

Control Module displays this warning message
Lamp Mains Fail Lamp ID Fault

	Possible Cause	Solution	
1	**Mains Fail issue:** The ballast does not receive enough power from the mains.		A certified electrician is required to verify that the Ballast Module receives the proper voltage feed from the Mains via the Connection Box.
2	One or more of the fuses in the Connection Box is burned out.		A certified electrician is required to replace the burned out fuse in the Connection Box. Contact Atlantium support for further instructions.
3	A combined problem inside the Ballast Module regarding power feed and temperature.	Contact Atlantium support for further instructions.	

11.11 Unexpected failure for an unknown reason

Control Module displays this warning message
Unexpected Failure Unknown Reason ID Fault

	Possible Cause	Solution
	Lamp Unknown: The lamp was unexpectedly shutdown due to one of the following reasons:	
1	There is a power interruption (for micro seconds) that caused the Ballast Module to stop working while the Connection Box Card still held power.	Turn the lamps back on using the On/Off button on the **Control Module**'s main screen.

	Possible Cause	Solution	
2	The Lamp burned out during operation.		Replace the Lamp. See ***Replacing a UV Lamp*** on page **123.**
3	The Lamp's connector(s) burned during operation.		Replace the connectors. Contact Atlantium support for further instructions.
4	Ballast Module malfunctioned during operation.		Replace the Ballast Module.

11.12 Lost communication with ballast module

Control Module displays this warning message
Lost Communication with Ballast Module

	Possible Cause	Solution	
1	**Communication Error:** The communication between the **Control Module**'s display card and Connection Box card was lost while the lamp was on or during the ignition process.		Verify that all Connection Box cards have power and the communication wire is connected properly on both ends. If the problem persists, contact Atlantium support for further instructions.
2	No power is feeding into the Connection Box.		Open the Connection boxes and verify that all Connection Box s have power.
3	A wire is loose.		With a small flat screwdriver go over all the communication screws in the **Control Module** and Connections Box s and make sure all wires are located properly in their place and are well tightened. .

	Possible Cause	Solution
4	A jumper is missing.	A communication jumper might need to be installed, contact Atlantium Support for further instruction.
5	Ground is not connected to the **Control Module**'s power inlet.	A certified electrician is required to verify that the **Control Module** is properly grounded.
6	There is a Hardware problem in the **Control Module**'s display card or in the Connection Box.	Contact Atlantium support for further instructions.

11.13 Clean system deposit

Control Module displays this warning message
Please Clean System Deposit

		Possible Cause
1	Excessive mineral deposit has collected on the quartz sleeve.	Perform the Cleaning in Place procedure. See ***Cleaning In Place (CIP)*** on page **111.**
2	The **UVT Min** value in the Technician screen is set too high.	Set the **UVT Min** value to **0**.

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Appendix A Modbus Communication Protocol

Modbus is an application layer messaging protocol for client/server communication between devices connected on different types of buses or networks.

Modbus is often used to connect a supervisory computer with a remote terminal unit (RTU) in supervisory control and data acquisition systems.

For information on how to configure the communication properties and make the cable connections, see ***Modbus Communication Connections*** on page **80**.

ATTENTION!

Modbus registers are meant to be used for specific commands in a command algorithm. Do not write to any register if you do not intend to use it.

A.1 Modbus Registry Map

The table on the following pages provides the information you need to help you configure your network to properly communicate with the Atlantium system.

Registry address	Registry name	Divide	Type	Comment	Units	Read/ Write
215	UVTS lamp 4	:100	Register		mA	R
205	UVTS lamp 3	:100	Register		mA	R
121	UVTS lamp 2	:100	Register		mA	R
113	UVTS lamp 1	:100	Register		mA	R
210	UVT lamp 4	:100	Register		%	R
200	UVT lamp 3	:100	Register		%	R
115	UVT lamp 2	:100	Register		%	R
107	UVT lamp 1	:100	Register		%	R
100	UVT	:100	Register		%	R
214	UVIS lamp 4	:100	Register		mA	R
204	UVIS lamp 3	:100	Register		mA	R
120	UVIS lamp 2	:100	Register		mA	R
112	UVIS lamp 1	:100	Register		mA	R
110	Temperature °C	:100	Register		°C	R
19	System Ready		Coil Bit	1 -> Ready		R
103	System Power	:1	Register		%	R/W

0	System On/Off		Coil Bit	0 -> Off, 1 -> On		R/W
217	Status lamp 4	:1	Register			R
216	Status lamp 3	:1	Register			R
114	Status lamp 2	:1	Register		%	R
106	Status lamp 1	:1	Register		%	R
125	Set for (dose)	:10	Register			RW
213	Power lamp 4	:100	Register		KW	R
203	Power lamp 3	:100	Register		KW	R
119	Power lamp 2	:100	Register		KW	R
111	Power lamp 1	:100	Register		KW	R
11	Over Temp		Coil Bit	1 -> Fault, 0 -> Ok		R
2	Operation Mode Dose/ Power		Coil Bit	1 -> Dose, 0 -> Power		R
127	Minimum Dose Alarm		Coil Bit	UNCHECK - 0 , CHECHED = 1		RW
126	Minimum Dose	:10	Register			RW
14	Min flow alarm		Coil Bit	1 -> Fault, 0 -> Ok		R
128	Maximum Dose	:10	Register			RW
26	Lamp 4 Enable/ Disable		Coil Bit	1 -> Enable, 0 -> Disable		R/W
25	Lamp 3 Enable/ Disable		Coil Bit	1 -> Enable, 0 -> Disable		R/W
24	Lamp 2 Enable/ Disable		Coil Bit	1 -> Enable, 0 -> Disable		R/W
23	Lamp 1 Enable/ Disable		Coil Bit	1 -> Enable, 0 -> Disable		R/W
16	General alarm		Coil Bit	1 -> OK		R
101	Flow	:10	Register		m^3/h	R
146	DPM 4 Counter Low	:1	Register			R
147	DPM 4 Counter High	:1	Register			R
144	DPM 3 Counter Low	:1	Register			R
145	DPM 3 Counter High	:1	Register			R
142	DPM 2 Counter Low	:1	Register			R
143	DPM 2 Counter High	:1	Register			R
140	DPM 1 Counter Low	:1	Register			R
141	DPM 1 Counter High	:1	Register			R
9	Dose < min dose		Coil Bit	1 -> Fault, 0 -> Ok		R
102	Dose	:1	Register		mJ/cm^2	R
18	Critical Over Temp		Coil Bit	1 -> OK		R
17	Cleaning System exist		Coil Bit	0 -> Exist		R
15	Cleaner System Off/ On		Coil Bit	1 -> OK, 0 -> Fault		R
6	Alarm/Ready output		Coil Bit	0 -> Alarm, 1 -> Ready		R

211	Age hours lamp 4 MSB	:1	Register		Hours	R
201	Age hours lamp 3 MSB	:1	Register		Hours	R
116	Age hours lamp 2 MSB	:1	Register		Hours	R
108	Age hours lamp 1 MSB	:1	Register		Hours	R

Appendix B Checking the System Parameters

You can use the checklist below to keep a record of your checks on Atlantium system parameters. **Print this page**.

Table B-1: Atlantium System Parameters Checklist

Date	Time	Lamp ID	Power	Flow Rate	UVT	UV Dose	Lamp Status	OK Y/N	Remarks/ Alarms/Events
			Add Expected Value _____	Add Expected Value _____	Add Expected Value _____	Add Expected Value _____	Add Expected Value _____		

Table B-1: Atlantium System Parameters Checklist

Date	Time	Lamp ID	Power	Flow Rate	UVT	UV Dose	Lamp Status	OK Y / N	Remarks / Alarms / Events
			Add Expected Value _____	Add Expected Value _____	Add Expected Value _____	Add Expected Value _____	Add Expected Value _____		

Appendix C System Messages

The table below details the message that can appear on the **Control Module**'s screen or in the Event logs. A number of parameters are User Defined and include setting a value and upon reaching, exceeding or falling below it, an Action is set to be triggered. The Action is also configurable. The options are:

- **None** - The event is not considered an error, so no corrective action is taken. No warning message is issued.
- **Caution** - The caution alerts you to an anomaly based on the user preferences.
- **Warning** - One or more warning messages are issued and it triggers a pop-up window message and a General Alarm (if configured, see ***Configuring the General Alarm*** on page **61**).
- **Shut-Down** - In case of a serious error or dangerous condition, the system shuts down and triggers a pop-up window message and a General Alarm (if configured, see ***Configuring the General Alarm*** on page **61**).

The other parameters are system defined. The type column in the table below details the type of parameter.

Table C-1: **Control Module**'s Messages

Key	Type	Event Log	Line Message / Popup Title	Popup Message
1	The value is User Defined	Over Max. Dose	The dose is above the maximum dose	The dose is above the maximum defined in the system settings.
2	The value is User Defined (The default is None)	Dose Too Low	The UV dose is too low	Dose too low! Below minimum!
4	Warning - The default temperature threshold is 5°C (9°F) below what is defined for Over Temp	Temp. Too High	Temperature is Too High	The lamp's temperature getting too high -- check water flow.
5	Shutdown - - The default temperature threshold is User Defined	Over Temp	Temperature is Critically High!	The lamp's temperature is too high and the unit will shut down.
6	Shutdown - Hardware malfunction	Temperature Sensor Fault	Temperature Sensor Fault	One of the temperature sensors has malfunctioned or the communication with the Connection Box card was lost.
9	Shutdown ---The Minimum Flow is User Defined	Low Flow	The rate of the water flow is too low	The water flow rate is lower than the minimum flow rate. Check the water flow.

Table C-1: **Control Module**'s Messages

Key	Type	Event Log	Line Message / Popup Title	Popup Message
10	The value is User Defined	High Flow	The rate of the water flow is too high	The water flow rate is higher than the maximum flow rate allowed. Check the water flow.
11	The value is User Defined	Transmission Low	UVT is too low!	UVT is below the required value
14	Warning	Lamp Ignition	UV Lamp Ignition Failed!	The UV lamp has failed to ignite due to an electronic fault. The UV lamp may be burnt out or a wire may be disconnected.
16	Appears in the Event Log only - pops up from additional info from Event Log	User On		The Control Module screen or an external signal turned on the system.
17	Appears in the Event Log only - pops up from additional info from Event Log	User Off		The Control Module screen or an external signal shut down the system.
18	Appears in the Event Log only - pops up from additional info from Event Log	Auto On		System power was restored or flow raise above minimum. The lamps reignite.
19	Appears in the Event Log only - pops up from additional info from Event Log	Auto Off		Flow rate below user low flow setting; automatic shutdown.
20	Appears in the Event Log only - pops up from additional info from Event Log	Lamps On		System Ready. All the enabled lamps are on.
24	Caution	Dose On Spec	Dose On Spec	The UV dose being delivered by the system is above the minimum validated value
25	Warning	Dose Off Spec	Dose Off Spec	The UV dose being delivered by the system is below the minimum validated value
26	Caution	UVT On Spec	UVT On Spec	The UVT value is above the minimum validated UVT
27	Warning	UVT Off Spec	UVT Off Spec	The UVT value is below the minimum validated UVT
28	Caution	Flow On Spec	Flow On Spec	The flow rate is below the maximum validated flow
29	Warning	Flow Off Spec	Flow Off Spec	The flow rate is above the maximum validated flow

Table C-1: **Control Module**'s Messages

Key	Type	Event Log	Line Message / Popup Title	Popup Message
33	Warning	UVT 4th calibration failed	UVT Analyzer 4th calibration check failed	UVT Analyzer calibration check failed 4 times in a row. System is off-spec
34	The value is User Defined (Default as None)	Validated Min UVT	Validated Min UVT	UVT is below Validated Min UVT
35	The value is User Defined (Default as None)	Validated Max Flow	Validated Max Flow	Flow is above Validated Max Flow
39	Appears in the Event Log only - pops up from additional info from Event Log	Power Failure	Power Failure	The system was turned off due to power failure
41	The value is User Defined	UV Dose	The UV Dose is below the set level	The UV Dose is below the Operate At dose
42	Warning - Sensor needs to be replaced or there may be a loose wire	Temp. Sensor Fault	Temperature Sensor Fault	Fault on lamp temperature sensor
43	Shutdown	Lamp Shutdown	The UV lamp unexpectedly shutdown	Lamp shutdown
45	Shutdown	PS Fault	Power supply fault	Ballast fault [value displayed], error code: [number displayed].
46	Warning	Communication Error	Communication Error	Lost communication with the Ballast Module
54	Caution	UVIS is calibrated successfully	UVIS is calibrated successfully	UVIS is calibrated successfully
55	Caution	UVIS calibration failed	UVIS Calibration Failed	UVIS calibration failed, Sensor is out of calibration. Correction Factor is implemented
56	Caution	UVT successful calibration check	UVT successful calibration check	UVT Analyzer successful calibration check
57	Caution	UVT check failed	UVT Analyzer check failed	UVT Analyzer check failed; Auto recalibration was done

Appendix D Glossary of Terms

Term	Definition
Auxiliaries	Valves, flow devices, etc. essential to the workings of the Atlantium system but not directly a part of the Atlantium system.
Ballast	The power supply that translates the electrical power to UV power.
Connection Box	The box that sits on top of each lamp unit and holds the circuit board for the Electrical Cabinet or Ballast Module electrical configurations. Not relevant for Ballast Module/Cable-style.
Control Module	Control Module that controls operations and dose measurement. Comes in two versions: Standard and Premium. The Standard has a monochromatic screen and several key features. The Premium has a color screen can accommodate a full complement of reporting and trending features.
DPM Cleaner	The Deposit Prevention Mechanism for cleaning the quartz sleeves.
External On/Off	This enables an external source (SCADA, PLC, Process) to turn the Atlantium system on and off.
Electrical configuration styles: ***Atlantium provides a choice of electrical configurations.***	
Electrical Cabinet-style (EC)	**EC** - Electrical Cabinet.
Ballast Module	Waterproof case for the ballast which connects via a consolidated plug from the circuit board which sits on top of each lamp unit.
General Alarm	This enables the Atlantium unit to send an alarm to a buzzer, cell phone or other destination.
Inlet/Outlet Valves	Isolating Valves on the Inlet/outlet sides.
Lamp Chamber	The chamber which holds the lamp.
Lifting Support	The lifting element which can also be used to help support the Unit.
Pinhole	This is a small disc that is fitted for each sensor port to ensure that the right amount of light penetrates.
Portable CIP System	The portable unit Atlantium sells that connects to the Unit to clean and disinfect it without disassembling. It includes several kits – a valve kit, an accessories kit and the CIP kit itself.
Quartz Chamber	The **quartz pipe** in which the disinfection occurs.

Term	Definition
System Ready Signal	The signal the Atlantium unit provides to tell the production system that the Atlantium System is ready to supply properly disinfected water.
System Tuning	This is process is performed when key components are replaced or changed to synchronize them.
Triangle Support Brackets	RZ-Series only- brackets that can be used to support and balance a unit.
Ultrasonic Cleaner	Ultrasonic cleaning system.
UV Intensity Sensor	The sensor that measures the UV intensity in the Lamp chamber and feeds the real time dose calculation.
UVT Analyzer	The sensor that measures the lamp UV intensity through the water. Together with the UV Intensity sensor and the Control Module, it analyzes, tracks and reports the water quality.